PT3S Documentation

Release 90.11.01.13.dev1

3S

Jul 10, 2019

CONTENTS

Use SIR 3S Modeldata and SIR 3S Calculationresults in pure Python. With pandas, matplotlib and others. For documentation, test, verification, analysis, reporting, prototyping, play.

CHAPTER

ONE

README

The mandatory README.rst File.

Version-String in conf.py and setup.py

sphinx-build -b html . _build was used to build html-doc

sphinx-build -b latex . _build was used to build latex

pdflatex –extra-mem-bot=10000000 PT3S.tex was used to build PDF (executed in _build; PT3S.PDF is generated in _build; copied to ..)

CHAPTER

TWO

MX

SIR 3S MX-Interface (short: MX)

MX is a file based, channel-oriented interface for SIR 3S' calculation results.

This module contains stuff to utilize SIR 3S' MX calculation results in pure Python.

SIR 3S MX calculation results overview:

- Binary .MXS-Files contain the SIR 3S calculations results.
- A Model calculation run creates at least one .MXS-File.
- There is one .MX1-File (an XML-File) for the corresponding .MXS-File(s).
- This .MX1-File defines (in XML) a sequence of MX-Channels in the corresponding .MXS-File(s).
- And - as a result - the Byte-Layout of a single Record in the corresponding .MXS-File(s).
- A single Record is called a MX3-Record. .MXS-File(s) contain a sequence of MX3-Records.
- A MX3-Record contains calculation results for one TIMESTAMP. TIMESTAMP ist Scenariotime.
- A corresponding .MX3-File contains one MX3-Record - the last Scenariotime calculated.
- Summary:
- .MXS-File(s): MX3-Records: MX-Channels
- MX3-Record Byte-Layout (the MX-Channels) defined in corresponding .MX1-File.

A MX-Channel can be:

- a single Value
- or a Vector: Sequence of Values of the same Type:
  - for all Objects of a certain Type or (called Vectorchannels)
  - number of interior Points for all Pipes (special Vectorchannels: Pipevectorchannels)
  - Vectors with ATTRTYPE in: {'SVEC', 'PVECMIN_INST', 'PVECMAX_INST'}

For Vectorchannels (including Pipevectorchannels) the sequence of Objects is defined in the .MX2-File.

```
>>> # ---
>>> # SETUP
>>> # ---
>>> import os
>>> import time
>>> import logging
>>> logger = logging.getLogger('PT3S.Mx')
```

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```
>>> # ---
>>> # path
>>> # ---
>>> if __name__ == "__main__":
...     try:
...         dummy=__file__
...         logger.debug("{0:s}{1:s}{2:s}".format('DOCTEST: __main__ Context: ','path =␣
↪os.path.dirname(__file__)'," ."))
...         path = os.path.dirname(__file__)
...     except NameError:
...         logger.debug("{0:s}{1:s}{2:s}".format('DOCTEST: __main__ Context: ","path =
↪'.' because __file__ not defined and: "," from Mx import Mx"))
...         path = '.'
...         from Mx import Mx
... else:
...     logger.debug("{0:s}{1:s}{2:s}{3:s}".format('DOCTEST: Not __main__ Context: ','_
↪_name__: ',__name__,"path = '."))
...     path = '.'
>>> # ---
>>> # testDir
>>> # ---
>>> # globs={'testDir':'testdata'}
>>> try:
...     dummy= testDir
... except NameError:
...     testDir='testdata'
>>> # ---
>>> # dotResolution
>>> # ---
>>> # globs={'dotResolution':''}
>>> try:
...     dummy= dotResolution
... except NameError:
...     dotResolution=''
>>> import zipfile
>>> import pandas as pd
>>> # ---
>>> # Init
>>> # ---
>>> h5File=os.path.join(path,os.path.join(testDir,'OneLPipe.h5'))
>>> mx1File=os.path.join(path,os.path.join(testDir,'WDOneLPipe\B1\V0\BZ1\M-1-0-1
↪'+dotResolution+'.MX1'))
>>> mx=Mx(mx1File=mx1File,NoH5Read=True,NoMxsRead=True)
>>> isinstance(mx.mx1Df,pd.core.frame.DataFrame) # MX1-Content
True
>>> isinstance(mx.df,type(None)) # MXS-Content
True
>>> # ---
>>> # Clean Up
>>> # ---
>>> mx.delFiles()
>>> if os.path.exists(mx.mxsZipFile):
...     os.remove(mx.mxsZipFile)
>>> mxsDumpFile=mx.mxsFile+'.dump'
>>> if os.path.exists(mxsDumpFile):
...     os.remove(mxsDumpFile)
>>> # ---
```

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```
>>> # 1st Read MXS
>>> # ---
>>> logger.debug("{0:s}: 1st Read MXS".format('DOCTEST'))
>>> mx.setResultsToMxsFile() # looks for M-1-0-1.MXS in same Dir
4
>>> isinstance(mx.df,pd.core.frame.DataFrame) # MXS-Content
True
>>> rowsDf,colsDf = mx.df.shape
>>> (firstTime,lastTime,rows)=mx._checkMxsVecsFile()
>>> rowsDf==rows
True
>>> mx.df.index[0]==firstTime
True
>>> # ---
>>> # Write H5
>>> # ---
>>> if os.path.exists(mx.h5File):
...     os.remove(mx.h5File)
>>> mx.ToH5() # M-1-0-1.h5 in same Dir
>>> os.path.exists(mx.h5File)
True
>>> # ---
>>> # Init with H5
>>> # ---
>>> mx=Mx(mx1File=mx1File) # looks for M-1-0-1.h5 in same Dir
>>> # and reads the .h5 if newer than .MX1 and newer than an existing .MXS
>>> isinstance(mx.mx1Df,pd.core.frame.DataFrame) # MX1-Content
True
>>> isinstance(mx.df,pd.core.frame.DataFrame) # MXS-Content
True
>>> # ---
>>> # 1st Read MXS Zip
>>> # ---
>>> # create the Zip first
>>> with zipfile.ZipFile(mx.mxsZipFile,'w') as myzip:
...     myzip.write(mx.mxsFile)
>>> logger.debug("{0:s}: 1st Read MXS Zip".format('DOCTEST'))
>>> mx.setResultsToMxsZipFile() # looks for M-1-0-1.ZIP in same Dir
4
>>> isinstance(mx.df,pd.core.frame.DataFrame) # MXS-Content
True
>>> rowsMxs,colsMxs = mx.df.shape
>>> mx.df.index.is_unique # all setResultsTo... will ensure this uniqueness under all␣
↪circumstances
True
>>> # uniqueness under all circumstances: also when add=True (setResultsTo... shall␣
↪add the MXS-Content) is used
>>> # ---
>>> # 1st Add same MXS (for testing ensuring uniqueness)
>>> # ---
>>> oldShape=mx.df.shape
>>> logger.debug("{0:s}: 1st Add same MXS (for testing ensuring uniqueness)".format(
↪'DOCTEST'))
>>> mx.setResultsToMxsFile(add=True) # looks for M-1-0-1.MXS in same Dir
0
>>> newShape=mx.df.shape
>>> newShape==oldShape
```

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```
True
>>> # ---
>>> # 1st Add same Zip (for testing ensuring uniqueness)
>>> # ---
>>> logger.debug("{0:s}: 1st Add same Zip (for testing ensuring uniqueness)".format(
↪'DOCTEST'))
>>> mx.setResultsToMxsZipFile(add=True) # looks for M-1-0-1.ZIP in same Dir
0
>>> newShape=mx.df.shape
>>> newShape==oldShape
True
>>> # ---
>>> # 1st Read MXS Zip with overlapping Timestamps (for testing ensuring uniqueness)
>>> # ---
>>> with zipfile.ZipFile(mx.mxsZipFile,'w') as myzip:
...     myzip.write(mx.mxsFile)
...     myzip.write(mx.mxsFile,arcname=mx.mxsFile+'.2')
>>> logger.debug("{0:s}: 1st Read MXS Zip with overlapping Timestamps (for testing␣
↪ensuring uniqueness)".format('DOCTEST'))
>>> mx.setResultsToMxsZipFile() # looks for M-1-0-1.ZIP in same Dir
4
>>> newShape=mx.df.shape
>>> newShape==oldShape
True
>>> # ---
>>> # shift to younger Timestamps (for testing purposes)
>>> # ---
>>> lastTimestamp=mx.df.index[-1]
>>> firstTimestamp=mx.df.index[0]
>>> timeSpan=lastTimestamp-firstTimestamp
>>> if len(mx.df.index)>1:
...     timeStep=mx.df.index[-1]-mx.df.index[-2]
... else:
...     timeStep=pd.to_timedelta('1 second')
>>> mx.df.index=mx.df.index-(timeSpan+timeStep)
>>> # ---
>>> # 1st Read MXS (with the original Timestamps)
>>> # ---
>>> logger.debug("{0:s}: 1st Read MXS (with the original Timestamps)".format('DOCTEST
↪'))
>>> mx.setResultsToMxsFile(add=True) # looks for M-1-0-1.MXS in same Dir
4
>>> rowsNew,colsNew=mx.df.shape
>>> rowsOld,colsOld=oldShape
>>> rowsNew==2*rowsOld
True
>>> colsNew==colsOld
True
>>> # ---
>>> # shift to older Timestamps (for testing purposes)
>>> # ---
>>> lastTimestamp=mx.df.index[-1]
>>> firstTimestamp=mx.df.index[0]
>>> timeSpan=lastTimestamp-firstTimestamp
>>> if len(mx.df.index)>1:
...     timeStep=mx.df.index[-1]-mx.df.index[-2]
... else:
```

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```
...       timeStep=pd.to_timedelta('1 second')
>>> mx.df.index=mx.df.index+(timeSpan+timeStep)
>>> # ---
>>> # 2nd Read MXS (with the original Timestamps)
>>> # ---
>>> logger.debug("{0:s}: 2nd Read MXS (with the original Timestamps)".format('DOCTEST
↪'))
>>> mx.setResultsToMxsFile(add=True) # looks for M-1-0-1.MXS in same Dir
4
>>> rowsNew,colsNew=mx.df.shape
>>> rowsNew==3*rowsOld
True
>>> colsNew==colsOld
True
>>> # ---
>>> # Write Dump
>>> # ---
>>> mx.dumpInMxsFormat() # dumps to .MXS.dump-File in same Dir
(12, 8)
>>> # ---
>>> # Read Dump
>>> # ---
>>> logger.debug("{0:s}: Read Dump".format('DOCTEST'))
>>> mx.setResultsToMxsFile(mxsFile=mxsDumpFile)
12
>>> with zipfile.ZipFile(mx.mxsZipFile,'w') as myzip:
...     myzip.write(mx.mxsFile)
...     myzip.write(mxsDumpFile)
>>> # ---
>>> # Read Zip with Orig and Dump
>>> # ---
>>> logger.debug("{0:s}: Read Zip with Orig and Dump".format('DOCTEST'))
>>> mx.setResultsToMxsZipFile()
12
>>> (rows,cols)=mx.df.shape
>>> rows
12
>>> mx.ToH5()
>>> # ---
>>> # Without MX1, MXS
>>> # ---
>>> os.rename(mx.mx1File,mx.mx1File+'.blind')
>>> os.rename(mx.mxsFile,mx.mxsFile+'.blind')
>>> mx=Mx(mx1File=mx1File)
>>> os.rename(mx.mx1File+'.blind',mx.mx1File)
>>> os.rename(mx.mxsFile+'.blind',mx.mxsFile)
>>> # ---
>>> sir3sIdTimestamp=mx.mx1Df['Sir3sID'].iloc[mx.idxTIMESTAMP]
>>> mx.mx1Df['Sir3sID'][mx.mx1Df['Sir3sID']==sir3sIdTimestamp].index[0]
0
>>> mx.mx1Df['unpackIdx'][mx.mx1Df['Sir3sID']==sir3sIdTimestamp].iloc[0]
0
>>> (rows,cols)=mx.df.shape
>>> rows
12
>>> isinstance(mx.df.index[0],pd.tslib.Timestamp)
True
```

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```
>>> str(mx.df.index[0])
'2018-03-03 00:00:00+00:00'
>>> sir3sId=mx.mx1Df['Sir3sID'][mx.mx1Df['Sir3sID'].str.contains('KNOT~\S*~~
↪5642914844465475844~QM')].iloc[0] #KNOT~I~~5642914844465475844~QM
>>> ts=mx.df[sir3sId]
>>> isinstance(ts,pd.core.series.Series)
True
>>> "{:06.2f}".format(round(ts.iloc[0],2))
'176.71'
>>> logger.debug("{:s}: CHANGEHISTORY: {:>10s}: {:>3d}: {:>6s}: {:s}".format('DOCTEST
↪','0.0.31',1,'New','_checkMxsVecsFile: (...,fullCheck=False,...)'))
>>> mx._checkMxsVecsFile()
(Timestamp('2018-03-03 00:00:00+0000', tz='UTC'), Timestamp('2018-03-03 00:00:11+0000
↪', tz='UTC'), 12)
>>> mx._checkMxsVecsFile(fullCheck=True)
(Timestamp('2018-03-03 00:00:00+0000', tz='UTC'), Timestamp('2018-03-03 00:00:11+0000
↪', tz='UTC'), 12)
>>> logger.debug("{:s}: CHANGEHISTORY: {:>10s}: {:>3d}: {:>6s}: {:s}".format('DOCTEST
↪','0.0.32',1,'Change','setResultsToMxsFile: finally: h5.close()'))
>>> logger.debug("{:s}: CHANGEHISTORY: {:>10s}: {:>3d}: {:>6s}: {:s}".format('DOCTEST
↪','0.0.32',2,'Change','setResultsToMxsZipFile: finally: h5.close()'))
>>> logger.debug("{:s}: CHANGEHISTORY: {:>10s}: {:>3d}: {:>6s}: {:s}".format('DOCTEST
↪','0.0.32',3,'Change','ToH5: finally: h5.close()'))
>>> logger.debug("{:s}: CHANGEHISTORY: {:>10s}: {:>3d}: {:>6s}: {:s}".format('DOCTEST
↪','0.0.32',4,'Change','FromH5: finally: h5.close()'))
>>> logger.debug("{:s}: CHANGEHISTORY: {:>10s}: {:>3d}: {:>6s}: {:s}".format('DOCTEST
↪','0.0.32',5,'Change','*: except Exception as e'))
>>> logger.debug("{:s}: CHANGEHISTORY: {:>10s}: {:>3d}: {:>6s}: {:s}".format('DOCTEST
↪','0.0.32',6,'Change','setResultsToMxsFile: finally: NewH5Vec=False'))
>>> logger.debug("{:s}: CHANGEHISTORY: {:>10s}: {:>3d}: {:>6s}: {:s}".format('DOCTEST
↪','0.0.32',7,'Change','__init__(...,NoH5Read=True,...)'))
>>> if os.path.exists(mx.h5FileVecs):
...     os.remove(mx.h5FileVecs)
>>> mx=Mx(mx1File=mx1File,NoH5Read=True)
>>> os.path.exists(mx.h5FileVecs) # h5 was written (h5 did not exist)
True
>>> h5VecsFileTime=os.path.getmtime(mx.h5FileVecs)
>>> mx=Mx(mx1File=mx1File,NoH5Read=True) # h5 will be written again (h5 exists)
>>> h5VecsFileTimeNow=os.path.getmtime(mx.h5FileVecs)
>>> logger.debug("1 h5VecsFileTime:{:s} < h5VecsFileTimeNow:{:s}".format(time.
↪strftime("%Y-%m-%d %H:%M:%S %z",time.gmtime(h5VecsFileTime)),time.strftime("%Y-%m-
↪%d %H:%M:%S %z",time.gmtime(h5VecsFileTimeNow))))
>>> h5VecsFileTime<h5VecsFileTimeNow # 1 h5 was written again (h5 did exist before)
True
>>> h5VecsFileTime=os.path.getmtime(mx.h5FileVecs)
>>> mx=Mx(mx1File=mx1File) # h5 is read - not written
>>> h5VecsFileTimeNow=os.path.getmtime(mx.h5FileVecs)
>>> logger.debug("2 h5VecsFileTime:{:s} == h5VecsFileTimeNow:{:s}".format(time.
↪strftime("%Y-%m-%d %H:%M:%S %z",time.gmtime(h5VecsFileTime)),time.strftime("%Y-%m-
↪%d %H:%M:%S %z",time.gmtime(h5VecsFileTimeNow))))
>>> mx.setResultsToMxsFile() # h5 will not be updated
0
>>> h5VecsFileTimeNow=os.path.getmtime(mx.h5FileVecs)
>>> logger.debug("3 h5VecsFileTime:{:s} == h5VecsFileTimeNow:{:s}".format(time.
↪strftime("%Y-%m-%d %H:%M:%S %z",time.gmtime(h5VecsFileTime)),time.strftime("%Y-%m-
↪%d %H:%M:%S %z",time.gmtime(h5VecsFileTimeNow))))
>>> mx.setResultsToMxsFile(NewH5Vec=True) # h5 is written
```

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```
4
>>> h5VecsFileTimeNow=os.path.getmtime(mx.h5FileVecs)
>>> logger.debug("h5VecsFileTime:{:s} == h5VecsFileTimeNow:{:s}".format(time.strftime(
↪"%Y-%m-%d %H:%M:%S %z",time.gmtime(h5VecsFileTime)),time.strftime("%Y-%m-%d %H:%M:
↪%S %z",time.gmtime(h5VecsFileTimeNow))))
>>> logger.debug("{:s}: CHANGEHISTORY: {:>10s}: {:>3d}: {:>6s}: {:s}".format('DOCTEST
↪','0.0.32',8,'Change','setResultsToMxsZipFile: finally: NewH5Vec=False'))
>>> mx.ToH5()
>>> mx=Mx(mx1File=mx1File,NoH5Read=True)
>>> os.path.exists(mx.h5File)
False
>>> pd.set_option('display.max_columns',None)
>>> pd.set_option('display.max_rows',None)
>>> pd.set_option('display.max_colwidth',666666)
>>> pd.set_option('display.width',666666666)
>>> print(mx._getMx1DfAsOneString().replace('\n','\n '))
ATTRTYPE  DATALENGTH DATATYPE  DATATYPELENGTH  FLAGS OBJTYPE           OBJTYPE_PK    ␣
↪           TITLE    UNIT
     PH           4     REAL               4   1265    KNOT  5289899964753656852    ␣
↪           Druck   [bar]
     QM           4     REAL               4   1265    KNOT  5289899964753656852 ␣
↪Externer Durchfluss  [m3/h]
     PH           4     REAL               4   1265    KNOT  5642914844465475844    ␣
↪           Druck   [bar]
     QM           4     REAL               4   1265    KNOT  5642914844465475844 ␣
↪Externer Durchfluss  [m3/h]
>>> print("'''{:s}'''".format(repr(mx.mx2Df).replace('\n','\n   ')))
'''       AttrType                                        Data  DataLength DataType ␣
↪DataTypeLength  NOfItems       ObjType
   0  tk            [5642914844465475844, 5289899964753656852]          40     CHAR  ␣
↪           20         2  KNOT
   1  pk                                  [5252810657060947333]          20     CHAR  ␣
↪           20         1  LFKT
   2  pk                                  [5502689500012692689]          20     CHAR  ␣
↪           20         1  PHI1
   3  pk                                  [5732781659713982525]          20     CHAR  ␣
↪           20         1  PUMD
   4  pk                                  [5163733225086798083]          20     CHAR  ␣
↪           20         1  PVAR
   5  pk                                  [4742976321174242828]          20     CHAR  ␣
↪           20         1  QVAR
   6  tk                                  [4737064599036143765]          20     CHAR  ␣
↪           20         1  ROHR
   7  N_OF_POINTS                                        (101,)           4     INT4  ␣
↪            4         1  ROHR
   8  pk                                  [5396761270498593493]          20     CHAR  ␣
↪           20         1  SWVT        '''
>>> print(mx._getDfAsOneString())
                         KNOT~~~5289899964753656852~PH KNOT~~~5289899964753656852~QM␣
↪KNOT~~~5642914844465475844~PH KNOT~~~5642914844465475844~QM
2018-03-03 00:00:00+00:00                          0.0                          -176.7␣
↪                         4.2                          176.7
2018-03-03 00:00:01+00:00                          0.0                          -176.7␣
↪                         4.2                          176.7
2018-03-03 00:00:02+00:00                          0.0                          -176.7␣
↪                         4.2                          176.7
2018-03-03 00:00:03+00:00                          0.0                          -176.7␣
↪                         4.2                          176.7
```

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```
>>> # ---
>>> # Clean Up OneLPipe
>>> # ---
>>> mx.delFiles()
>>> if os.path.exists(mx.mxsZipFile):
...     os.remove(mx.mxsZipFile)
>>> mxsDumpFile=mx.mxsFile+'.dump'
>>> if os.path.exists(mxsDumpFile):
...     os.remove(mxsDumpFile)
>>> # ---
>>> # LocalHeatingNetwork
>>> # ---
>>> mx1File=os.path.join(path,os.path.join(testDir,'WDLocalHeatingNetwork\B1\V0\BZ1\M-
↪1-0-1'+dotResolution+'.MX1'))
>>> mx=Mx(mx1File=mx1File,NoH5Read=True,NoMxsRead=True)
>>> # ---
>>> # Clean Up LocalHeatingNetwork
>>> # ---
>>> mx.delFiles()
>>> if os.path.exists(mx.mxsZipFile):
...     os.remove(mx.mxsZipFile)
>>> mxsDumpFile=mx.mxsFile+'.dump'
>>> if os.path.exists(mxsDumpFile):
...     os.remove(mxsDumpFile)
>>> mx.setResultsToMxsFile(maxRecords=1)
1
>>> print(mx._getDfAsOneString())
                          KNOT~~~5356267303828212700~PH KNOT~~~5397990465339071638~QM␣
↪KNOT~~~5736262931552588702~PH
2004-09-22 08:30:00+00:00                           2.0                           0.0␣
↪                          4.1
>>> logger.debug("{:s}: CHANGEHISTORY: {:>10s}: {:>3d}: {:>6s}: {:s}".format('DOCTEST
↪','0.0.41',1,'New',"getMxsVecsFileData"))
>>> timesReq=[]
>>> timesReq.append(mx.df.index[0])
>>> plotTimeDfs=mx.getMxsVecsFileData(timesReq=timesReq)
>>> len(plotTimeDfs)
1
>>> isinstance(plotTimeDfs[0],pd.core.frame.DataFrame)
True
>>> print(mx._getDfVecAsOneString(df=plotTimeDfs[0],regex='KNOT~\S*~\S*~\S*~T$'))
                         KNOT~~~~T
2004-09-22 08:30:00+00:00      60.00
>>> timesReq[0]=timesReq[0]-pd.to_timedelta('1 second')
>>> plotTimeDfs=mx.getMxsVecsFileData(timesReq=timesReq)
>>> len(plotTimeDfs)
0
>>> # ---
>>> # Clean Up LocalHeatingNetwork
>>> # ---
>>> mx.delFiles()
>>> if os.path.exists(mx.mxsZipFile):
...     os.remove(mx.mxsZipFile)
>>> mxsDumpFile=mx.mxsFile+'.dump'
>>> if os.path.exists(mxsDumpFile):
...     os.remove(mxsDumpFile)
>>> # ---
```

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```
>>> # TinyWDN
>>> # ---
>>> mx1File=os.path.join(path,os.path.join(testDir,'WDTinyWDN\B1\V0\BZ1\M-1-0-1
↪'+dotResolution+'.MX1'))
>>> mx=Mx(mx1File=mx1File,NoH5Read=True,NoMxsRead=True)
>>> print(mx._getMx1DfAsOneString().replace('\n','\n '))
ATTRTYPE  DATALENGTH DATATYPE  DATATYPELENGTH  FLAGS OBJTYPE           OBJTYPE_PK␣
↪TITLE    UNIT
     PH           4     REAL               4      1    KNOT  4711309381204507891    ␣
↪    [bar]
     PH           4     REAL               4      1    KNOT  5179406559406617933    ␣
↪    [bar]
     QM           4     REAL               4      1    KNOT  5179406559406617933    ␣
↪   [m3/h]
>>> print("'''{:s}'''".format(repr(mx.mx2Df).replace('\n','\n   ')))
'''        AttrType                                                                ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                                          Data  DataLength DataType ␣
↪DataTypeLength  NOfItems       ObjType
   0   tk                                                                          ␣
↪                                        [5179406559406617933, 5706345341889312301,␣
↪5028754475676510796, 4880261452311588026, 4711309381204507891, 5697271655044179265,␣
↪5165939128645634755, 4790715298669926433, 5629305599838467353, 5388350113283448399,␣
↪4702108116010765829, 5095105260880965618, 5113971272348625691, 5042575626021291052,␣
↪4780213881308610359, 4832703654265095420, 5498009282312522569]         340     CHAR␣
↪             20        17  KNOT
   1   pk                                                                          ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                        [4673068329861421062]          20     CHAR ␣
↪            20         1  LFKT
   2   tk                                                                          ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                        [4914542339545953765]          20     CHAR ␣
↪            20         1  OBEH
   3   pk                                                                          ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                        [5295290952622367198]          20     CHAR ␣
↪            20         1  PHI1
   4   pk                                                                          ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                        [5623478865255556069]          20     CHAR ␣
↪            20         1  PUMD
```

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```
  5   pk                                                                    ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                        [4728046487162557851]          20     CHAR ␣
↪            20         1  PVAR
  6   pk                                                                    ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                        [5377391575499483345]          20     CHAR ␣
↪            20         1  QVAR
  7   tk            [4820249423461369400, 5508607066407516090, 5095827458323717990,␣
↪5700891034663173564, 5729543164571807746, 5067763789937102564, 4799443678518131207,␣
↪5605831797210762998, 5126294008398890918, 5419686039699182683, 5312551747276023841,␣
↪5296152509037292880, 5670035893444309530, 5433880705192526755, 4648047345314768819,␣
↪4984438795139137900, 5644872080928983958, 5148090523913666712, 5461179577260327606,␣
↪4978978527327130204, 5497762617222653432, 5076321356874807093]         440     CHAR␣
↪             20        22  ROHR
  8   N_OF_POINTS                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                          (5,␣
↪6, 5, 3, 4, 3, 3, 3, 3, 3, 3, 3, 3, 2, 5, 4, 3, 3, 4, 3, 2, 2)          88     INT4␣
↪              4        22  ROHR
  9   pk                                                                    ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                        [4955975670134047353]          20     CHAR ␣
↪            20         1  SWVT
  10  pk                                                                    ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                        [4821755615165519990]          20     CHAR ␣
↪            20         1  TEVT
  11  tk                                                                    ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                        [5466655470152247657]          20     CHAR ␣
↪            20         1  VENT
  12  pk                                                                    ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                        [5736734929574151957]          20     CHAR ␣
↪            20         1  WEVT        '''
>>> mx.setResultsToMxsFile(maxRecords=1)
```

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```
1
>>> print(mx._getDfAsOneString())
                        KNOT~~~4711309381204507891~PH KNOT~~~5179406559406617933~PH␣
↪KNOT~~~5179406559406617933~QM
2002-05-22 16:16:16+00:00                        1.0                          3.0␣
↪                       140.0
>>> # ---
>>> # Clean Up Tiny WDN
>>> # ---
>>> mx.delFiles()
>>> # ---
>>> # GPipe
>>> # ---
>>> mx1File=os.path.join(path,os.path.join(testDir,'WDGPipe\B1\V0\BZ1\M-1-0-1
↪'+dotResolution+'.MX1'))
>>> mx=Mx(mx1File=mx1File)
>>> plotTimeDfs=mx.getMxsVecsFileData()
>>> len(plotTimeDfs[0]['ROHR~*~*~*~PVEC'].iloc[0]) == mx.mx2Df[mx.mx2Df['AttrType'].
↪str.contains('N_OF_POINTS')].iloc[0].Data[0]
True
>>> plotTimeDfs[0]['ROHR~*~*~*~PVEC'].iloc[0][0]
41.0
>>> # ---
>>> # Clean Up GPipe
>>> # ---
>>> mx.delFiles()
```

class Mx.**Mx**(*mx1File*, *NoH5Read=False*, *NoMxsRead=False*)

Reading SIR 3S' MX-Files.

Summary:

- (mx1File): use this to profit from previous reads finalized with ToH5()
- (mx1File,NoH5Read=True): use this for a fresh start with implicit .MXS-File read; finalize with ToH5()
- (mx1File,NoH5Read=True,NoMxsRead=True): use this for a fresh start; call setResultsTo...() explicit; finalize with ToH5()
- note that base.Y.h5-File has to be dunped explicitely with ToH5()
- and base.Y.vec.h5-File is written implicitely while (implicit or explicit calls to) setResultsTo...()
- and is deleted explicitely (mx1File,NoH5Read=True) or implicitely (because it is i.e. to old)

Args:

- mx1File (str): base.MX1-File (an XML-File) (base.Y.MX1-File from 90-10 on)
- **NoH5Read (bool):**

  False (default - use this to profit from previous reads finalized with ToH5()):

  - **If a base.Y.h5-File**
    - exists
    - and is newer (>) than an .MX1-File (base.Y.Mx1-File from 90-10 on)
    - and is newer (>) than an .MXS-File (base.Y.MXS-File from 90-10 on):
      - The base.Y.h5-File is read instead of the .MX1-File.

True (use this for a fresh start):

- An base.Y.h5-File is deleted if existing.
- The base.Y.Mx1-File is read.
- The base.Y.vec.h5-File is newly created in case of an .MXS-File read.

- **NoMxsRead (bool):**

  True:

  - a base.Y.MXS-File is not read
  - a base.Y.vec.h5-File is not touched

  False (default):

  - **If a base.Y.MXS-File**
    - exists
    - and is newer (>=) than base.Y.Mx1-File
    - and base.Y.h5-File is not read:
      - The base.Y.MXS-File is read.
      - NoH5Read=True will delete base.Y.vec.h5-File.

Attributes:

- **states**
  - h5Read: True, if read from H5
- **fileNames**
  - .mx1File: base.Y.MX1-File

  derived from mx1File

  - .mx2File: base.MX2-File
  - .mxsFile: base.Y.MXS-File
  - .mxsZipFile base.ZIP
  - constructed from MX during Init and Usage:
  -
  - .h5File: base.Y.h5-File
  - .h5FileVecs: base.Y.vec.h5-File: MXS-H5Dump written implicitely
  - .h5FileMx1FmtString: base.Y.h5-File.metadata written implicitely

- **.mxRecordStructFmtString**
  - usage: struct.unpack(self.mxRecordStructFmtString,a_MXS_Record)
  - **.h5FileMx1FmtString:**
    - it was not possible to store mxRecordStructFmtString in H5-Format as Metadata
    - therefore mxRecordStructFmtString is stored in a file named .h5FileMx1FmtString
    - in .h5File the Link to this file is stored as Metadata

        * as pointed out with usage above mxRecordStructFmtString has nothing to do with writing to or reading from H5
        * mxRecordStructFmtString is only about reading from (and writing to for test purposes) MXS
        * the .mxRecordStructFmtString/.h5FileMx1FmtString stuff is only about performance:
        * if after reading from H5 only a(nother) MXS shall be read again ...
        * ... the stuff avoids the time-consuming reconstruction of mxRecordStructFmtString
    - **dataFrames**
      - .mx1Df
      - .mx2Df
      - **.df**
        * the base.Y.MXS-File(s) Content
        * non Vectordata only
        * index: TIMESTAMP (scenario time)
        * **columns: Values**
          · The following (String-)ID - called Sir3sID - is used as Columnlabel:
          · this Sir3sID consists of ~ (Mx.reSir3sIDSep) separated .MX1-File terms:
          · OBJTYPE~NAME1~NAME2~OBJTYPE_PK~ATTRTYPE
          · Sir3sID regExp Example: Mx.reSir3sID: (?P<OBJTYPE>S+)~(?P<NAME1>[S]*)~(?P<NAME1>S*)~(?P<OBJTYPE_PK>d+)~(?P<ATTRTYPE>S+)'
      - **.dfVecAggs**
        * some base.Y.vec.h5-File Aggregates as df
        * **MultiIndex:**
          · TYPE: SNAPSHOTTYPEs: TIME,TMIN,TMAX (from _readMxsFile) or Aggregates from 2 Times: MIN,MAX,... (from getVecAggs)
          · Sir3sID
          · TIMESTAMPL
          · TIMESTAMPR
        * Cols: mx2Idx

Raises: MxError

delFiles()

Deletes Files constructed by MX during Init and Usage.

_initWithMx1()

(Re-)initialize .mx1Df, .mx2Df, .mxRecordStructFmtString and related stuff with .mx1File.

Calls:

- ._parseMx1()
- ._parseMx2()
- ._buildMxRecordStructUnpackFmtString()

- ._buildMxRecordStructUnpackFmtStringPost()

Raises: MxError

_parseMx1()

Parses .mx1File.

Sets

- .mx1Df

Raises: MxError

_parseMx2()

Parses .mx2File.

Sets

- .mx2Df

Raises: MxError

_buildMxRecordStructUnpackFmtString()

(Re-)builds .mxRecordStructFmtString and releated stuff.

Sets

- .mxRecordStructFmtString
- .mx1Df['unpackIdx']

Raises: MxError

_buildMxRecordStructUnpackFmtStringPost()

Stuff todo after buildMxRecordStructUnpackFmtString.

Sets

- .idxCVERSO (idx of CVERSO in MX1)
- .unpackIdxCVERSO (idx of CVERSO in recordData)
- .idxTIMESTAMP (idx of TIMESTAMP in MX1)
- .unpackIdxTIMESTAMP (idx of TIMESTAMP in recordData)
- .mxColumnNames=[] (of non Vectordata without TIMESTAMP in MX1-Sequence)
- .mxColumnNamesVecs=[] (of Vectordata without TIMESTAMP in MX1-Sequence)
- .idxUnpackNonVectorChannels[] (idx in recordData)
- .idxUnpackVectorChannels[] (idx in recordData of the 1st ([0]) Element of the Vector)
- .idxOfNonVectorChannels[] (idx in MX1 without TIMESTAMP)
- .idxVectorChannels[] (idx in MX1)

Raises: MxError

_readMxsFile(*mxsFilePtr*, *mxsVecsH5StorePtr*, *firstTime=None*, *maxRecords=None*)

Args:

- mxsFilePtr: .MXS-File
- mxsVecsH5StorePtr: .vec.h5-File
- **firstTime: used to calculate h5Key for mxsVecsH5Store**

  - None (default): firstTime is set to 1st TIMESTAMP in .MXS-File
  - **else:**
    * caller sets firstTime - in general the youngest TIMESTAMP in the df to be extended
- maxRecords

Returns: df: the mxsFile (non Vectordata only) as DataFrame

- index: TIMESTAMP
- df.index.is_unique
  - might be False
  - because of SIR 3S'
  - 1st Time twice (SNAPSHOTTYPE: STAT+TIME) and Last Time triple (SNAPSHOTTYPE: TIME+TMIN/TMAX)

timesWrittenToMxsVecs: the times written to the mxsVecsH5Store

mxsVecsH5StorePtr

- is used to update the mxsVecsH5Store with mxsFile-Content
  - **Key: microseconds from firstTime**
    * can be all negative if firstTime (in general the youngest TIMESTAMP in the df to be extended) is older than the oldest TIMESTAMP in the .MXS-File
  - **dfVecs**
    * index: TIMESTAMP
    * values: Vectordata for the TIMESTAMP
- the Vectordata for a TIMESTAMP is only written when the Key does _not already exist

Raises: MxError

```
>>> mx=mxs['LocalHeatingNetwork']
>>> mx.delFiles()
>>> mx.setResultsToMxsFile() # reads 5 TIMESTAMPS and constructs .vec.h5␣
↪while reading
5
>>> mx.dfVecAggs.loc[(['TIME','TMIN','TMAX'],'KNOT~*~*~*~PH',slice(None),
↪slice(None)),0:22]
                                                                    0         1␣
↪        2         3         4         5         6         7         8       ␣
↪  9         10        11        12        13   14        15        16       ␣
↪ 17        18   19        20        21        22
TYPE Sir3sID       TIMESTAMPL          TIMESTAMPR                              ␣
↪                                                                              ␣
↪                                                                              ␣
↪
TIME KNOT~*~*~*~PH 2004-09-22 08:30:00 2004-09-22 08:30:00  2.302971  3.
↪985846  4.083384  4.121495  2.043288  2.283566  2.004937  4.311307  4.
↪126019  2.309655  4.291591  2.000133  2.141440  3.825970  2.0  3.819467  2.
↪314658  3.816599  2.312659  2.0  3.845104  4.125885  3.814690
TMIN KNOT~*~*~*~PH 2004-09-22 08:30:00 2004-09-22 08:31:00  2.052100  2.
↪183028  2.200011  2.206647  2.007717  2.048865  2.000910  2.248923  2.
↪207463  2.053240  2.234365  2.000021  2.025138  2.156905  2.0  2.155822  2.
↪054124  2.155325  2.053771  2.0  2.160025  2.207441  2.154995
```
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```
TMAX KNOT~*~*~*~PH 2004-09-22 08:30:00 2004-09-22 08:31:00  2.302971  4.
↪085822  4.183360  4.221471  2.043288  2.283566  2.004937  4.411284  4.
↪225996  2.309655  4.391567  2.000133  2.141440  3.925947  2.0  3.919444  2.
↪314658  3.916576  2.312659  2.0  3.945080  4.225861  3.914667
>>> mx.dfVecAggs
                                                                                 ␣
↪0           1             2           3           4           5           6␣
↪          7           8           9           10          11          12    ␣
↪      13          14          15          16          17          18        ␣
↪  19          20          21          22          23          24          ␣
↪25          26          27          28          29          30          31
TYPE Sir3sID                 TIMESTAMPL          TIMESTAMPR                   ␣
↪                                                                              ␣
↪                                                                              ␣
↪                                                                              ␣
↪                                                                              ␣
↪
TIME ROHR~*~*~*~QMI          2004-09-22 08:30:00 2004-09-22 08:30:00   -8.
↪509475   19.059780 -1.537890e+01    8.509476  -22.987946   22.987946   22.
↪987947   -3.928166   22.987947   15.378901    3.928167  -22.987946  -19.
↪059778   -3.928166    3.928167  -22.987946         NaN         NaN         ␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN     ␣
↪    NaN
    KNOT~*~*~*~P            2004-09-22 08:30:00 2004-09-22 08:30:00    3.
↪302971    4.985846  5.083384e+00    5.121495    3.043288    3.283566    3.
↪004937    5.311307    5.126019    3.309655    5.291591    3.000133    3.
↪141440    4.825970    3.000000    4.819467    3.314658    4.816599    3.
↪312659    3.000000    4.845104    5.125885    4.814690         NaN         ␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    FWES~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:30:00   22.
↪987947         NaN           NaN         NaN         NaN         NaN         ␣
↪ NaN         NaN         NaN         NaN         NaN         NaN         ␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN     ␣
↪    NaN         NaN         NaN         NaN         NaN         NaN         ␣
↪NaN
    KNOT~*~*~*~HMAX_INST    2004-09-22 08:30:00 2004-09-22 08:30:00    2.
↪302971    3.985846  4.083384e+00    4.121495    2.043288    2.283566    2.
↪004937    4.311307    4.126019    2.309655    4.291591    2.000133    2.
↪141440    3.825970    2.000000    3.819467    2.314658    3.816599    2.
↪312659    2.000000    3.845104    4.125885    3.814690         NaN         ␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    ROHR~*~*~*~PVECMIN_INST 2004-09-22 08:30:00 2004-09-22 08:30:00    3.
↪302971    3.309655  4.985845e+00    4.845103    3.283566    3.302971    4.
↪825970    4.819467    3.000133    3.004937    5.083383    4.985845    5.
↪125884    5.121495    3.312659    3.314658    5.121495    5.083383    4.
↪845103    4.825970    4.819467    4.816599    3.043288    3.141441    3.
↪141441    3.283566    3.309655    3.312659    4.816599    4.814690    3.
↪004937    3.043288
    VENT~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:30:00   22.
↪987947   22.987946  2.199973e-06         NaN         NaN         NaN         ␣
↪ NaN         NaN         NaN         NaN         NaN         NaN         ␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN     ␣
↪    NaN         NaN         NaN         NaN         NaN         NaN         ␣
↪NaN
```

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```
    KNOT~*~*~*~H            2004-09-22 08:30:00 2004-09-22 08:30:00    2.
↪302971    3.985846  4.083384e+00    4.121495    2.043288    2.283566    2.
↪004937    4.311307    4.126019    2.309655    4.291591    2.000133    2.
↪141440    3.825970    2.000000    3.819467    2.314658    3.816599    2.
↪312659    2.000000    3.845104    4.125885    3.814690         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    ROHR~*~*~*~SVEC         2004-09-22 08:30:00 2004-09-22 08:30:00    0.
↪000000   88.019997  0.000000e+00  405.959991    0.000000   83.550003    0.
↪000000   88.019997    0.000000   73.419998    0.000000  195.529999    0.
↪000000   68.599998    0.000000  109.769997    0.000000   76.400002    0.
↪000000   83.550003    0.000000  164.910004    0.000000  195.529999    0.
↪000000  405.959991    0.000000  164.910004    0.000000  109.769997    0.
↪000000   76.400002
    ROHR~*~*~*~PVEC         2004-09-22 08:30:00 2004-09-22 08:30:00    3.
↪302971    3.309655  4.985845e+00    4.845103    3.283566    3.302971    4.
↪825970    4.819467    3.000133    3.004937    5.083383    4.985845    5.
↪125884    5.121495    3.312659    3.314659    5.121495    5.083383    4.
↪845103    4.825970    4.819467    4.816599    3.043288    3.141441    3.
↪141441    3.283566    3.309655    3.312659    4.816599    4.814690    3.
↪004937    3.043288
    ROHR~*~*~*~VI           2004-09-22 08:30:00 2004-09-22 08:30:00   -0.
↪266728    0.608561 -4.820491e-01    0.271700   -0.321646    0.733984    0.
↪327641   -0.123128    0.733984    0.491034    0.125423   -0.720553   -0.
↪597426   -0.123128    0.125423   -0.720553         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    KNOT~*~*~*~T            2004-09-22 08:30:00 2004-09-22 08:30:00   60.
↪000000   90.000000  9.000000e+01   90.000000   60.000000   60.000000   60.
↪000000   60.000000   90.000000   60.000000   60.000000   60.000000   60.
↪000000   90.000000   60.000000   90.000000   60.000000   90.000000   60.
↪000000   60.000000   90.000000   90.000000   90.000000         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    KLAP~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:30:00   22.
↪987947         NaN           NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    ROHR~*~*~*~RHOVEC       2004-09-22 08:30:00 2004-09-22 08:30:00  983.
↪700012  983.700012  9.657000e+02  965.700012  983.700012  983.700012  965.
↪700012  965.700012  983.700012  983.700012  965.700012  965.700012  965.
↪700012  965.700012  983.700012  983.700012  965.700012  965.700012  965.
↪700012  965.700012  965.700012  965.700012  983.700012  983.700012  983.
↪700012  983.700012  983.700012  983.700012  965.700012  965.700012  983.
↪700012  983.700012
    ROHR~*~*~*~TVEC         2004-09-22 08:30:00 2004-09-22 08:30:00   60.
↪000000   60.000000  9.000000e+01   90.000000   60.000000   60.000000   90.
↪000000   90.000000   60.000000   60.000000   90.000000   90.000000   90.
↪000000   90.000000   60.000000   60.000000   90.000000   90.000000   90.
↪000000   90.000000   90.000000   90.000000   60.000000   60.000000   60.
↪000000   60.000000   60.000000   60.000000   90.000000   90.000000   60.
↪000000   60.000000
    FWES~*~*~*~V            2004-09-22 08:30:00 2004-09-22 08:30:00    1.
↪291413         NaN           NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
```

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```
    ROHR~*~*~*~MVEC         2004-09-22 08:30:00 2004-09-22 08:30:00   -2.
↪363743   -2.363743  5.294384e+00    5.294384   -4.271917   -4.271917    2.
↪363743    2.363743   -6.385540   -6.385540    6.385540    6.385540    6.
↪385541    6.385541   -1.091157   -1.091157    6.385540    6.385540    4.
↪271917    4.271917    1.091157    1.091157   -6.385540   -6.385540   -5.
↪294383   -5.294383   -1.091157   -1.091157    1.091157    1.091157   -6.
↪385540   -6.385540
    ROHR~*~*~*~VK           2004-09-22 08:30:00 2004-09-22 08:30:00   -0.
↪266728    0.608561 -4.820491e-01    0.271700   -0.321646    0.733984    0.
↪327641   -0.123128    0.733984    0.491034    0.125423   -0.720553   -0.
↪597426   -0.123128    0.125423   -0.720553         NaN         NaN␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN␣
↪    NaN
    PUMP~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:30:00   22.
↪987947         NaN           NaN         NaN         NaN         NaN␣
↪ NaN         NaN         NaN         NaN         NaN         NaN␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN␣
↪    NaN         NaN         NaN         NaN         NaN         NaN␣
↪NaN
    KNOT~*~*~*~PDAMPF       2004-09-22 08:30:00 2004-09-22 08:30:00    0.
↪199200    0.701100  7.011000e-01    0.701100    0.199200    0.199200    0.
↪199200    0.199200    0.701100    0.199200    0.199200    0.199200    0.
↪199200    0.701100    0.199200    0.701100    0.199200    0.701100    0.
↪199200    0.199200    0.701100    0.701100    0.701100         NaN␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    ROHR~*~*~*~PVECMAX_INST 2004-09-22 08:30:00 2004-09-22 08:30:00    3.
↪302971    3.309655  4.985845e+00    4.845104    3.283566    3.302971    4.
↪825970    4.819468    3.000133    3.004937    5.083383    4.985845    5.
↪125884    5.121495    3.312659    3.314659    5.121495    5.083383    4.
↪845104    4.825970    4.819468    4.816599    3.043288    3.141441    3.
↪141441    3.283566    3.309655    3.312659    4.816599    4.814690    3.
↪004937    3.043288
    VENT~*~*~*~V            2004-09-22 08:30:00 2004-09-22 08:30:00    0.
↪374182    0.367335  3.163897e-07         NaN         NaN         NaN␣
↪ NaN         NaN         NaN         NaN         NaN         NaN␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN␣
↪    NaN         NaN         NaN         NaN         NaN         NaN␣
↪NaN
    KNOT~*~*~*~HMIN_INST    2004-09-22 08:30:00 2004-09-22 08:30:00    2.
↪302971    3.985846  4.083384e+00    4.121495    2.043288    2.283565    2.
↪004937    4.311307    4.126019    2.309655    4.291591    2.000133    2.
↪141440    3.825970    2.000000    3.819467    2.314658    3.816599    2.
↪312659    2.000000    3.845104    4.125885    3.814690         NaN␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    ROHR~*~*~*~QMK          2004-09-22 08:30:00 2004-09-22 08:30:00   -8.
↪509475   19.059780 -1.537890e+01    8.509476  -22.987946   22.987946   22.
↪987947   -3.928166   22.987947   15.378901    3.928167  -22.987946  -19.
↪059778   -3.928166    3.928167  -22.987946         NaN         NaN␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN␣
↪    NaN
    KNOT~*~*~*~PMIN_INST    2004-09-22 08:30:00 2004-09-22 08:30:00    3.
↪302971    4.985846  5.083384e+00    5.121495    3.043288    3.283565    3.
↪004937    5.311307    5.126019    3.309655    5.291591    3.000133    3.
↪141440    4.825970    3.000000    4.819467    3.314658    4.816599    3.
↪312659    3.000000    4.845104    5.125885    4.814690         NaN␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
```

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```
    KNOT~*~*~*~RHO          2004-09-22 08:30:00 2004-09-22 08:30:00  983.
↪700012  965.700012  9.657000e+02  965.700012  983.700012  983.700012  983.
↪700012  983.700012  965.700012  983.700012  983.700012  983.700012  983.
↪700012  965.700012  983.700012  965.700012  983.700012  965.700012  983.
↪700012  983.700012  965.700012  965.700012  965.700012         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    ROHR~*~*~*~ZVEC         2004-09-22 08:30:00 2004-09-22 08:30:00   20.
↪000000   20.000000  2.000000e+01   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000
    KNOT~*~*~*~PMAX_INST    2004-09-22 08:30:00 2004-09-22 08:30:00    3.
↪302971    4.985846  5.083384e+00    5.121495    3.043288    3.283566    3.
↪004937    5.311307    5.126019    3.309655    5.291591    3.000133    3.
↪141440    4.825970    3.000000    4.819468    3.314658    4.816599    3.
↪312659    3.000000    4.845104    5.125885    4.814691         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    KLAP~*~*~*~V            2004-09-22 08:30:00 2004-09-22 08:30:00    1.
↪291413         NaN           NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    KNOT~*~*~*~WALTER       2004-09-22 08:30:00 2004-09-22 08:30:00    0.
↪209004    0.161072  8.707356e-02    0.058160    0.403910    0.207487    0.
↪433363    0.496769    0.000000    0.286059    0.496769    0.496769    0.
↪328532    0.393637    0.000000    0.483626    0.000000    0.848857    0.
↪247643    0.496769    0.346373    0.000000    1.091969         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    KNOT~*~*~*~PH           2004-09-22 08:30:00 2004-09-22 08:30:00    2.
↪302971    3.985846  4.083384e+00    4.121495    2.043288    2.283566    2.
↪004937    4.311307    4.126019    2.309655    4.291591    2.000133    2.
↪141440    3.825970    2.000000    3.819467    2.314658    3.816599    2.
↪312659    2.000000    3.845104    4.125885    3.814690         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    FWVB~*~*~*~W            2004-09-22 08:30:00 2004-09-22 08:30:00  160.
↪000000  200.000000  1.600000e+02  160.000000  120.000000         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    FWVB~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:30:00    3.
↪928166    6.869426  4.581308e+00    3.928166    3.680879         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    ROHR~*~*~*~QMAV         2004-09-22 08:30:00 2004-09-22 08:30:00   -8.
↪509475   19.059780 -1.537890e+01    8.509476  -22.987946   22.987948   22.
↪987947   -3.928166   22.987947   15.378901    3.928167  -22.987946  -19.
↪059778   -3.928166    3.928167  -22.987946         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
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```
    ROHR~*~*~*~VAV          2004-09-22 08:30:00 2004-09-22 08:30:00   -0.
↪266728    0.608561 -4.820491e-01    0.271700   -0.321646    0.733984    0.
↪327641   -0.123128    0.733984    0.491034    0.125423   -0.720553   -0.
↪597426   -0.123128    0.125423   -0.720553         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    KNOT~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:30:00    0.
↪000000    0.000000  0.000000e+00    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    ROHR~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:30:00    0.
↪000000    0.000000  0.000000e+00    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    VENT~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:30:00    0.
↪000000    0.000000  0.000000e+00         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    KLAP~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:30:00    0.
↪000000         NaN         NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    PUMP~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:30:00    0.
↪000000         NaN         NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    FWVB~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:30:00    0.
↪000000    0.000000  0.000000e+00    0.000000    0.000000         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    FWES~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:30:00    0.
↪000000         NaN         NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
TMIN ROHR~*~*~*~QMI          2004-09-22 08:30:00 2004-09-22 08:31:00   -8.
↪509475    7.394749 -1.537890e+01    3.256006  -22.987946    9.266181   -9.
↪266181   -3.928166    9.266181    5.923044    1.496261  -22.987946  -19.
↪059778   -3.928166    1.496260  -22.987946         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
```

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```
    KNOT~*~*~*~P            2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪052100    3.183028  3.200011e+00    3.206647    3.007717    3.048865    3.
↪000910    3.248923    3.207463    3.053240    3.234365    3.000021    3.
↪025138    3.156905    3.000000    3.155822    3.054124    3.155325    3.
↪053771    3.000000    3.160025    3.207441    3.154995         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    FWES~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:31:00    9.
↪266181         NaN           NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    KNOT~*~*~*~HMAX_INST    2004-09-22 08:30:00 2004-09-22 08:31:00    2.
↪302971    3.985846  4.083384e+00    4.121495    2.043288    2.283565    2.
↪004937    4.311307    4.126019    2.309655    4.291591    2.000133    2.
↪141440    3.825970    2.000000    3.819467    2.314658    3.816599    2.
↪312659    2.000000    3.845104    4.125885    3.814690         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    ROHR~*~*~*~PVECMIN_INST 2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪052100    3.053240  3.183028e+00    3.160025    3.048864    3.052100    3.
↪156905    3.155822    3.000022    3.000910    3.200011    3.183028    3.
↪207441    3.206647    3.053771    3.054124    3.206647    3.200011    3.
↪160025    3.156905    3.155822    3.155326    3.007717    3.025139    3.
↪025139    3.048864    3.053240    3.053771    3.155326    3.154995    3.
↪000910    3.007717
    VENT~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:31:00    9.
↪266181    9.266179  2.199715e-06         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    KNOT~*~*~*~H            2004-09-22 08:30:00 2004-09-22 08:31:00    2.
↪052100    2.183028  2.200011e+00    2.206647    2.007717    2.048865    2.
↪000910    2.248923    2.207463    2.053240    2.234365    2.000021    2.
↪025138    2.156905    2.000000    2.155822    2.054124    2.155325    2.
↪053771    2.000000    2.160025    2.207441    2.154995         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    ROHR~*~*~*~SVEC         2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000   88.019997  0.000000e+00  405.959991    0.000000   83.550003    0.
↪000000   88.019997    0.000000   73.419998    0.000000  195.529999    0.
↪000000   68.599998    0.000000  109.769997    0.000000   76.400002    0.
↪000000   83.550003    0.000000  164.910004    0.000000  195.529999    0.
↪000000  405.959991    0.000000  164.910004    0.000000  109.769997    0.
↪000000   76.400002
    ROHR~*~*~*~PVEC         2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪052100    3.053240  3.183028e+00    3.160025    3.048864    3.052100    3.
↪156905    3.155822    3.000022    3.000910    3.200011    3.183028    3.
↪207441    3.206647    3.053771    3.054124    3.206647    3.200011    3.
↪160025    3.156905    3.155822    3.155326    3.007717    3.025139    3.
↪025139    3.048864    3.053240    3.053771    3.155326    3.154995    3.
↪000910    3.007717
    ROHR~*~*~*~VI           2004-09-22 08:30:00 2004-09-22 08:31:00   -0.
↪266728    0.236108 -4.820491e-01    0.103961   -0.321646    0.295861   -0.
↪132068   -0.123128    0.295861    0.189117    0.047774   -0.720553   -0.
↪597426   -0.123128    0.047774   -0.720553         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
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    KNOT~*~*~*~T            2004-09-22 08:30:00 2004-09-22 08:31:00   60.
↪000000   90.000000  9.000000e+01   90.000000   60.000000   60.000000   60.
↪000000   60.000000   90.000000   60.000000   60.000000   60.000000   60.
↪000000   90.000000   60.000000   90.000000   60.000000   90.000000   60.
↪000000   60.000000   90.000000   90.000000   90.000000         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    KLAP~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:31:00    9.
↪266181         NaN           NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    ROHR~*~*~*~RHOVEC       2004-09-22 08:30:00 2004-09-22 08:31:00  983.
↪700012  983.700012  9.657000e+02  965.700012  983.700012  983.700012  965.
↪700012  965.700012  983.700012  983.700012  965.700012  965.700012  965.
↪700012  965.700012  983.700012  983.700012  965.700012  965.700012  965.
↪700012  965.700012  965.700012  965.700012  983.700012  983.700012  983.
↪700012  983.700012  983.700012  983.700012  965.700012  965.700012  983.
↪700012  983.700012
    ROHR~*~*~*~TVEC         2004-09-22 08:30:00 2004-09-22 08:31:00   60.
↪000000   60.000000  9.000000e+01   90.000000   60.000000   60.000000   90.
↪000000   90.000000   60.000000   60.000000   90.000000   90.000000   90.
↪000000   90.000000   60.000000   60.000000   90.000000   90.000000   90.
↪000000   90.000000   90.000000   90.000000   60.000000   60.000000   60.
↪000000   60.000000   60.000000   60.000000   90.000000   90.000000   60.
↪000000   60.000000
    FWES~*~*~*~V            2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪520554         NaN           NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    ROHR~*~*~*~MVEC         2004-09-22 08:30:00 2004-09-22 08:31:00   -2.
↪363743   -2.363743  2.054097e+00    2.054097   -4.271917   -4.271917    0.
↪904446    0.904446   -6.385540   -6.385540    2.573939    2.573939    2.
↪573939    2.573939   -1.091157   -1.091157    2.573939    2.573939    1.
↪645290    1.645290    0.415628    0.415628   -6.385540   -6.385540   -5.
↪294383   -5.294383   -1.091157   -1.091157    0.415628    0.415628   -6.
↪385540   -6.385540
    ROHR~*~*~*~VK           2004-09-22 08:30:00 2004-09-22 08:31:00   -0.
↪266728    0.236108 -4.820491e-01    0.103961   -0.321646    0.295861    0.
↪132068   -0.123128    0.295861    0.189117    0.047774   -0.720553   -0.
↪597426   -0.123128    0.047774   -0.720553         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    PUMP~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:31:00    9.
↪266181         NaN           NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    KNOT~*~*~*~PDAMPF       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪199200    0.701100  7.011000e-01    0.701100    0.199200    0.199200    0.
↪199200    0.199200    0.701100    0.199200    0.199200    0.199200    0.
↪199200    0.701100    0.199200    0.701100    0.199200    0.701100    0.
↪199200    0.199200    0.701100    0.701100    0.701100         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
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    ROHR~*~*~*~PVECMAX_INST 2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪302971    3.309655  4.985845e+00    4.845104    3.283566    3.302971    4.
↪825970    4.819468    3.000133    3.004937    5.083383    4.985845    5.
↪125884    5.121495    3.312659    3.314658    5.121495    5.083383    4.
↪845104    4.825970    4.819468    4.816599    3.043288    3.141441    3.
↪141441    3.283566    3.309655    3.312659    4.816599    4.814690    3.
↪004937    3.043288
    VENT~*~*~*~V            2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪150829    0.148069  3.163526e-07         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    KNOT~*~*~*~HMIN_INST    2004-09-22 08:30:00 2004-09-22 08:31:00    2.
↪052100    2.183028  2.200011e+00    2.206647    2.007717    2.048865    2.
↪000910    2.248923    2.207463    2.053240    2.234365    2.000021    2.
↪025138    2.156905    2.000000    2.155822    2.054124    2.155325    2.
↪053771    2.000000    2.160025    2.207441    2.154995         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    ROHR~*~*~*~QMK          2004-09-22 08:30:00 2004-09-22 08:31:00   -8.
↪509475    7.394749 -1.537890e+01    3.256006  -22.987946    9.266180    9.
↪266181   -3.928166    9.266181    5.923044    1.496261  -22.987946  -19.
↪059778   -3.928166    1.496260  -22.987946         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    KNOT~*~*~*~PMIN_INST    2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪052100    3.183028  3.200011e+00    3.206647    3.007717    3.048865    3.
↪000910    3.248923    3.207463    3.053240    3.234365    3.000021    3.
↪025138    3.156905    3.000000    3.155822    3.054124    3.155325    3.
↪053771    3.000000    3.160025    3.207441    3.154995         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    KNOT~*~*~*~RHO          2004-09-22 08:30:00 2004-09-22 08:31:00  983.
↪700012  965.700012  9.657000e+02  965.700012  983.700012  983.700012  983.
↪700012  983.700012  965.700012  983.700012  983.700012  983.700012  983.
↪700012  965.700012  983.700012  965.700012  983.700012  965.700012  983.
↪700012  983.700012  965.700012  965.700012  965.700012         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    ROHR~*~*~*~ZVEC         2004-09-22 08:30:00 2004-09-22 08:31:00   20.
↪000000   20.000000  2.000000e+01   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000
    KNOT~*~*~*~PMAX_INST    2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪302971    4.985846  5.083384e+00    5.121495    3.043288    3.283565    3.
↪004937    5.311307    5.126019    3.309655    5.291591    3.000133    3.
↪141440    4.825970    3.000000    4.819468    3.314658    4.816599    3.
↪312659    3.000000    4.845104    5.125885    4.814691         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    KLAP~*~*~*~V            2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪520554         NaN         NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
```

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```
    KNOT~*~*~*~WALTER       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪209004    0.161072  8.707356e-02    0.058160    0.403910    0.207487    0.
↪433363    0.496769    0.000000    0.286059    0.496769    0.496769    0.
↪328532    0.393637    0.000000    0.483626    0.000000    0.848857    0.
↪247643    0.496769    0.346373    0.000000    1.091969         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    KNOT~*~*~*~PH           2004-09-22 08:30:00 2004-09-22 08:31:00    2.
↪052100    2.183028  2.200011e+00    2.206647    2.007717    2.048865    2.
↪000910    2.248923    2.207463    2.053240    2.234365    2.000021    2.
↪025138    2.156905    2.000000    2.155822    2.054124    2.155325    2.
↪053771    2.000000    2.160025    2.207441    2.154995         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    FWVB~*~*~*~W            2004-09-22 08:30:00 2004-09-22 08:31:00   76.
↪226158   77.649529  6.145824e+01   60.944885   47.978928         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    FWVB~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:31:00    1.
↪871431    2.667038  1.759745e+00    1.496260    1.471705         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    ROHR~*~*~*~QMAV         2004-09-22 08:30:00 2004-09-22 08:31:00   -8.
↪509475    7.394749 -1.537890e+01    3.256006  -22.987946    9.266180    9.
↪266181   -3.928166    9.266181    5.923044    1.496261  -22.987946  -19.
↪059778   -3.928166    1.496260  -22.987946         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    ROHR~*~*~*~VAV          2004-09-22 08:30:00 2004-09-22 08:31:00   -0.
↪266728    0.236108 -4.820491e-01    0.103961   -0.321646    0.295861    0.
↪132068   -0.123128    0.295861    0.189117    0.047774   -0.720553   -0.
↪597426   -0.123128    0.047774   -0.720553         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    KNOT~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000    0.000000  0.000000e+00    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    ROHR~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000    0.000000  0.000000e+00    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    VENT~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000    0.000000  0.000000e+00         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
```

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```
    KLAP~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000         NaN           NaN         NaN         NaN         NaN        ␣
↪ NaN         NaN         NaN         NaN         NaN         NaN         ␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN    ␣
↪    NaN         NaN         NaN         NaN         NaN         NaN        ␣
↪NaN
    PUMP~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000         NaN           NaN         NaN         NaN         NaN        ␣
↪ NaN         NaN         NaN         NaN         NaN         NaN         ␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN    ␣
↪    NaN         NaN         NaN         NaN         NaN         NaN        ␣
↪NaN
    FWVB~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000    0.000000  0.000000e+00    0.000000    0.000000         NaN        ␣
↪ NaN         NaN         NaN         NaN         NaN         NaN         ␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN    ␣
↪    NaN         NaN         NaN         NaN         NaN         NaN        ␣
↪NaN
    FWES~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000         NaN           NaN         NaN         NaN         NaN        ␣
↪ NaN         NaN         NaN         NaN         NaN         NaN         ␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN    ␣
↪    NaN         NaN         NaN         NaN         NaN         NaN        ␣
↪NaN
TMAX ROHR~*~*~*~QMI          2004-09-22 08:30:00 2004-09-22 08:31:00   -3.
↪256005   19.059780 -5.923043e+00    8.509476   -9.266179   22.987946   22.
↪987947   -1.496260   22.987947   15.378901    3.928167   -9.266179   -7.
↪394748   -1.496260    3.928167   -9.266179         NaN         NaN        ␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN    ␣
↪    NaN
    KNOT~*~*~*~P            2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪302971    5.085822  5.183360e+00    5.221471    3.043288    3.283566    3.
↪004937    5.411284    5.225996    3.309655    5.391567    3.000133    3.
↪141440    4.925947    3.000000    4.919444    3.314658    4.916576    3.
↪312659    3.000000    4.945080    5.225861    4.914667         NaN        ␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    FWES~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:31:00   22.
↪987947         NaN           NaN         NaN         NaN         NaN        ␣
↪ NaN         NaN         NaN         NaN         NaN         NaN         ␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN    ␣
↪    NaN         NaN         NaN         NaN         NaN         NaN        ␣
↪NaN
    KNOT~*~*~*~HMAX_INST    2004-09-22 08:30:00 2004-09-22 08:31:00    2.
↪302971    4.085822  4.183360e+00    4.221471    2.043288    2.283566    2.
↪004937    4.411284    4.225996    2.309655    4.391567    2.000133    2.
↪141440    3.925947    2.000000    3.919444    2.314658    3.916576    2.
↪312659    2.000000    3.945080    4.225861    3.914667         NaN        ␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    ROHR~*~*~*~PVECMIN_INST 2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪302971    3.309655  4.985845e+00    4.845104    3.283566    3.302971    1.
↪825970    4.819468    3.000133    3.004937    5.083383    4.985845    5.
↪125884    5.121495    3.312659    3.314658    5.121495    5.083383    4.
↪845104    4.825970    4.819468    4.816599    3.043288    3.141441    3.
↪141441    3.283566    3.309655    3.312659    4.816599    4.814690    3.
↪004937    3.043288
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```
    VENT~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:31:00   22.
↪987947   22.987946  2.199973e-06         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    KNOT~*~*~*~H            2004-09-22 08:30:00 2004-09-22 08:31:00    2.
↪302971    4.085822  4.183360e+00    4.221471    2.043288    2.283566    2.
↪004937    4.411284    4.225996    2.309655    4.391567    2.000133    2.
↪141440    3.925947    2.000000    3.919444    2.314658    3.916576    2.
↪312659    2.000000    3.945080    4.225861    3.914667         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    ROHR~*~*~*~SVEC         2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000   88.019997  0.000000e+00  405.959991    0.000000   83.550003    0.
↪000000   88.019997    0.000000   73.419998    0.000000  195.529999    0.
↪000000   68.599998    0.000000  109.769997    0.000000   76.400002    0.
↪000000   83.550003    0.000000  164.910004    0.000000  195.529999    0.
↪000000  405.959991    0.000000  164.910004    0.000000  109.769997    0.
↪000000   76.400002
    ROHR~*~*~*~PVEC         2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪302971    3.309655  5.085822e+00    4.945080    3.283566    3.302971    4.
↪925946    4.919444    3.000133    3.004937    5.183360    5.085822    5.
↪225861    5.221471    3.312659    3.314659    5.221471    5.183360    4.
↪945080    4.925946    4.919444    4.916576    3.043288    3.141441    3.
↪141441    3.283566    3.309655    3.312659    4.916576    4.914667    3.
↪004937    3.043288
    ROHR~*~*~*~VI           2004-09-22 08:30:00 2004-09-22 08:31:00   -0.
↪102059    0.608561 -1.856568e-01    0.271700   -0.129652    0.733984    0.
↪327641   -0.046900    0.733984    0.491034    0.125423   -0.290447   -0.
↪231787   -0.046900    0.125423   -0.290447         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    KNOT~*~*~*~T            2004-09-22 08:30:00 2004-09-22 08:31:00   60.
↪000000   90.000000  9.000000e+01   90.000000   60.000000   60.000000   60.
↪000000   60.000000   90.000000   60.000000   60.000000   60.000000   60.
↪000000   90.000000   60.000000   90.000000   60.000000   90.000000   60.
↪000000   60.000000   90.000000   90.000000   90.000000         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    KLAP~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:31:00   22.
↪987947         NaN           NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    ROHR~*~*~*~RHOVEC       2004-09-22 08:30:00 2004-09-22 08:31:00  983.
↪700012  983.700012  9.657000e+02  965.700012  983.700012  983.700012  965.
↪700012  965.700012  983.700012  983.700012  965.700012  965.700012  965.
↪700012  965.700012  983.700012  983.700012  965.700012  965.700012  965.
↪700012  965.700012  965.700012  965.700012  983.700012  983.700012  983.
↪700012  983.700012  983.700012  983.700012  965.700012  965.700012  983.
↪700012  983.700012
    ROHR~*~*~*~TVEC         2004-09-22 08:30:00 2004-09-22 08:31:00   60.
↪000000   60.000000  9.000000e+01   90.000000   60.000000   60.000000   90.
↪000000   90.000000   60.000000   60.000000   90.000000   90.000000   90.
↪000000   90.000000   60.000000   60.000000   90.000000   90.000000   90.
↪000000   90.000000   90.000000   90.000000   60.000000   60.000000   60.
↪000000   60.000000   60.000000   60.000000   90.000000   90.000000   60.
↪000000   60.000000
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    FWES~*~*~*~V            2004-09-22 08:30:00 2004-09-22 08:31:00    1.
↪291413         NaN           NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    ROHR~*~*~*~MVEC         2004-09-22 08:30:00 2004-09-22 08:31:00   -0.
↪904446   -0.904446  5.294384e+00    5.294384   -1.645290   -1.645290    2.
↪363743    2.363743   -2.573939   -2.573939    6.385540    6.385540    6.
↪385541    6.385541   -0.415628   -0.415628    6.385540    6.385540    4.
↪271917    4.271917    1.091157    1.091157   -2.573939   -2.573939   -2.
↪054097   -2.054097   -0.415628   -0.415628    1.091157    1.091157   -2.
↪573939   -2.573939
    ROHR~*~*~*~VK           2004-09-22 08:30:00 2004-09-22 08:31:00   -0.
↪102059    0.608561 -1.856568e-01    0.271700   -0.129652    0.733984    0.
↪327641   -0.046900    0.733984    0.491034    0.125423   -0.290447   -0.
↪231787   -0.046900    0.125423   -0.290447         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    PUMP~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:31:00   22.
↪987947         NaN           NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    KNOT~*~*~*~PDAMPF       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪199200    0.701100  7.011000e-01    0.701100    0.199200    0.199200    0.
↪199200    0.199200    0.701100    0.199200    0.199200    0.199200    0.
↪199200    0.701100    0.199200    0.701100    0.199200    0.701100    0.
↪199200    0.199200    0.701100    0.701100    0.701100         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    ROHR~*~*~*~PVECMAX_INST 2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪302971    3.309655  5.085822e+00    4.945080    3.283566    3.302971    4.
↪925946    4.919444    3.000133    3.004937    5.183360    5.085822    5.
↪225861    5.221471    3.312659    3.314659    5.221471    5.183360    4.
↪945080    4.925946    4.919444    4.916576    3.043288    3.141441    3.
↪141441    3.283566    3.309655    3.312659    4.916576    4.914667    3.
↪004937    3.043288
    VENT~*~*~*~V            2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪374182    0.367335  3.163897e-07         NaN         NaN         NaN
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↪NaN
    KNOT~*~*~*~HMIN_INST    2004-09-22 08:30:00 2004-09-22 08:31:00    2.
↪302971    3.985846  4.083384e+00    4.121495    2.043288    2.283565    2.
↪004937    4.311307    4.126019    2.309655    4.291591    2.000133    2.
↪141440    3.825970    2.000000    3.819467    2.314658    3.816599    2.
↪312659    2.000000    3.845104    4.125885    3.814690         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    ROHR~*~*~*~QMK          2004-09-22 08:30:00 2004-09-22 08:31:00   -3.
↪256005   19.059780 -5.923043e+00    8.509476   -9.266179   22.987948   22.
↪987947   -1.496260   22.987947   15.378901    3.928167   -9.266179   -7.
↪394748   -1.496260    3.928167   -9.266179         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
```

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```
    KNOT~*~*~*~PMIN_INST    2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪302971    4.985846  5.083384e+00    5.121495    3.043288    3.283565    3.
↪004937    5.311307    5.126019    3.309655    5.291591    3.000133    3.
↪141440    4.825970    3.000000    4.819468    3.314658    4.816599    3.
↪312659    3.000000    4.845104    5.125885    4.814691         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    KNOT~*~*~*~RHO          2004-09-22 08:30:00 2004-09-22 08:31:00  983.
↪700012  965.700012  9.657000e+02  965.700012  983.700012  983.700012  983.
↪700012  983.700012  965.700012  983.700012  983.700012  983.700012  983.
↪700012  965.700012  983.700012  965.700012  983.700012  965.700012  983.
↪700012  983.700012  965.700012  965.700012  965.700012         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    ROHR~*~*~*~ZVEC         2004-09-22 08:30:00 2004-09-22 08:31:00   20.
↪000000   20.000000  2.000000e+01   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000
    KNOT~*~*~*~PMAX_INST    2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪302971    5.085822  5.183360e+00    5.221471    3.043288    3.283566    3.
↪004937    5.411284    5.225996    3.309655    5.391567    3.000133    3.
↪141440    4.925947    3.000000    4.919444    3.314658    4.916576    3.
↪312659    3.000000    4.945080    5.225861    4.914667         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    KLAP~*~*~*~V            2004-09-22 08:30:00 2004-09-22 08:31:00    1.
↪291413         NaN           NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    KNOT~*~*~*~WALTER       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪542668    0.399595  2.160159e-01    0.144285    1.002038    0.534794    1.
↪075106    1.232407    0.000000    0.747607    1.232407    1.232407    0.
↪815037    0.999922    0.000000    1.235105    0.000000    2.193956    0.
↪650142    1.232407    0.877202    0.000000    2.832201         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    KNOT~*~*~*~PH           2004-09-22 08:30:00 2004-09-22 08:31:00    2.
↪302971    4.085822  4.183360e+00    4.221471    2.043288    2.283566    2.
↪004937    4.411284    4.225996    2.309655    4.391567    2.000133    2.
↪141440    3.925947    2.000000    3.919444    2.314658    3.916576    2.
↪312659    2.000000    3.945080    4.225861    3.914667         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    FWVB~*~*~*~W            2004-09-22 08:30:00 2004-09-22 08:31:00  160.
↪000000  200.000000  1.600000e+02  160.000000  120.000000         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    FWVB~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪928166    6.869426  4.581308e+00    3.928166    3.680879         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
```

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```
    ROHR~*~*~*~QMAV         2004-09-22 08:30:00 2004-09-22 08:31:00   -3.
↪256005   19.059780 -5.923043e+00    8.509476   -9.266179   22.987946   22.
↪987947   -1.496260   22.987947   15.378901    3.928167   -9.266179   -7.
↪394748   -1.496260    3.928167   -9.266179         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    ROHR~*~*~*~VAV          2004-09-22 08:30:00 2004-09-22 08:31:00   -0.
↪102059    0.608561 -1.856568e-01    0.271700   -0.129652    0.733984    0.
↪327641   -0.046900    0.733984    0.491034    0.125423   -0.290447   -0.
↪231787   -0.046900    0.125423   -0.290447         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    KNOT~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000    0.000000  0.000000e+00    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    ROHR~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000    0.000000  0.000000e+00    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    VENT~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000    0.000000  0.000000e+00         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    KLAP~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000         NaN         NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    PUMP~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000         NaN         NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    FWVB~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000    0.000000  0.000000e+00    0.000000    0.000000         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    FWES~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000         NaN         NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
```

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_checkMxsVecsFile (*fullCheck=False*)

Returns (firstTime,lastTime,NOfTimes).

Args:

- **fullCheck (bool)**
  - False (default): only 1st and last h5Keys are read
  - True: all h5Keys are read

Raises: MxError

_handleMxsVecsFileDeletion (*mxsFile*, *newMxsVecsFile=False*)

Handles the deletion of mxsVecsFile.

Args:

- mxsFile
- newMxsVecsFile (default: False)

mxsVecsFile is DELETED! if:

- existing and older than mxsFile
- or newMxsVecsFile is True

Raises: MxError

setResultsToMxsFile (*mxsFile=None*, *add=False*, *NewH5Vec=False*, *maxRecords=None*)

Sets (default) or adds mxsFile-Content to .df.

Args:

- **mxsFile (str)**
  - None (default): .mxsFile is used
- add (bool): default: False: sets df to mxsFile-Content
- NewH5Vec: False (default); if True, an existing mxsVecsFile will be deleted even if it is newer than mxsFile
- maxRecords

Returns:

- timesWrittenToMxsVecs

.df

- index: TIMESTAMP
- self.df.index.is_unique will be True
- because in SIR 3S'
- 1st Time twice (SNAPSHOTTYPE: STAT+TIME) and Last Time triple (SNAPSHOTTYPE: TIME+TMIN/TMAX)
  - **+TIME** is dropped (STAT is used in df)
  - +TMIN/TMAX are dropped (not used in df)

- and because resulting overlapping TIMESTAMPs due to intersections (add=True) are also dropped

.h5FileVecs

- is updated with mxsFile-Content
- **is DELETED! before if existing and**
  - older than mxsFile
  - or newMxsVecsFile

Raises: MxError

```
>>> mx=mxs['GPipes']
>>> mx.df
>>> mx.setResultsToMxsFile(NewH5Vec=True)
1
```

setResultsToMxsZipFile(*mxsZipFile=None*, *add=False*, *NewH5Vec=False*, *maxRecords=None*)

Sets (default) or adds mxsZipFile-Content to .df.

Args:

- **mxsZipFile (str)**
  - None (default): .mxsFile is used
- add (bool): default: False: sets df to mxsZipFile-Content
- NewH5Vec
- maxRecords

Returns:

- timesWrittenToMxsVecsFromZip

Raises: MxError

ToH5(*h5File=None*)

Stores .mx1Df, .mx2Df, .df, .dfVecAggs to h5File.

Args: h5File(str): default: None: self.h5File is used

.h5File

- is !DELETED! before if existing

Keys used

- /MX1
- /MX2
- /MXS

Raises: MxError

FromH5(*h5File=None*)

Sets .mx1Df, .mx2Df, .df, .dfVecAggs to h5File-Content.

Args:

- **h5File(str)**

- None (default): .h5File is used

Keys expected

- /MX1
- /MX2
- /MXS
- /VecAggs

Raises: MxError

getMxsVecsFileData (*timesReq=None*, *fastMode=False*)

Returns List of dfs with mxsVecsFileData. One TIMESTAMP (index) per df.

Args:

- **timesReq: List of TIMESTAMPs**
  - if None: a List with a single time only, the 1st Time, is constructed as timesReq
- fastMode (default: False): H5-Access with no Checks

Returns:

- List of dfs with mxsVecsFileData
- empty List if no TIMESTAMP could be found
- one df per TIMESTAMP
- index: TIMESTAMP

Raises: MxError

```
>>> mx=mxs['LocalHeatingNetwork']
>>> mx.delFiles()
>>> mx.setResultsToMxsFile() # reads 5 TIMESTAMPS and constructs .vec.h5␣
↪while reading
5
>>> mxVecsFileDataLst=mx.getMxsVecsFileData()
>>> len(mxVecsFileDataLst)
1
>>> mxVecsFileData=mxVecsFileDataLst[0]
>>> type(mxVecsFileData)
<class 'pandas.core.frame.DataFrame'>
>>> mxVecsFileData.index[0]
Timestamp('2004-09-22 08:30:00+0000', tz='UTC')
>>> vecsFileDataOneCol=mxVecsFileData['ROHR~*~*~*~SVEC']
>>> vecsFileDataOneColResult=vecsFileDataOneCol[0]
>>> vecsFileDataOneColResult[-1]
76.4000015258789
>>> mxVecsFileDataLst=mx.getMxsVecsFileData(fastMode=True)
>>> mxVecsFileData=mxVecsFileDataLst[0]
>>> vecsFileDataOneCol=mxVecsFileData['ROHR~*~*~*~SVEC']
>>> vecsFileDataOneColResult=vecsFileDataOneCol[0]
>>> vecsFileDataOneColResult[-1]
76.4000015258789
```

unPackMxsVecsFileDataDf (*mxVecsFileData*, *mIndex*, *returnMultiIndex=True*)

Unpacks mxVecsFileData-Content into a returned df.

Args:

- mxVecsFileData: the stuff returned by getMxsVecsFileData is: List of dfs with mxsVecsFileData. One TIMESTAMP (index) per df. One of these dfs.
- **mIndex: MultiIndex to be used**
  - 1st Level: Timestamps
  - 2nd Level: cols (Sir3sIds to be unpacked)
- **returnMultiIndex (default: True):**
  - if True, the data is unpacked row-wise (stored in cols) and the index is mIndex and the col-Labels are mx2Idx
  - if False, the data is unpacked col-wise (stored in rows) and the index is mx2Idx and th col-Labels are the cols (the Sir3sIds)

Returns:

- df

Raises: MxError

```
>>> mx=mxs['LocalHeatingNetwork']
>>> mx.delFiles()
>>> mx.setResultsToMxsFile() # reads 5 TIMESTAMPS and constructs .vec.h5␣
↪while reading
5
>>> timesReq=list(mx.df.index[:]) # all times
>>> mxVecsFileDataLst=mx.getMxsVecsFileData(timesReq=timesReq)
>>> len(mxVecsFileDataLst)
5
>>> mxVecsFileData=mxVecsFileDataLst[0]
>>> # construct MultiIndex Start ... ---
>>> colsToBeUnpacked=['ROHR~*~*~*~SVEC','ROHR~*~*~*~QMAV','KNOT~*~*~*~PH'] #␣
↪mxVecsFileData.columns.tolist() for all columns
>>> arrays=[[mxVecsFileData.index[0]]*len(colsToBeUnpacked),colsToBeUnpacked]
>>> tuples = list(zip(*arrays))
>>> import pandas as pd
>>> mIndex = pd.MultiIndex.from_tuples(tuples, names=['Timestamp', 'Sir3sID'])
>>> mIndex
MultiIndex(levels=[[2004-09-22 08:30:00+00:00], ['KNOT~*~*~*~PH', 'ROHR~*~*~*~
↪QMAV', 'ROHR~*~*~*~SVEC']],
           labels=[[0, 0, 0], [2, 1, 0]],
           names=['Timestamp', 'Sir3sID'])
>>> colIdx=mIndex.get_level_values('Sir3sID')
>>> colIdx
Index(['ROHR~*~*~*~SVEC', 'ROHR~*~*~*~QMAV', 'KNOT~*~*~*~PH'], dtype='object',
↪ name='Sir3sID')
>>> colIdx.values.tolist()
['ROHR~*~*~*~SVEC', 'ROHR~*~*~*~QMAV', 'KNOT~*~*~*~PH']
>>> # construct MultiIndex End ... ---
>>> mx.unPackMxsVecsFileDataDf(mxVecsFileData,mIndex,returnMultiIndex=False)
    ROHR~*~*~*~SVEC  ROHR~*~*~*~QMAV  KNOT~*~*~*~PH
0          0.000000        -8.509475       2.302971
1         88.019997        19.059780       3.985846
2          0.000000       -15.378901       4.083384
3        405.959991         8.509476       4.121495
4          0.000000       -22.987946       2.043288
5         83.550003        22.987946       2.283565
6          0.000000        22.987947       2.004937
```

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```
7         88.019997        -3.928166       4.311307
8          0.000000        22.987947       4.126019
9         73.419998        15.378901       2.309655
10         0.000000         3.928167       4.291591
11       195.529999       -22.987946       2.000133
12         0.000000       -19.059778       2.141440
13        68.599998        -3.928166       3.825970
14         0.000000         3.928167       2.000000
15       109.769997       -22.987946       3.819467
16         0.000000              NaN       2.314658
17        76.400002              NaN       3.816599
18         0.000000              NaN       2.312659
19        83.550003              NaN       2.000000
20         0.000000              NaN       3.845104
21       164.910004              NaN       4.125885
22         0.000000              NaN       3.814690
23       195.529999              NaN            NaN
24         0.000000              NaN            NaN
25       405.959991              NaN            NaN
26         0.000000              NaN            NaN
27       164.910004              NaN            NaN
28         0.000000              NaN            NaN
29       109.769997              NaN            NaN
30         0.000000              NaN            NaN
31        76.400002              NaN            NaN
>>> mx.unPackMxsVecsFileDataDf(mxVecsFileData,mIndex)
                                                 0          1          2
↪     3          4          5          6          7          8          9
↪     10          11         12         13         14         15         16
↪      17         18         19         20         21         22         23
↪24          25   26          27   28          29   30          31
Timestamp                 Sir3sID
↪
↪
↪
↪
2004-09-22 08:30:00+00:00 ROHR~*~*~*~SVEC  0.000000  88.019997   0.000000
↪405.959991   0.000000  83.550003   0.000000  88.019997   0.000000  73.
↪419998  0.000000  195.529999   0.000000  68.599998  0.000000  109.769997  0.
↪000000  76.400002  0.000000  83.550003  0.000000  164.910004  0.00000  195.
↪529999  0.0  405.959991  0.0  164.910004  0.0  109.769997  0.0  76.400002
                          ROHR~*~*~*~QMAV -8.509475  19.059780 -15.378901
↪8.509476 -22.987946  22.987946  22.987947  -3.928166  22.987947  15.378901
↪3.928167  -22.987946 -19.059778  -3.928166  3.928167  -22.987946       NaN
↪      NaN       NaN        NaN       NaN         NaN      NaN         NaN
↪NaN         NaN  NaN         NaN  NaN         NaN  NaN        NaN
                          KNOT~*~*~*~PH    2.302971   3.985846   4.083384
↪4.121495   2.043288   2.283565   2.004937   4.311307   4.126019   2.309655
↪4.291591    2.000133   2.141440   3.825970  2.000000    3.819467  2.314658
↪ 3.816599  2.312659   2.000000  3.845104    4.125885  3.81469         NaN
↪NaN         NaN  NaN         NaN  NaN         NaN  NaN        NaN
>>> dfs=[]
>>> for idx,mxVecsFileData in enumerate(mxVecsFileDataLst):
...     arrays=[[mxVecsFileData.index[0]]*len(colsToBeUnpacked),
↪colsToBeUnpacked]
...     tuples = list(zip(*arrays))
...     mIndex = pd.MultiIndex.from_tuples(tuples, names=['Timestamp',
↪'Sir3sID'])
```

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```
...     dfs.append(mx.unPackMxsVecsFileDataDf(mxVecsFileData,mIndex))
>>> df=pd.concat(dfs)
>>> idx=pd.IndexSlice
>>> dfOneVecChannel=df.loc[(idx[:],'KNOT~*~*~*~PH'),0:22] # df.loc[(idx[:],
↪idx[:]),idx[:]]: everything
>>> dfOneVecChannel
                                                 0         1         2
↪3         4         5         6         7         8         9         10
↪    11        12        13   14        15        16        17        18
↪19        20        21        22
Timestamp                 Sir3sID
↪
↪
↪
2004-09-22 08:30:00+00:00 KNOT~*~*~*~PH  2.302971  3.985846  4.083384  4.
↪121495  2.043288  2.283565  2.004937  4.311307  4.126019  2.309655  4.
↪291591  2.000133  2.141440  3.825970  2.0  3.819467  2.314658  3.816599  2.
↪312659  2.0  3.845104  4.125885  3.814690
2004-09-22 08:30:15+00:00 KNOT~*~*~*~PH  2.272133  3.034820  3.123339  3.
↪157926  2.039330  2.255054  2.004493  3.331398  3.162040  2.277968  3.
↪311887  2.000120  2.128487  2.892818  2.0  2.887151  2.282320  2.884661  2.
↪280581  2.0  2.909634  3.161918  2.883003
2004-09-22 08:30:30+00:00 KNOT~*~*~*~PH  2.144123  2.528542  2.576208  2.
↪594833  2.021343  2.135238  2.002467  2.694164  2.597075  2.147196  2.
↪676144  2.000063  2.069655  2.455464  2.0  2.452505  2.149524  2.451183  2.
↪148594  2.0  2.464144  2.597010  2.450303
2004-09-22 08:30:45+00:00 KNOT~*~*~*~PH  2.052100  2.183028  2.200011  2.
↪206647  2.007717  2.048865  2.000910  2.248923  2.207463  2.053240  2.
↪234365  2.000021  2.025138  2.156905  2.0  2.155822  2.054124  2.155325  2.
↪053771  2.0  2.160025  2.207441  2.154995
2004-09-22 08:31:00+00:00 KNOT~*~*~*~PH  2.302971  4.085822  4.183360  4.
↪221471  2.043288  2.283566  2.004937  4.411284  4.225996  2.309655  4.
↪391567  2.000133  2.141440  3.925947  2.0  3.919444  2.314658  3.916576  2.
↪312659  2.0  3.945080  4.225861  3.914667
>>> dfOneVecChannel.min()
0     2.052100
1     2.183028
2     2.200011
3     2.206647
4     2.007717
5     2.048865
6     2.000910
7     2.248923
8     2.207463
9     2.053240
10    2.234365
11    2.000021
12    2.025138
13    2.156905
14    2.000000
15    2.155822
16    2.054124
17    2.155325
18    2.053771
19    2.000000
20    2.160025
21    2.207441
```

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```
22    2.154995
dtype: float64
>>> df.min(level=1)
                        0          1          2          3          4
↪ 5          6          7          8          9         10         11
↪12         13         14         15         16         17         18
↪19         20         21         22         23   24         25   26
↪  27   28         29   30         31
Sir3sID
↪
↪
↪
↪
ROHR~*~*~*~SVEC  0.000000  88.019997   0.000000  405.959991   0.000000  83.
↪550003  0.000000  88.019997  0.000000  73.419998  0.000000  195.529999   0.
↪000000  68.599998  0.00000  109.769997  0.000000  76.400002  0.000000  83.
↪550003  0.000000  164.910004  0.000000  195.529999  0.0  405.959991  0.0
↪164.910004  0.0  109.769997  0.0  76.400002
ROHR~*~*~*~QMAV -8.509475   7.394749 -15.378901    3.256006 -22.987946   9.
↪266180  9.266181  -3.928166  9.266181   5.923044  1.496261  -22.987946 -19.
↪059778  -3.928166  1.49626  -22.987946       NaN        NaN       NaN
↪ NaN       NaN         NaN       NaN         NaN  NaN         NaN  NaN
↪   NaN  NaN         NaN  NaN        NaN
KNOT~*~*~*~PH    2.052100   2.183028   2.200011    2.206647   2.007717   2.
↪048865  2.000910   2.248923  2.207463   2.053240  2.234365    2.000021   2.
↪025138   2.156905  2.00000    2.155822  2.054124   2.155325  2.053771   2.
↪000000  2.160025    2.207441  2.154995         NaN  NaN         NaN  NaN
↪     NaN  NaN         NaN  NaN        NaN
```

getVecAggs (*time1st=None*, *time1stIncluded=True*, *time2nd=None*, *time2ndIncluded=True*)

Gets (or calcs) Aggregates (MIN, MAX, . . .) of mxsVecsFileData between the 2 Times.

- Newly calced Aggregates are stored in dfVecAggs.

Args:

- time1st: TIMESTAMP (first if None)
- time2nd: TIMESTAMP (last if None)

Returns:

- **dfs with MultiIndex:**
  - Level 0: 'MIN', 'MAX', . . .
  - Level 1: col (Sir3sID)
  - cols: mx2Idx
- TIMESTAMPL: left ScenTimeStamp included in calculating the Aggregate
- TIMESTAMPR: right ScenTimeStamp included in calculating the Aggregate

Raises: MxError

```
>>> mx=mxs['LocalHeatingNetwork']
>>> mx.delFiles()
>>> mx.setResultsToMxsFile() # reads 5 TIMESTAMPS and constructs .vec.h5
↪while reading
```

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```
5
>>> # check dfVecAggs to demonstrate how getVecAggs stores to / reads from␣
↪getVecAggs
>>> Sir3sIDs=mx.dfVecAggs.index.unique(level=1).values
>>> len(Sir3sIDs)
41
>>> mx.dfVecAggs.index.unique(level=0).values
array(['TIME', 'TMIN', 'TMAX'], dtype=object)
>>> len(mx.dfVecAggs.columns.tolist())
32
>>> mx.dfVecAggs.shape # (3*41,32)
(123, 32)
>>> df,tL,tR=mx.getVecAggs()
>>> mx.dfVecAggs.index.unique(level=0).values
array(['TIME', 'TMIN', 'TMAX', 'MIN', 'MAX', 'DIF'], dtype=object)
>>> mx.dfVecAggs.shape
(246, 32)
>>> import pandas as pd
>>> #idx=pd.IndexSlice
>>> df.loc[(['MIN','MAX','DIF'],'KNOT~*~*~*~PH'),0:22].round(6) ## df.
↪loc[(slice(None),'KNOT~*~*~*~PH'),slice(None)] # df.loc[(idx[:],'KNOT~*~*~*~
↪PH'),idx[:]]
                          0         1         2         3         4         5␣
↪        6         7         8         9        10        11        12      ␣
↪  13   14        15        16        17        18   19        20        21 ␣
↪      22
TYPE Sir3sID                                                                 ␣
↪                                                                            ␣
↪                                                                            ␣
↪
MIN  KNOT~*~*~*~PH  2.052100  2.183028  2.200011  2.206647  2.007717  2.
↪048865  2.000910  2.248923  2.207463  2.053240  2.234365  2.000021  2.
↪025138  2.156905  2.0  2.155822  2.054124  2.155325  2.053771  2.0  2.
↪160025  2.207441  2.154995
MAX  KNOT~*~*~*~PH  2.302971  4.085822  4.183360  4.221471  2.043288  2.
↪283566  2.004937  4.411284  4.225996  2.309655  4.391567  2.000133  2.
↪141440  3.925947  2.0  3.919444  2.314658  3.916576  2.312659  2.0  3.
↪945080  4.225861  3.914667
DIF  KNOT~*~*~*~PH  0.000000  0.099977  0.099977  0.099977  0.000000  0.
↪000000  0.000000  0.099977  0.099977  0.000000  0.099977  0.000000  0.
↪000000  0.099977  0.0  0.099977  0.000000  0.099977  0.000000  0.0  0.
↪099977  0.099977  0.099977
>>> # demonstrate how to transform an getVecAggs()-df-Result for Xm...
>>> dfT=df.loc[('MIN',df.index.get_level_values(1).tolist()),:].
↪transpose(copy=True) ## dfT=df.loc[('MIN',slice(None)),:].
↪transpose(copy=True)   # dfT=df.loc[('MIN',idx[:]),idx[:]].
↪transpose(copy=True)
>>> colIndex=dfT.columns.droplevel(level=0)
>>> colIndex.name=None
>>> pd.DataFrame(dfT.values,columns=colIndex)[['ROHR~*~*~*~SVEC', 'ROHR~*~*~*~
↪QMAV', 'KNOT~*~*~*~PH']]
    ROHR~*~*~*~SVEC  ROHR~*~*~*~QMAV  KNOT~*~*~*~PH
0          0.000000        -8.509475       2.052100
1         88.019997         7.394749       2.183028
2          0.000000       -15.378901       2.200011
3        405.959991         3.256006       2.206647
4          0.000000       -22.987946       2.007717
```

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```
5         83.550003         9.266180       2.048865
6          0.000000         9.266181       2.000910
7         88.019997        -3.928166       2.248923
8          0.000000         9.266181       2.207463
9         73.419998         5.923044       2.053240
10         0.000000         1.496261       2.234365
11       195.529999       -22.987946       2.000021
12         0.000000       -19.059778       2.025138
13        68.599998        -3.928166       2.156905
14         0.000000         1.496260       2.000000
15       109.769997       -22.987946       2.155822
16         0.000000              NaN       2.054124
17        76.400002              NaN       2.155325
18         0.000000              NaN       2.053771
19        83.550003              NaN       2.000000
20         0.000000              NaN       2.160025
21       164.910004              NaN       2.207441
22         0.000000              NaN       2.154995
23       195.529999              NaN            NaN
24         0.000000              NaN            NaN
25       405.959991              NaN            NaN
26         0.000000              NaN            NaN
27       164.910004              NaN            NaN
28         0.000000              NaN            NaN
29       109.769997              NaN            NaN
30         0.000000              NaN            NaN
31        76.400002              NaN            NaN
>>> df,tL,tR=mx.getVecAggs()
>>> mx.dfVecAggs.shape
(246, 32)
>>> # demonstrate how to transform an AggEntry for Xm...
>>> df=mx.dfVecAggs.loc[('TMIN',slice(None),mx.df.index[0],mx.df.index[-1]),:]
>>> dfT=df.transpose(copy=True)
>>> colIndex=dfT.columns.droplevel(level=0)
>>> colIndex=colIndex.droplevel(level=1)
>>> colIndex=colIndex.droplevel(level=1)
>>> colIndex.name=None
>>> pd.DataFrame(dfT.values,columns=colIndex)[['ROHR~*~*~*~SVEC', 'ROHR~*~*~*~
↪QMAV', 'KNOT~*~*~*~PH']]
    ROHR~*~*~*~SVEC  ROHR~*~*~*~QMAV  KNOT~*~*~*~PH
0          0.000000        -8.509475       2.052100
1         88.019997         7.394749       2.183028
2          0.000000       -15.378901       2.200011
3        405.959991         3.256006       2.206647
4          0.000000       -22.987946       2.007717
5         83.550003         9.266180       2.048865
6          0.000000         9.266181       2.000910
7         88.019997        -3.928166       2.248923
8          0.000000         9.266181       2.207463
9         73.419998         5.923044       2.053240
10         0.000000         1.496261       2.234365
11       195.529999       -22.987946       2.000021
12         0.000000       -19.059778       2.025138
13        68.599998        -3.928166       2.156905
14         0.000000         1.496260       2.000000
15       109.769997       -22.987946       2.155822
16         0.000000              NaN       2.054124
```

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```
17        76.400002             NaN       2.155325
18         0.000000             NaN       2.053771
19        83.550003             NaN       2.000000
20         0.000000             NaN       2.160025
21       164.910004             NaN       2.207441
22         0.000000             NaN       2.154995
23       195.529999             NaN            NaN
24         0.000000             NaN            NaN
25       405.959991             NaN            NaN
26         0.000000             NaN            NaN
27       164.910004             NaN            NaN
28         0.000000             NaN            NaN
29       109.769997             NaN            NaN
30         0.000000             NaN            NaN
31        76.400002             NaN            NaN
```

dumpInMxsFormat (*mxsDumpFile=None*)

Dumps in MXS-Format to mxsDumpFile (for testing purposes).

Returns:

- **(TimeStampsDumped, TimeStampsFoundInH5)**
  - normally: TimeStampsDumped=TimeStampsFoundInH5
  - if TimesStamps in self.df are manipulated ...
  - ... the H5-Content remains unchanged
  - in effect the H5-Content can be different from self.df-Content ...
  - während in self.df die Zeiten (Index) immer geordnet und voneinander verschieden sind
  - sind beim H5-Content nur die Keys voneinander verschieden
  - um pruefen zu koennen, ob alle Zeiten in self.df im H5-Content auch gefunden wurden, wird TimeStampsFoundInH5 mit ausgegeben
  - gedumped werden immer alle Zeiten aus self.df
  - * fuer jede im H5-Content nicht gefundene Zeit wird das Ergebnis der zuletzt zuvor gefundenen Zeit ausgegeben

Raises: MxError

_getMx1DfAsOneString (*regex='KNOT~\\S*~\\S*~\\d+~[P|Q]{1}[H|M]{1}$'*)

Returns .mx1Df-Content as one String (for Doctest-Purposes).

Raises: MxError

_getDfAsOneString (*regex='KNOT~\\S*~\\S*~\\d+~[P|Q]{1}[H|M]{1}$'*)

Returns .df-Content as one String (for Doctest-Purposes).

Raises: MxError

_getDfVecAsOneString (*df=None, regex='KNOT~\\S*~\\S*~\\S*~[P|Q]{1}[H|M]{1}$'*)

Returns dfVec-Content as one String (for Doctest-Purposes).

Raises: MxError

Mx.**getMicrosecondsFromRefTime** (*refTime, time*)

Returns time in microseconds since refTime.

Args:

- refTime
- time

Raises: MxError

```
>>> import pandas as pd
>>> timeReadFromMx=b'2019-01-01 00:00:12.500000      '
>>> timeRefMx=b'2019-01-01 00:00:00.000000      '
>>> timeStampTimeReadFromMx=pd.to_datetime(timeReadFromMx.decode(),utc=True)
>>> timeStampTimeRefFromMx=pd.to_datetime(timeRefMx.decode(),utc=True)
>>> timeDelta=timeStampTimeReadFromMx-timeStampTimeRefFromMx
>>> timeDelta.total_seconds()
12.5
>>> import Mx
>>> Mx.getMicrosecondsFromRefTime(timeStampTimeRefFromMx,timeStampTimeReadFromMx)
12500
```

exception Mx.**MxError**(*value*)

MxError.

CHAPTER

THREE

XM

```
>>> # ---
>>> # Imports
>>> # ---
>>> import os
>>> import logging
>>> logger = logging.getLogger('PT3S.Xm')
>>> # ---
>>> # path
>>> # ---
>>> if __name__ == "__main__":
...   try:
...      dummy=__file__
...      logger.debug("{0:s}{1:s}{2:s}".format('DOCTEST: __main__ Context: ','path =␣
↪os.path.dirname(__file__)'," ."))
...      path = os.path.dirname(__file__)
...   except NameError:
...      logger.debug("{0:s}{1:s}{2:s}".format('DOCTEST: __main__ Context: ","path =
↪'.' because __file__ not defined: ","from Xm import Xm follows ..."))
...      path = '.'
...      from Xm import Xm
... else:
...    logger.debug("{0:s}{1:s}{2:s}{3:s}".format('DOCTEST: Not __main__ Context: ','_
↪__name__: ',__name__,"path = '."))
...    path = '.'
>>> try:
...    from PT3S import Mx
... except ImportError:
...    logger.debug("{0:s}{1:s}".format("DOCTEST: ImportError: from PT3S import Mx: ",
↪"- trying import Mx instead ... maybe pip install -e . is active ..."))
...    import Mx
>>> # ---
>>> # testDir
>>> # ---
>>> # globs={'testDir':'testdata'}
>>> try:
...    dummy= testDir
... except NameError:
...    testDir='testdata'
>>> import pandas as pd
>>> # ---
>>> # Clean Up
>>> # ---
>>> h5File=os.path.join(os.path.join(path,testDir),'OneLPipe.h5')
>>> if os.path.exists(h5File):
```

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```
...     os.remove(h5File)
>>> # ---
>>> # Init
>>> # ---
>>> xmlFile=os.path.join(os.path.join(path,testDir),'OneLPipe.XML')
>>> xm=Xm(xmlFile=xmlFile)
>>> # ---
>>> # a View
>>> # ---
>>> v='vKNOT'
>>> v in xm.dataFrames
True
>>> isinstance(xm.dataFrames[v],pd.core.frame.DataFrame)
True
>>> # ---
>>> # ToH5
>>> # ---
>>> xm.ToH5()
>>> os.path.exists(xm.h5File)
True
>>> # ---
>>> # force Read H5 instead of Xml
>>> # ---
>>> os.rename(xm.xmlFile,xm.xmlFile+'.blind')
>>> xm=Xm(xmlFile=xmlFile)
>>> os.rename(xm.xmlFile+'.blind',xm.xmlFile)
>>> # ---
>>> vKNOT=xm.dataFrames['vKNOT']
>>> vKNOT[(vKNOT.KTYP.isin(['QKON','PKON'])) & (vKNOT.BESCHREIBUNG.fillna('').str.
↪startswith('Template Element')==False)].shape
(2, 41)
>>> vROHR=xm.dataFrames['vROHR']
>>> vROHR.shape
(1, 74)
>>> isinstance(vROHR['pXCors'],pd.core.series.Series)
True
>>> vROHR['pXCors'][0]
[0.0, 500.0]
>>> vROHR.pYCors[0]
[0.0, 0.0]
>>> # ---
>>> # getWDirModelDirModelName()
>>> # ---
>>> (wDir,modelDir,modelName,mx1File)=xm.getWDirModelDirModelName()
>>> modelName
'M-1-0-1'
>>> # ---
>>> # H5-Deletion if NoH5Read=True
>>> # ---
>>> if os.path.exists(xm.h5File):
...     os.remove(xm.h5File)
>>> xm=Xm(xmlFile=xmlFile)
>>> xm.ToH5()
>>> os.path.exists(xm.h5File)
True
>>> xm=Xm(xmlFile=xmlFile,NoH5Read=True)
>>> os.path.exists(xm.h5File)
```

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```
False
>>> # ---
>>> # print-Options
>>> # ---
>>> pd.set_option('display.max_columns',None)
>>> pd.set_option('display.max_rows',None)
>>> pd.set_option('display.max_colwidth',666666)
>>> pd.set_option('display.width',666666666)
>>> # ---
>>> # vKNOT
>>> # ---
>>> print(xm._getvXXXXAsOneString(vXXXX='vKNOT',dropColList=['LFKT_NAME','LF','LF_min
↪','LF_max','PVAR_NAME','PH','PH_min','PH_max','PZON_NAME','FSTF_NAME','STOF_NAME',
↪'GMIX_NAME','UTMP_NAME','2L_NAME','2L_KVR','fkHYDR','fkFQPS']))
  NAME BESCHREIBUNG              IDREFERENZ      CONT CONT_ID  CONT_LFDNR  CONT_VKNO ␣
↪KTYP LFAKT    QM_EIN QVAR_NAME  QM  QM_min  QM_max KVR  TE  TM XKOR YKOR ZKOR      ␣
↪            pk                   tk  pXCor  pYCor
0    I          NaN  3S5642914844465475844  OneLPipe    1001         NaN        NaN ␣
↪QKON     1  176.7146       NaN NaN     NaN     NaN   0 NaN  10  300  600   10 ␣
↪5642914844465475844  5642914844465475844    0.0    0.0
1    K          NaN  3S5289899964753656852  OneLPipe    1001         NaN        NaN ␣
↪PKON     1         0       NaN NaN     NaN     NaN   0 NaN  10  800  600   10 ␣
↪5289899964753656852  5289899964753656852  500.0    0.0
>>> # ---
>>> # vROHR
>>> # ---
>>> print(xm._getvXXXXAsOneString(vXXXX='vROHR',dropColList=['NAME_i_2L','NAME_k_2L
↪']))
  BESCHREIBUNG             IDREFERENZ BAUJAHR HAL IPLANUNG KENNUNG      L LZU   RAU␣
↪ZAUS ZEIN ZUML JLAMBS LAMBDA0 ASOLL INDSCHALL KVR AUSFALLZEIT DA   DI   DN KT PN␣
↪REHABILITATION REPARATUR  S WSTEIG WTIEFE LTGR_NAME  LTGR_BESCHREIBUNG SICHTBARKEIT␣
↪VERLEGEART DTRO_NAME                        DTRO_BESCHREIBUNG        E fkSTRASSE␣
↪fkSRAT                  pk                  tk IRTRENN LECKSTART LECKEND␣
↪LECKMENGE LECKORT LECKSTATUS QSVB ZVLIMPTNZ KANTENZV      CONT CONT_ID  CONT_LFDNR␣
↪NAME_i KVR_i TM_i XKOR_i YKOR_i ZKOR_i NAME_k KVR_k TM_k XKOR_k YKOR_k ZKOR_k ␣
↪pXCor_i  pYCor_i  pXCor_k  pYCor_k        pXCors      pYCors    pWAYPXCors ␣
↪pWAYPYCors                              WAYP
0          NaN  3S4737064599036143765    2017   0        1       0  10000   0  0.25  ␣
↪ 0    0    0      1   0.025  1000         0   0            0  0  250  250  0  0    ␣
↪        0         0  0      0      0   STDROHR                NaN            1    ␣
↪999999   STDROHR  Standard-Druckrohre mit di = DN (DIN 2402)  2.1E+11        -1    ␣
↪-1  4737064599036143765  4737064599036143765       0         0       0         0   ␣
↪   0          0    0         0        0  OneLPipe    1001         NaN      I     0 ␣
↪ 10    300    600     10      K     0   10    800    600     10      0.0      0.0  ␣
↪ 500.0      0.0  [0.0, 500.0]  [0.0, 0.0]  [0.0, 500.0]  [0.0, 0.0]  [(300.0, 600.
↪0), (800.0, 600.0)]
>>> # ---
>>> # Clean Up
>>> # ---
>>> xm.delFiles()
>>> # ---
>>> # LocalHeatingNetwork
>>> # ---
>>> xmlFile=os.path.join(os.path.join(path,testDir),'LocalHeatingNetwork.XML')
>>> xm=Xm(xmlFile=xmlFile)
>>> # ---
>>> # vKNOT
```

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```
>>> # ---
>>> print(xm._getvXXXXAsOneString(vXXXX='vKNOT',dropColList=['LFKT_NAME','LF','LF_min
↪','LF_max','PVAR_NAME','PH','PH_min','PH_max','PZON_NAME','FSTF_NAME','STOF_NAME',
↪'GMIX_NAME','UTMP_NAME','2L_NAME','2L_KVR','fkHYDR','fkFQPS']))
           NAME                        BESCHREIBUNG IDREFERENZ                                 ␣
↪           CONT CONT_ID CONT_LFDNR CONT_VKNO  KTYP LFAKT QM_EIN QVAR_NAME  QM  QM_
↪min  QM_max KVR  TE  TM     XKOR     YKOR ZKOR                   pk                  ␣
↪  tk   pXCor  pYCor
0        R-K004                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60  2541539  5706361   20  4638663808856251977 ␣
↪4638663808856251977   799.0  152.0
1        V-K002                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   1  10  90  2541059  5706265   20  4731792362611615619 ␣
↪4731792362611615619   319.0   56.0
2        V-K001                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   1  10  90  2540867  5706228   20  4756962427318766791 ␣
↪4756962427318766791   127.0   19.0
3        V-K000                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   1  10  90  2540793  5706209   20  4766681917240867943 ␣
↪4766681917240867943    53.0    0.0
4        R-K001                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60  2540867  5706228   20  4807712987325933680 ␣
↪4807712987325933680   127.0   19.0
5        R-K003                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60  2541457  5706345   20  4891048046264179170 ␣
↪4891048046264179170   717.0  136.0
6        R-K000                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60  2540793  5706209   20  4979785838440534851 ␣
↪4979785838440534851    53.0    0.0
7        R-K005                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60  2541627  5706363   20  5183147862966701025 ␣
↪5183147862966701025   887.0  154.0
8           R-L                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1      BHKW  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60  2540740  5706225   20  5356267303828212700 ␣
↪5356267303828212700     0.0   16.0
9        R-K002                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60  2541059  5706265   20  5364712333175450942 ␣
↪5364712333175450942   319.0   56.0
10       V-K004                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   1  10  90  2541539  5706361   20  5370423799772591808 ␣
↪5370423799772591808   799.0  152.0
11       V-K005                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   1  10  90  2541627  5706363   20  5444644492819213978 ␣
↪5444644492819213978   887.0  154.0
12       R-K007                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60  2541899  5706325   20  5508992300317633799 ␣
↪5508992300317633799  1159.0  116.0
```

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```
13       V-K006                            None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   1  10  90  2541790  5706338   20  5515313800585145571 ␣
↪5515313800585145571  1050.0  129.0
14       R-K006                            None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60  2541790  5706338   20  5543326527366090679 ␣
↪5543326527366090679  1050.0  129.0
15       V-K003                            None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   1  10  90  2541457  5706345   20  5646671866542823796 ␣
↪5646671866542823796   717.0  136.0
16          V-L                            None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1      BHKW  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   1  10  90  2540740  5706240   20  5736262931552588702 ␣
↪5736262931552588702     0.0   31.0
17       V-K007                            None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   1  10  90  2541899  5706325   20  5741235692335544560 ␣
↪5741235692335544560  1159.0  116.0
18           R2                            None         -1
↪          BHKW    1002         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60      170       20   20  5002109894154139899 ␣
↪5002109894154139899   170.0   20.0
19          V-1                            None         -1
↪          BHKW    1002         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   1  10  90      140      160   20  5049461676240771430 ␣
↪5049461676240771430   140.0  160.0
20           R3                            None         -1
↪          BHKW    1002         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60      140       20   20  5219230031772497417 ␣
↪5219230031772497417   140.0   20.0
21  PKON-Knoten  Druckhaltung - 2 bar Ruhedruck         -1
↪          BHKW    1002         -1       NaN  PKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  60  60      200       40   20  5397990465339071638 ␣
↪5397990465339071638   200.0   40.0
22          R-1          Anbindung Druckhaltung         -1
↪          BHKW    1002         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60      195       20   20  5557222628687032084 ␣
↪5557222628687032084   195.0   20.0
>>> # ---
>>> # vROHR
>>> # ---
>>> print(xm._getvXXXXAsOneString(vXXXX='vROHR',dropColList=['NAME_i_2L','NAME_k_2L
↪']))
   BESCHREIBUNG IDREFERENZ BAUJAHR HAL IPLANUNG KENNUNG       L LZU  RAU ZAUS ZEIN␣
↪ZUML JLAMBS LAMBDA0 ASOLL INDSCHALL KVR  AUSFALLZEIT     DA     DI   DN     KT  PN ␣
↪REHABILITATION  REPARATUR    S WSTEIG WTIEFE LTGR_NAME            LTGR_BESCHREIBUNG␣
↪SICHTBARKEIT VERLEGEART DTRO_NAME                        DTRO_BESCHREIBUNG        E␣
↪fkSTRASSE fkSRAT                    pk                    tk IRTRENN LECKSTART␣
↪LECKEND LECKMENGE LECKORT LECKSTATUS QSVB ZVLIMPTNZ KANTENZV                      ␣
↪              CONT CONT_ID CONT_LFDNR  NAME_i KVR_i TM_i   XKOR_i   YKOR_i ZKOR_i ␣
↪NAME_k KVR_k TM_k   XKOR_k   YKOR_k ZKOR_k  pXCor_i  pYCor_i  pXCor_k  pYCor_k    ␣
↪      pXCors          pYCors                                pWAYPXCors           ␣
↪                 pWAYPYCors                                                       ␣
↪                                     WAYP
0          None         -1    None   0        1       0   88.02   0  0.1    0    0  ␣
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.925  NaN  ␣
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe  ␣
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11  ␣
↪    -1     -1  4613782368750024999  4613782368750024999       0         0       0  ␣
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  R-K004     2   60  2541539  5706361     20  R-
↪K005     2   60  2541627  5706363     20    799.0    152.0    887.0    154.0    ␣
```

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```
1          None         -1    None   0        1       0  405.96   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4614949065966596185  4614949065966596185       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  V-K002     1   90  2541059  5706265     20  V-
↪K003     1   90  2541457  5706345     20    319.0     56.0    717.0    136.0
↪[319.0, 717.0]   [56.0, 136.0]                [319.0, 716.9500000001863]
↪   [56.049999999813735, 136.0]
↪[(2541059.0, 5706265.05), (2541456.95, 5706345.0)]
2          None         -1    None   0        1       0   83.55   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4637102239750163477  4637102239750163477       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  R-K003     2   60  2541457  5706345     20  R-
↪K004     2   60  2541539  5706361     20    717.0    136.0    799.0    152.0
↪[717.0, 799.0]  [136.0, 152.0]    [725.8500000000931, 807.8999999999069]  [124.
↪04999999981374, 140.09999999962747]
↪[(2541465.85, 5706333.05), (2541547.9, 5706349.1)]
3          None         -1    None   0        1       0   88.02   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4713733238627697042  4713733238627697042       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  V-K004     1   90  2541539  5706361     20  V-
↪K005     1   90  2541627  5706363     20    799.0    152.0    887.0    154.0
↪[799.0, 887.0]  [152.0, 154.0]                [799.0, 887.0499999998137]
↪                [152.0, 154.0]
↪[(2541539.0, 5706361.0), (2541627.05, 5706363.0)]
4          None         -1    None   0        1       0  195.53   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4789218195240364437  4789218195240364437       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  V-K001     1   90  2540867  5706228     20  V-
↪K002     1   90  2541059  5706265     20    127.0     19.0    319.0     56.0
↪[127.0, 319.0]    [19.0, 56.0]                              [127.0, 319.0]
↪    [19.0, 56.049999999813735]
↪[(2540867.0, 5706228.0), (2541059.0, 5706265.05)]
5          None         -1    None   0        1       0  109.77   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4945727430885351042  4945727430885351042       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  R-K006     2   60  2541790  5706338     20  R-
↪K007     2   60  2541899  5706325     20   1050.0    129.0   1159.0    116.0  [1050.
↪0, 1159.0]  [129.0, 116.0]  [1058.8500000000931, 1167.8999999999069]
↪[117.0, 104.09999999962747]
↪[(2541798.85, 5706326.0), (2541907.9, 5706313.1)]
6          None         -1    None   0        1       0    76.4   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4984202422877610920  4984202422877610920       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  V-K000     1   90  2540793  5706209     20  V-
↪K001     1   90  2540867  5706228     20     53.0      0.0    127.0     19.0
↪[53.0, 127.0]     [0.0, 19.0]                [53.049999999813735, 127.0]
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7          None         -1    None   0        1       0   83.55   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5037777106796980248  5037777106796980248       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  V-K003     1   90  2541457  5706345     20  V-
↪K004     1   90  2541539  5706361     20    717.0    136.0    799.0    152.0
↪[717.0, 799.0]  [136.0, 152.0]                [716.9500000001863, 799.0]
↪                [136.0, 152.0]
↪[(2541456.95, 5706345.0), (2541539.0, 5706361.0)]
8          None         -1    None   0        1       0  164.91   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5123819811204259837  5123819811204259837       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  V-K005     1   90  2541627  5706363     20  V-
↪K006     1   90  2541790  5706338     20    887.0    154.0   1050.0    129.0   [887.
↪0, 1050.0]  [154.0, 129.0]   [887.0499999998137, 1049.9500000001863]
↪[154.0, 128.95000000018626]
↪[(2541627.05, 5706363.0), (2541789.95, 5706337.95)]
9          None         -1    None   0        1       0  195.53   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5266224553324203132  5266224553324203132       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  R-K001     2   60  2540867  5706228     20  R-
↪K002     2   60  2541059  5706265     20    127.0     19.0    319.0     56.0
↪[127.0, 319.0]    [19.0, 56.0]  [135.89999999990687, 327.89999999990687]   [7.
↪0499999998137355, 44.09999999962747]
↪  [(2540875.9, 5706216.05), (2541067.9, 5706253.1)]
10         None         -1    None   0        1       0  405.96   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5379365049009065623  5379365049009065623       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  R-K002     2   60  2541059  5706265     20  R-
↪K003     2   60  2541457  5706345     20    319.0     56.0    717.0    136.0
↪[319.0, 717.0]   [56.0, 136.0]   [327.89999999990687, 725.8500000000931]   [44.
↪09999999962747, 124.04999999981374]
↪[(2541067.9, 5706253.1), (2541465.85, 5706333.05)]
11         None         -1    None   0        1       0  164.91   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5611703699850694889  5611703699850694889       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  R-K005     2   60  2541627  5706363     20  R-
↪K006     2   60  2541790  5706338     20    887.0    154.0   1050.0    129.0   [887.
↪0, 1050.0]  [154.0, 129.0]   [895.9500000001863, 1058.8500000000931]
↪[142.04999999981374, 117.0]
↪[(2541635.95, 5706351.05), (2541798.85, 5706326.0)]
12         None         -1    None   0        1       0  109.77   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5620197984230756681  5620197984230756681       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  V-K006     1   90  2541790  5706338     20  V-
↪K007     1   90  2541899  5706325     20   1050.0    129.0   1159.0    116.0  [1050.
↪0, 1159.0]  [129.0, 116.0]              [1049.9500000001863, 1159.0]  [128.
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```
13         None         -1    None   0        1       0    76.4   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5647213228462830353  5647213228462830353       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  R-K000     2   60  2540793  5706209     20  R-
↪K001     2   60  2540867  5706228     20     53.0      0.0    127.0     19.0
↪[53.0, 127.0]     [0.0, 19.0]  [61.950000000186265, 135.89999999990687]
↪  [-12.0, 7.0499999998137355]
↪[(2540801.95, 5706197.0), (2540875.9, 5706216.05)]
14         None         -1    None   0        1       0   73.42   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  168.3  160.3  150   0.45 NaN
↪        NaN        NaN    4    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4769996343148550485  4769996343148550485       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1     R-L     2   60  2540740  5706225     20  R-
↪K000     2   60  2540793  5706209     20      0.0     16.0     53.0      0.0
↪[0.0, 53.0]     [16.0, 0.0]     [0.0, 24.0, 45.0, 61.950000000186265]
↪ [16.0, 16.0, -12.0, -12.0]  [(2540740.0, 5706225.0), (2540764.0, 5706225.0),␣
↪(2540785.0, 5706197.0), (2540801.95, 5706197.0)]
15         None         -1    None   0        1       0    68.6   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  168.3  160.3  150   0.45 NaN
↪        NaN        NaN    4    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4939422678063487923  4939422678063487923       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1     V-L     1   90  2540740  5706240     20  V-
↪K000     1   90  2540793  5706209     20      0.0     31.0     53.0      0.0
↪[0.0, 53.0]     [31.0, 0.0]          [0.0, 30.0, 53.049999999813735]       [31.0,
↪31.0, -0.049999999813735485]                        [(2540740.0, 5706240.0),␣
↪(2540770.0, 5706240.0), (2540793.05, 5706208.95)]
>>> # ---
>>> # vWBLZ
>>> # ---
>>> print("'''{:s}'''".format(repr(xm.dataFrames['vWBLZ']).replace('\n','\n    ')))
'''   AKTIV BESCHREIBUNG IDIM       NAME OBJTYPE                OBJID
↪  pk
   0      1  Wärmebilanz    0      BLNZ1    KNOT  4731792362611615619
↪5579937562601803472
   1      1  Wärmebilanz    0      BLNZ1    KNOT  5364712333175450942
↪5579937562601803472
   2      1  Wärmebilanz    0    BLNZ1u5    KNOT  5183147862966701025
↪5187647097142898375
   3      1  Wärmebilanz    0    BLNZ1u5    KNOT  5444644492819213978
↪5187647097142898375
   4      1  Wärmebilanz    0    BLNZ1u5    KNOT  4731792362611615619
↪5187647097142898375
   5      1  Wärmebilanz    0    BLNZ1u5    KNOT  5364712333175450942
↪5187647097142898375
   6      1  Wärmebilanz    0  BLNZ1u5u7    KNOT  5183147862966701025
↪4694700216019268978
   7      1  Wärmebilanz    0  BLNZ1u5u7    KNOT  5444644492819213978
↪4694700216019268978
   8      1  Wärmebilanz    0  BLNZ1u5u7    KNOT  4731792362611615619
↪4694700216019268978
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   9      1  Wärmebilanz     0  BLNZ1u5u7    KNOT  5364712333175450942 ␣
↪4694700216019268978
   10     1  Wärmebilanz     0  BLNZ1u5u7    KNOT  5508992300317633799 ␣
↪4694700216019268978
   11     1  Wärmebilanz     0  BLNZ1u5u7    KNOT  5741235692335544560 ␣
↪4694700216019268978
   12     1  Wärmebilanz     0      BLNZ5    KNOT  5183147862966701025 ␣
↪5581152085151655438
   13     1  Wärmebilanz     0      BLNZ5    KNOT  5444644492819213978 ␣
↪5581152085151655438'''
>>> # ---
>>> # vAGSN
>>> # ---
>>> print(xm._getvXXXXAsOneString(vXXXX='vAGSN',end=7,dropColList=['nrObjIdTypeInAgsn
↪']))
  LFDNR                                     NAME AKTIV OBJTYPE                OBJID ␣
↪                 pk                    tk  nrObjIdInAgsn  Layer nextNODE  compNr
0     1  Netzdruckdiagramm VL/RL: BHKW - Netzende   101    ROHR  4939422678063487923 ␣
↪5252525269080005909  5252525269080005909              1      1   V-K000       1
1     1  Netzdruckdiagramm VL/RL: BHKW - Netzende   101    ROHR  4984202422877610920 ␣
↪5252525269080005909  5252525269080005909              2      1   V-K001       1
2     1  Netzdruckdiagramm VL/RL: BHKW - Netzende   101    ROHR  4789218195240364437 ␣
↪5252525269080005909  5252525269080005909              3      1   V-K002       1
3     1  Netzdruckdiagramm VL/RL: BHKW - Netzende   101    ROHR  4614949065966596185 ␣
↪5252525269080005909  5252525269080005909              4      1   V-K003       1
4     1  Netzdruckdiagramm VL/RL: BHKW - Netzende   101    ROHR  5037777106796980248 ␣
↪5252525269080005909  5252525269080005909              5      1   V-K004       1
5     1  Netzdruckdiagramm VL/RL: BHKW - Netzende   101    ROHR  4713733238627697042 ␣
↪5252525269080005909  5252525269080005909              6      1   V-K005       1
6     1  Netzdruckdiagramm VL/RL: BHKW - Netzende   101    ROHR  5123819811204259837 ␣
↪5252525269080005909  5252525269080005909              7      1   V-K006       1
>>> # ---
>>> # vFWVB
>>> # ---
>>> print("'''{:s}'''".format(repr(xm.dataFrames['vFWVB']).replace('\n','\n    ')))
'''  BESCHREIBUNG IDREFERENZ   W0  LFK  W0LFK  TVL0  TRS0  LFKT      W  W_min  W_max ␣
↪INDTR  TRSK  VTYP  DPHAUS  IMBG  IRFV                    pk                   tk ␣
↪NAME_i KVR_i TM_i   XKOR_i   YKOR_i ZKOR_i  pXCor_i  pYCor_i  NAME_k KVR_k TM_k  ␣
↪XKOR_k   YKOR_k ZKOR_k  pXCor_k  pYCor_k                                      CONT␣
↪CONT_ID CONT_LFDNR                         WBLZ
   0           1         -1  200  0.8  160.0    90    50  LFKT  160.0  160.0  160.0 ␣
↪    1    55    14     0.7     0   0.0  4643800032883366034  4643800032883366034  V-
↪K002     1   90  2541059  5706265     20    319.0     56.0  R-K002     2   60 ␣
↪2541059  5706265     20    319.0     56.0  Nahwärmenetz mit 1000 kW Anschlussleistu␣
↪   1001         -1  [BLNZ1, BLNZ1u5, BLNZ1u5u7]
   1           3         -1  200  1.0  200.0    90    65  LFKT  200.0  200.0  200.0 ␣
↪    1    65    14     0.7     0   0.0  4704603947372595298  4704603947372595298  V-
↪K004     1   90  2541539  5706361     20    799.0    152.0  R-K004     2   60 ␣
↪2541539  5706361     20    799.0    152.0  Nahwärmenetz mit 1000 kW Anschlussleistu␣
↪   1001         -1                            []
   2           4         -1  200  0.8  160.0    90    60  LFKT  160.0  160.0  160.0 ␣
↪    1    60    14     0.7     0   0.0  5121101823283893406  5121101823283893406  V-
↪K005     1   90  2541627  5706363     20    887.0    154.0  R-K005     2   60 ␣
↪2541627  5706363     20    887.0    154.0  Nahwärmenetz mit 1000 kW Anschlussleistu␣
↪   1001         -1  [BLNZ1u5, BLNZ1u5u7, BLNZ5]
   3           5         -1  200  0.8  160.0    90    55  LFKT  160.0  160.0  160.0 ␣
↪    1    55    14     0.7     0   0.0  5400405917816384862  5400405917816384862  V-
↪K007     1   90  2541899  5706325     20   1159.0    116.0  R-K007     2   60 ␣
↪2541899  5706325     20   1159.0    116.0  Nahwärmenetz mit 1000 kW Anschlussleistu␣
↪   1001         -1                   [BLNZ1u5u7]
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   4            2         -1  200  0.6  120.0    90    60  LFKT  120.0  120.0  120.0 ␣
↪    1    62    14     0.7     0   0.0  5695730293103267172  5695730293103267172  V-
↪K003     1   90  2541457  5706345     20    717.0    136.0  R-K003     2   60 ␣
↪2541457  5706345     20    717.0    136.0  Nahwärmenetz mit 1000 kW Anschlussleistu␣
↪   1001         -1                          []'''
>>> # ---
>>> # vLAYR
>>> # ---
>>> # print("'''{:s}'''".format(repr(xm.dataFrames['vLAYR'].sort_values(['LFDNR','NAME
↪','OBJTYPE','OBJID'],ascending=True)).replace('\n','\n    ')))
>>> print(xm._getvXXXXAsOneString(vXXXX='vLAYR'))
   LFDNR           NAME OBJTYPE                OBJID                    pk            ␣
↪      tk  nrObjInGroup  nrObjtypeInGroup
0      1        Vorlauf    FWES  5638756766880678918  5206516471428693478 ␣
↪5206516471428693478             1                  1
1      1        Vorlauf    KNOT  4731792362611615619  5206516471428693478 ␣
↪5206516471428693478             1                  1
2      1        Vorlauf    KNOT  4756962427318766791  5206516471428693478 ␣
↪5206516471428693478             1                  2
3      1        Vorlauf    KNOT  4766681917240867943  5206516471428693478 ␣
↪5206516471428693478             1                  3
4      1        Vorlauf    KNOT  5049461676240771430  5206516471428693478 ␣
↪5206516471428693478             1                  4
5      1        Vorlauf    KNOT  5370423799772591808  5206516471428693478 ␣
↪5206516471428693478             1                  5
6      1        Vorlauf    KNOT  5444644492819213978  5206516471428693478 ␣
↪5206516471428693478             1                  6
7      1        Vorlauf    KNOT  5515313800585145571  5206516471428693478 ␣
↪5206516471428693478             1                  7
8      1        Vorlauf    KNOT  5646671866542823796  5206516471428693478 ␣
↪5206516471428693478             1                  8
9      1        Vorlauf    KNOT  5736262931552588702  5206516471428693478 ␣
↪5206516471428693478             1                  9
10     1        Vorlauf    KNOT  5741235692335544560  5206516471428693478 ␣
↪5206516471428693478             1                 10
11     1        Vorlauf    ROHR  4614949065966596185  5206516471428693478 ␣
↪5206516471428693478             1                  1
12     1        Vorlauf    ROHR  4713733238627697042  5206516471428693478 ␣
↪5206516471428693478             1                  2
13     1        Vorlauf    ROHR  4789218195240364437  5206516471428693478 ␣
↪5206516471428693478             1                  3
14     1        Vorlauf    ROHR  4939422678063487923  5206516471428693478 ␣
↪5206516471428693478             1                  4
15     1        Vorlauf    ROHR  4984202422877610920  5206516471428693478 ␣
↪5206516471428693478             1                  5
16     1        Vorlauf    ROHR  5037777106796980248  5206516471428693478 ␣
↪5206516471428693478             1                  6
17     1        Vorlauf    ROHR  5123819811204259837  5206516471428693478 ␣
↪5206516471428693478             1                  7
18     1        Vorlauf    ROHR  5620197984230756681  5206516471428693478 ␣
↪5206516471428693478             1                  8
19     1        Vorlauf    VENT  4678923650983295610  5206516471428693478 ␣
↪5206516471428693478             1                  1
20     2       Rücklauf    KLAP  4801110583764519435  4693347477612662930 ␣
↪4693347477612662930             1                  1
21     2       Rücklauf    KNOT  4638663808856251977  4693347477612662930 ␣
↪4693347477612662930             1                  1
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22     2       Rücklauf    KNOT  4807712987325933680  4693347477612662930 ␣
↪4693347477612662930             1                  2
23     2       Rücklauf    KNOT  4891048046264179170  4693347477612662930 ␣
↪4693347477612662930             1                  3
24     2       Rücklauf    KNOT  4979785838440534851  4693347477612662930 ␣
↪4693347477612662930             1                  4
25     2       Rücklauf    KNOT  5002109894154139899  4693347477612662930 ␣
↪4693347477612662930             1                  5
26     2       Rücklauf    KNOT  5183147862966701025  4693347477612662930 ␣
↪4693347477612662930             1                  6
27     2       Rücklauf    KNOT  5219230031772497417  4693347477612662930 ␣
↪4693347477612662930             1                  7
28     2       Rücklauf    KNOT  5356267303828212700  4693347477612662930 ␣
↪4693347477612662930             1                  8
29     2       Rücklauf    KNOT  5364712333175450942  4693347477612662930 ␣
↪4693347477612662930             1                  9
30     2       Rücklauf    KNOT  5397990465339071638  4693347477612662930 ␣
↪4693347477612662930             1                 10
31     2       Rücklauf    KNOT  5508992300317633799  4693347477612662930 ␣
↪4693347477612662930             1                 11
32     2       Rücklauf    KNOT  5543326527366090679  4693347477612662930 ␣
↪4693347477612662930             1                 12
33     2       Rücklauf    KNOT  5557222628687032084  4693347477612662930 ␣
↪4693347477612662930             1                 13
34     2       Rücklauf    PUMP  5481331875203087055  4693347477612662930 ␣
↪4693347477612662930             1                  1
35     2       Rücklauf    ROHR  4613782368750024999  4693347477612662930 ␣
↪4693347477612662930             1                  1
36     2       Rücklauf    ROHR  4637102239750163477  4693347477612662930 ␣
↪4693347477612662930             1                  2
37     2       Rücklauf    ROHR  4769996343148550485  4693347477612662930 ␣
↪4693347477612662930             1                  3
38     2       Rücklauf    ROHR  4945727430885351042  4693347477612662930 ␣
↪4693347477612662930             1                  4
39     2       Rücklauf    ROHR  5266224553324203132  4693347477612662930 ␣
↪4693347477612662930             1                  5
40     2       Rücklauf    ROHR  5379365049009065623  4693347477612662930 ␣
↪4693347477612662930             1                  6
41     2       Rücklauf    ROHR  5611703699850694889  4693347477612662930 ␣
↪4693347477612662930             1                  7
42     2       Rücklauf    ROHR  5647213228462830353  4693347477612662930 ␣
↪4693347477612662930             1                  8
43     2       Rücklauf    VENT  4897018421024717974  4693347477612662930 ␣
↪4693347477612662930             1                  1
44     2       Rücklauf    VENT  5525310316015533093  4693347477612662930 ␣
↪4693347477612662930             1                  2
45     3  Kundenanlagen    FWVB  4643800032883366034  5003333277973347346 ␣
↪5003333277973347346             1                  1
46     3  Kundenanlagen    FWVB  4704603947372595298  5003333277973347346 ␣
↪5003333277973347346             1                  2
47     3  Kundenanlagen    FWVB  5121101823283893406  5003333277973347346 ␣
↪5003333277973347346             1                  3
48     3  Kundenanlagen    FWVB  5400405917816384862  5003333277973347346 ␣
↪5003333277973347346             1                  4
49     3  Kundenanlagen    FWVB  5695730293103267172  5003333277973347346 ␣
↪5003333277973347346             1                  5
50     4           BHKW    BSYM  5043395081363401573  5555393404073362943 ␣
↪5555393404073362943             1                  1
```

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```
51      4            BHKW     TEXT  5056836766824229789  5555393404073362943 ␣
↪5555393404073362943              1                   1
52      4            BHKW     TEXT  5329748935118523443  5555393404073362943 ␣
↪5555393404073362943              1                   2
53      5           Texte     ARRW  4664845735864571219  5394410243594912680 ␣
↪5394410243594912680              1                   1
54      5           Texte     ARRW  4902474974831811106  5394410243594912680 ␣
↪5394410243594912680              1                   2
55      5           Texte     ARRW  5026846801782366678  5394410243594912680 ␣
↪5394410243594912680              1                   3
56      5           Texte     ARRW  5688313372729413840  5394410243594912680 ␣
↪5394410243594912680              1                   4
57      5           Texte     NRCV  4681213816714574464  5394410243594912680 ␣
↪5394410243594912680              1                   1
58      5           Texte     NRCV  4857294696992797631  5394410243594912680 ␣
↪5394410243594912680              1                   2
59      5           Texte     NRCV  4914949875368816179  5394410243594912680 ␣
↪5394410243594912680              1                   3
60      5           Texte     NRCV  4946584950744559030  5394410243594912680 ␣
↪5394410243594912680              1                   4
61      5           Texte     NRCV  4968703141722117357  5394410243594912680 ␣
↪5394410243594912680              1                   5
62      5           Texte     NRCV  5091374651838464239  5394410243594912680 ␣
↪5394410243594912680              1                   6
63      5           Texte     NRCV  5097127385155151127  5394410243594912680 ␣
↪5394410243594912680              1                   7
64      5           Texte     NRCV  5179988968597313889  5394410243594912680 ␣
↪5394410243594912680              1                   8
65      5           Texte     NRCV  5281885868749421521  5394410243594912680 ␣
↪5394410243594912680              1                   9
66      5           Texte     NRCV  5410904806390050339  5394410243594912680 ␣
↪5394410243594912680              1                  10
67      5           Texte     NRCV  5476262878682325254  5394410243594912680 ␣
↪5394410243594912680              1                  11
68      5           Texte     NRCV  5557806245003742769  5394410243594912680 ␣
↪5394410243594912680              1                  12
69      5           Texte     RECT  4994817837124479818  5394410243594912680 ␣
↪5394410243594912680              1                   1
70      5           Texte     RPFL  5158870568935841216  5394410243594912680 ␣
↪5394410243594912680              1                   1
71      5           Texte     TEXT  4628671704393700430  5394410243594912680 ␣
↪5394410243594912680              1                   1
72      5           Texte     TEXT  4654104397990769217  5394410243594912680 ␣
↪5394410243594912680              1                   2
73      5           Texte     TEXT  4666644549022031339  5394410243594912680 ␣
↪5394410243594912680              1                   3
74      5           Texte     TEXT  4693143208412077585  5394410243594912680 ␣
↪5394410243594912680              1                   4
75      5           Texte     TEXT  4768731522550494423  5394410243594912680 ␣
↪5394410243594912680              1                   5
76      5           Texte     TEXT  4770844990228490264  5394410243594912680 ␣
↪5394410243594912680              1                   6
77      5           Texte     TEXT  4782197969172967134  5394410243594912680 ␣
↪5394410243594912680              1                   7
78      5           Texte     TEXT  4855692488683645764  5394410243594912680 ␣
↪5394410243594912680              1                   8
79      5           Texte     TEXT  4965628942555351751  5394410243594912680 ␣
↪5394410243594912680              1                   9
```

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```
80     5          Texte    TEXT  4995961504641886710  5394410243594912680 ␣
↪5394410243594912680             1                 10
81     5          Texte    TEXT  5017907661719368413  5394410243594912680 ␣
↪5394410243594912680             1                 11
82     5          Texte    TEXT  5028052147238787802  5394410243594912680 ␣
↪5394410243594912680             1                 12
83     5          Texte    TEXT  5036153631350515544  5394410243594912680 ␣
↪5394410243594912680             1                 13
84     5          Texte    TEXT  5054433315422452796  5394410243594912680 ␣
↪5394410243594912680             1                 14
85     5          Texte    TEXT  5108336975548011049  5394410243594912680 ␣
↪5394410243594912680             1                 15
86     5          Texte    TEXT  5262441422409836340  5394410243594912680 ␣
↪5394410243594912680             1                 16
87     5          Texte    TEXT  5297832234834839298  5394410243594912680 ␣
↪5394410243594912680             1                 17
88     5          Texte    TEXT  5370727463979416592  5394410243594912680 ␣
↪5394410243594912680             1                 18
89     5          Texte    TEXT  5421223289472778073  5394410243594912680 ␣
↪5394410243594912680             1                 19
90     5          Texte    TEXT  5501963349880613918  5394410243594912680 ␣
↪5394410243594912680             1                 20
91     5          Texte    TEXT  5502619581048467908  5394410243594912680 ␣
↪5394410243594912680             1                 21
92     5          Texte    TEXT  5540395812045688781  5394410243594912680 ␣
↪5394410243594912680             1                 22
93     5          Texte    TEXT  5550982489075668484  5394410243594912680 ␣
↪5394410243594912680             1                 23
94     5          Texte    TEXT  5610916400841895317  5394410243594912680 ␣
↪5394410243594912680             1                 24
95     5          Texte    TEXT  5646820849868629537  5394410243594912680 ␣
↪5394410243594912680             1                 25
96     5          Texte    TEXT  5696590398594231893  5394410243594912680 ␣
↪5394410243594912680             1                 26
97     5          Texte    TEXT  5697088036451277538  5394410243594912680 ␣
↪5394410243594912680             1                 27
>>> # ---
>>> # vGTXT
>>> # ---
>>> print(xm._getvXXXXAsOneString(vXXXX='vGTXT',sortList=['CONT_ID','pk'],index=False,
↪header=False))
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                              ␣
↪numerische Anzeige:  4614148870174765680  4614148870174765680                    ␣
↪    (219.0, -278.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                              ␣
↪Georeferenzpunkt 2  4628671704393700430  4628671704393700430              (1115.
↪9500000001863, -323.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                              ␣
↪    Block  4666644549022031339  4666644549022031339                        (-58.
↪0, -77.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                              ␣
↪numerische Anzeige  4693143208412077585  4693143208412077585                   ␣
↪    (1211.0, -9.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                           Knoten␣
↪und Rohre  4995961504641886710  4995961504641886710                          (570.
↪0, -49.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                              ␣
↪Vorlaufstrang  5017907661719368413  5017907661719368413  (358.20699999999600, 120.2
↪39499999955297)
```

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Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                          ␣
↪LocalHeatingNetwork  5028052147238787802  5028052147238787802                ␣
↪    (1163.0, 536.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1   Tel. 05131 - 4980-0 ; Fax. 05131␣
↪- 4980-15  5054433315422452796  5054433315422452796                       (-230.0,
↪ -1143.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                   Kontrolle: DH-
↪Massenstrom  5100960407865990868  5100960407865990868                        (-
↪60.0, -160.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                          ␣
↪Wärmebilanz: 3 Kunden  5150752151066924202  5150752151066924202              ␣
↪      (219.0, -318.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1  eMail. info@3SConsult.de ; www.
↪3SConsult.de  5370727463979416592  5370727463979416592                       (-
↪230.0, -1204.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1               Differenzdruck VL-/␣
↪RL-Knoten  5502619581048467908  5502619581048467908                         (1211.
↪0, -49.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                          ␣
↪Kundenanlage  5540395812045688781  5540395812045688781  (1131.9500000001863, 283.
↪95000000018626)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                          ␣
↪Fernwärmeverbraucher  5550982489075668484  5550982489075668484              ␣
↪     (1050.0, 239.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                          ␣
↪Rücklaufstrang  5610916400841895317  5610916400841895317                    ␣
↪ (570.0, -9.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                         Knoten␣
↪und Rohre  5646820849868629537  5646820849868629537  (358.20699999993667, 174.
↪39499999955297)
                                  BHKW  1002  -1                            ␣
↪Fernwärmeeinspeiser  4654104397990769217  4654104397990769217              ␣
↪      (115.0, 80.0)
                                  BHKW  1002  -1                            ␣
↪    Pumpe  4768731522550494423  4768731522550494423                        ␣
↪(175.0, 25.0)
                                  BHKW  1002  -1                            ␣
↪Wärmebilanz Netz:  4770844990228490264  4770844990228490264                 ␣
↪    (90.0, 160.0)
                                  BHKW  1002  -1                            ␣
↪Speicherung  4782197969172967134  4782197969172967134                      ␣
↪(110.0, 140.0)
                                  BHKW  1002  -1                            ␣
↪Richtungspfeil  4855692488683645764  4855692488683645764                    ␣
↪(220.0, 105.0)
                                  BHKW  1002  -1                            ␣
↪ Verluste  4965628942555351751  4965628942555351751                        (110.
↪0, 145.0)
                                  BHKW  1002  -1                      (Element␣
↪verbinden)  5036153631350515544  5036153631350515544                       ␣
↪(150.0, 90.0)
                                  BHKW  1002  -1                BHKW Modul 1000␣
↪kW therm.  5056836766824229789  5056836766824229789                         ␣
↪(35.0, 55.0)
                                  BHKW  1002  -1                            ␣
↪   Ventil  5108336975548011049  5108336975548011049                         ␣
↪(205.0, 25.0)
```

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                                    BHKW  1002  -1                                                  ␣
↪Verbrauch  5262441422409836340  5262441422409836340                                (110.
↪0, 150.0)
                                    BHKW  1002  -1                                                   ␣
↪Einspeisung  5297832234834839298  5297832234834839298                                ␣
↪(110.0, 155.0)
                                    BHKW  1002  -1                                          ␣
↪Druckhaltung 2 bar  5329748935118523443  5329748935118523443                                ␣
↪     (180.0, 65.0)
                                    BHKW  1002  -1                                          ␣
↪Numerische Anzeige  5421223289472778073  5421223289472778073                                ␣
↪    (190.0, 115.0)
                                    BHKW  1002  -1                                             ␣
↪Verbindungslinie  5501963349880613918  5501963349880613918                                ␣
↪   (150.0, 95.0)
                                    BHKW  1002  -1                                                ␣
↪   (Text)  5696590398594231893  5696590398594231893                                ␣
↪(35.0, 50.0)
                                    BHKW  1002  -1                                                ␣
↪   Klappe  5697088036451277538  5697088036451277538                                ␣
↪(145.0, 25.0)
>>> # ---
>>> # vNRCV
>>> # ---
>>> print(xm._getvXXXXAsOneString(vXXXX='vNRCV',end=14,dropColList=['DPGR','CONT_LFDNR
↪','pk_ROWS'],sortList=['OBJTYPE','fkOBJTYPE','ATTRTYPE','cRefLfdNr']))
   cRefLfdNr                                    CONT CONT_ID OBJTYPE                ␣
↪fkOBJTYPE ATTRTYPE              tk_ROWS                   pk                   tk   ␣
↪                              pXYLB
0          1                                    BHKW    1002    FWES ␣
↪5638756766880678918        W  5762106696740202356  4857294696992797631 ␣
↪4857294696992797631                          (90.0, 65.0)
1          1                                    BHKW    1002    KNOT ␣
↪5049461676240771430        T  4723443975311885965  5097127385155151127 ␣
↪5097127385155151127                          (90.0, 95.0)
2          1                                    BHKW    1002    KNOT ␣
↪5219230031772497417        T  5602301870151014230  5557806245003742769 ␣
↪5557806245003742769                          (90.0, 35.0)
3          1                                    BHKW    1002    KNOT ␣
↪5356267303828212700       PH  5000989080893535213  4968703141722117357 ␣
↪4968703141722117357                         (220.0, 25.0)
4          1  Nahwärmenetz mit 1000 kW Anschlussleistu    1001    KNOT ␣
↪5397990465339071638       QM  5134531789044068877  5410059595276504750 ␣
↪5410059595276504750                         (91.0, -94.0)
5          2                                    BHKW    1002    KNOT ␣
↪5397990465339071638       QM  5134531789044068877  5357021981944933535 ␣
↪5357021981944933535  (184.999999464624, 57.99999953107601)
6          1                                    BHKW    1002    KNOT ␣
↪5736262931552588702       PH  4754881272083464445  4681213816714574464 ␣
↪4681213816714574464                         (220.0, 85.0)
7          1  Nahwärmenetz mit 1000 kW Anschlussleistu    1001    KNOT ␣
↪5741235692335544560       DP  4949183695502554728  4914949875368816179 ␣
↪4914949875368816179                        (1234.0, 83.0)
8          1                                    BHKW    1002    PUMP ␣
↪5481331875203087055        N  5563842594211689762  5091374651838464239 ␣
↪5091374651838464239                         (170.0, 45.0)
9          1                                    BHKW    1002    VENT ␣
↪4678923650983295610       QM  5126307362398248950  5410904806390050339 ␣
↪5410904806390050339                        (200.0, 110.0)
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10          1  Nahwärmenetz mit 1000 kW Anschlussleistu    1001    WBLZ ␣
↪4694700216019268978      WVB  4778244458749966216  4991097791264453745 ␣
↪4991097791264453745                        (354.0, -225.0)
11          1                                   BHKW    1002    WBLZ ␣
↪5262603207038486299      WES  5690691957596882133  5179988968597313889 ␣
↪5179988968597313889                          (90.0, 155.0)
12          1                                   BHKW    1002    WBLZ ␣
↪5262603207038486299    WSPEI  5153847813311339683  4946584950744559030 ␣
↪4946584950744559030                          (90.0, 140.0)
13          1                                   BHKW    1002    WBLZ ␣
↪5262603207038486299      WVB  5214984699859365639  5281885868749421521 ␣
↪5281885868749421521                          (90.0, 150.0)
>>> # ---
>>> # MxSync() - without Mx-Object
>>> # ---
>>> vROHR=xm.dataFrames['vROHR']
>>> vROHR.shape
(16, 74)
>>> 'vNRCV_Mx1' in xm.dataFrames
False
>>> mx=xm.MxSync()
>>> 'vNRCV_Mx1' in xm.dataFrames
True
>>> vROHR.shape
(16, 76)
>>> print(xm._getvXXXXAsOneString(vXXXX='vROHR',dropColList=['NAME_i_2L','NAME_k_2L
↪']))
   BESCHREIBUNG IDREFERENZ BAUJAHR HAL IPLANUNG KENNUNG       L LZU  RAU ZAUS ZEIN␣
↪ZUML JLAMBS LAMBDA0 ASOLL INDSCHALL KVR  AUSFALLZEIT     DA     DI   DN     KT  PN ␣
↪REHABILITATION  REPARATUR    S WSTEIG WTIEFE LTGR_NAME           LTGR_BESCHREIBUNG␣
↪SICHTBARKEIT VERLEGEART DTRO_NAME                      DTRO_BESCHREIBUNG        E␣
↪fkSTRASSE fkSRAT                   pk                    tk IRTRENN LECKSTART␣
↪LECKEND LECKMENGE LECKORT LECKSTATUS QSVB ZVLIMPTNZ KANTENZV                    ␣
↪              CONT CONT_ID CONT_LFDNR  NAME_i KVR_i TM_i   XKOR_i   YKOR_i ZKOR_i ␣
↪NAME_k KVR_k TM_k   XKOR_k   YKOR_k ZKOR_k  pXCor_i  pYCor_i  pXCor_k  pYCor_k    ␣
↪      pXCors          pYCors                                pWAYPXCors           ␣
↪                 pWAYPYCors                                                       ␣
↪                                      WAYP  mx2NofPts  mx2Idx
0          None         -1    None   0        1       0   88.02   0  0.1    0    0 ␣
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN  ␣
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe  ␣
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11  ␣
↪    -1     -1  4613782368750024999  4613782368750024999       0         0       0 ␣
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  R-K004     2   60  2541539  5706361     20  R-
↪K005     2   60  2541627  5706363     20    799.0    152.0    887.0    154.0    ␣
↪[799.0, 887.0]  [152.0, 154.0]    [807.8999999999069, 895.9500000001863]  [140.
↪09999999962747, 142.04999999981374]                                             ␣
↪[(2541547.9, 5706349.1), (2541635.95, 5706351.05)]          2       0
1          None         -1    None   0        1       0  405.96   0  0.1    0    0 ␣
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN  ␣
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe  ␣
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11  ␣
↪    -1     -1  4614949065966596185  4614949065966596185       0         0       0 ␣
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  V-K002     1   90  2541059  5706265     20  V-
↪K003     1   90  2541457  5706345     20    319.0     56.0    717.0    136.0    ␣
↪[319.0, 717.0]   [56.0, 136.0]                [319.0, 716.9500000001863]         ␣
↪   [56.049999999813735, 136.0]                                                   ␣
↪[(2541059.0, 5706265.05), (2541456.95, 5706345.0)]          2       1
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2          None         -1    None   0        1       0   83.55   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4637102239750163477  4637102239750163477       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  R-K003     2   60  2541457  5706345     20  R-
↪K004     2   60  2541539  5706361     20    717.0    136.0    799.0    152.0
↪[717.0, 799.0]  [136.0, 152.0]    [725.8500000000931, 807.8999999999069]  [124.
↪04999999981374, 140.09999999962747]
↪[(2541465.85, 5706333.05), (2541547.9, 5706349.1)]          2       2
3          None         -1    None   0        1       0   88.02   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4713733238627697042  4713733238627697042       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  V-K004     1   90  2541539  5706361     20  V-
↪K005     1   90  2541627  5706363     20    799.0    152.0    887.0    154.0
↪[799.0, 887.0]  [152.0, 154.0]                [799.0, 887.0499999998137]
↪                [152.0, 154.0]
↪[(2541539.0, 5706361.0), (2541627.05, 5706363.0)]          2       3
4          None         -1    None   0        1       0  195.53   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4789218195240364437  4789218195240364437       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  V-K001     1   90  2540867  5706228     20  V-
↪K002     1   90  2541059  5706265     20    127.0     19.0    319.0     56.0
↪[127.0, 319.0]    [19.0, 56.0]                            [127.0, 319.0]
↪    [19.0, 56.049999999813735]
↪[(2540867.0, 5706228.0), (2541059.0, 5706265.05)]          2       5
5          None         -1    None   0        1       0  109.77   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4945727430885351042  4945727430885351042       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  R-K006     2   60  2541790  5706338     20  R-
↪K007     2   60  2541899  5706325     20   1050.0    129.0   1159.0    116.0  [1050.
↪0, 1159.0]  [129.0, 116.0]  [1058.8500000000931, 1167.8999999999069]
↪[117.0, 104.09999999962747]
↪[(2541798.85, 5706326.0), (2541907.9, 5706313.1)]          2       7
6          None         -1    None   0        1       0    76.4   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4984202422877610920  4984202422877610920       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  V-K000     1   90  2540793  5706209     20  V-
↪K001     1   90  2540867  5706228     20     53.0      0.0    127.0     19.0
↪[53.0, 127.0]     [0.0, 19.0]                [53.049999999813735, 127.0]
↪[-0.049999999813735485, 19.0]
↪[(2540793.05, 5706208.95), (2540867.0, 5706228.0)]          2       8
7          None         -1    None   0        1       0   83.55   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5037777106796980248  5037777106796980248       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  V-K003     1   90  2541457  5706345     20  V-
↪K004     1   90  2541539  5706361     20    717.0    136.0    799.0    152.0
↪[717.0, 799.0]  [136.0, 152.0]                [716.9500000001863, 799.0]
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8          None         -1    None   0        1       0  164.91   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5123819811204259837  5123819811204259837       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  V-K005     1   90  2541627  5706363     20  V-
↪K006     1   90  2541790  5706338     20    887.0    154.0   1050.0    129.0   [887.
↪0, 1050.0]  [154.0, 129.0]   [887.0499999998137, 1049.9500000001863]
↪[154.0, 128.95000000018626]
↪[(2541627.05, 5706363.0), (2541789.95, 5706337.95)]          2      10
9          None         -1    None   0        1       0  195.53   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5266224553324203132  5266224553324203132       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  R-K001     2   60  2540867  5706228     20  R-
↪K002     2   60  2541059  5706265     20    127.0     19.0    319.0     56.0
↪[127.0, 319.0]    [19.0, 56.0]  [135.89999999990687, 327.89999999990687]   [7.
↪0499999998137355, 44.09999999962747]
↪  [(2540875.9, 5706216.05), (2541067.9, 5706253.1)]          2      11
10         None         -1    None   0        1       0  405.96   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5379365049009065623  5379365049009065623       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  R-K002     2   60  2541059  5706265     20  R-
↪K003     2   60  2541457  5706345     20    319.0     56.0    717.0    136.0
↪[319.0, 717.0]   [56.0, 136.0]   [327.89999999990687, 725.8500000000931]   [44.
↪09999999962747, 124.04999999981374]
↪[(2541067.9, 5706253.1), (2541465.85, 5706333.05)]          2      12
11         None         -1    None   0        1       0  164.91   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5611703699850694889  5611703699850694889       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  R-K005     2   60  2541627  5706363     20  R-
↪K006     2   60  2541790  5706338     20    887.0    154.0   1050.0    129.0   [887.
↪0, 1050.0]  [154.0, 129.0]   [895.9500000001863, 1058.8500000000931]
↪[142.04999999981374, 117.0]
↪[(2541635.95, 5706351.05), (2541798.85, 5706326.0)]          2      13
12         None         -1    None   0        1       0  109.77   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5620197984230756681  5620197984230756681       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  V-K006     1   90  2541790  5706338     20  V-
↪K007     1   90  2541899  5706325     20   1050.0    129.0   1159.0    116.0  [1050.
↪0, 1159.0]  [129.0, 116.0]             [1049.9500000001863, 1159.0]  [128.
↪95000000018626, 116.04999999981374]
↪[(2541789.95, 5706337.95), (2541899.0, 5706325.05)]          2      14
13         None         -1    None   0        1       0    76.4   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5647213228462830353  5647213228462830353       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  R-K000     2   60  2540793  5706209     20  R-
↪K001     2   60  2540867  5706228     20     53.0      0.0    127.0     19.0
↪[53.0, 127.0]     [0.0, 19.0]  [61.950000000186265, 135.89999999990687]
```

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14         None         -1    None   0        1       0   73.42   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  168.3  160.3  150   0.45 NaN
↪        NaN        NaN    4    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4769996343148550485  4769996343148550485       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1     R-L     2   60  2540740  5706225     20  R-
↪K000     2   60  2540793  5706209     20      0.0     16.0     53.0      0.0
↪[0.0, 53.0]     [16.0, 0.0]     [0.0, 24.0, 45.0, 61.950000000186265]
↪ [16.0, 16.0, -12.0, -12.0]  [(2540740.0, 5706225.0), (2540764.0, 5706225.0),␣
↪(2540785.0, 5706197.0), (2540801.95, 5706197.0)]          2       4
15         None         -1    None   0        1       0    68.6   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  168.3  160.3  150   0.45 NaN
↪        NaN        NaN    4    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4939422678063487923  4939422678063487923       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1     V-L     1   90  2540740  5706240     20  V-
↪K000     1   90  2540793  5706209     20      0.0     31.0     53.0      0.0
↪[0.0, 53.0]     [31.0, 0.0]           [0.0, 30.0, 53.049999999813735]       [31.0,␣
↪31.0, -0.049999999813735485]                        [(2540740.0, 5706240.0),␣
↪(2540770.0, 5706240.0), (2540793.05, 5706208.95)]          2       6
>>> # ---------
>>> # MxSync()
>>> # ---------
>>> xm=Xm(xmlFile=xmlFile)
>>> vROHR=xm.dataFrames['vROHR']
>>> vROHR.shape
(16, 74)
>>> 'vNRCV_Mx1' in xm.dataFrames
False
>>> (wDir,modelDir,modelName,mx1File)=xm.getWDirModelDirModelName()
>>> mx=Mx.Mx(mx1File=mx1File)
>>> xm.MxSync(mx=mx)
>>> vROHR.shape
(16, 76)
>>> 'vNRCV_Mx1' in xm.dataFrames
True
>>> # ---
>>> # vNRCV_Mx1
>>> # ---
>>> import re
>>> f=lambda s: re.match('(\S+)~(\S*)~(\S*)~(\d+)~(\S+)',s).group(1)+'~~~'+re.match(
↪'(\S+)~(\S*)~(\S*)~(\d+)~(\S+)',s).group(4)+'~'+re.match('(\S+)~(\S*)~(\S*)~(\d+)~
↪(\S+)',s).group(5)
>>> print(xm._getvXXXXAsOneString(vXXXX='vNRCV_Mx1',end=14,dropColList=['DPGR','CONT_
↪LFDNR','pk_ROWS'],mapFunc={'Sir3sID':f},sortList=['Sir3sID'],fmtFunc={'Sir3sID':f},
↪index=False,header=False))
FWES~~~5638756766880678918~W  1                                        BHKW  1002  FWES␣
↪ 5638756766880678918      W  5762106696740202356  4857294696992797631 ␣
↪4857294696992797631     (90.0, 65.0)
    KNOT~~~5049461676240771430~T  1                                        BHKW  1002 ␣
↪KNOT  5049461676240771430      T  4723443975311885965  5097127385155151127 ␣
↪5097127385155151127     (90.0, 95.0)
    KNOT~~~5219230031772497417~T  1                                        BHKW  1002 ␣
↪KNOT  5219230031772497417      T  5602301870151014230  5557806245003742769 ␣
↪5557806245003742769     (90.0, 35.0)
```

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   KNOT~~~5356267303828212700~PH  1                                      BHKW  1002 ␣
↪KNOT  5356267303828212700     PH  5000989080893535213  4968703141722117357 ␣
↪4968703141722117357    (220.0, 25.0)
   KNOT~~~5397990465339071638~QM  1  Nahwärmenetz mit 1000 kW Anschlussleistu  1001 ␣
↪KNOT  5397990465339071638     QM  5134531789044068877  5410059595276504750 ␣
↪5410059595276504750    (91.0, -94.0)
   KNOT~~~5736262931552588702~PH  1                                      BHKW  1002 ␣
↪KNOT  5736262931552588702     PH  4754881272083464445  4681213816714574464 ␣
↪4681213816714574464    (220.0, 85.0)
   KNOT~~~5741235692335544560~DP  1  Nahwärmenetz mit 1000 kW Anschlussleistu  1001 ␣
↪KNOT  5741235692335544560     DP  4949183695502554728  4914949875368816179 ␣
↪4914949875368816179   (1234.0, 83.0)
    PUMP~~~5481331875203087055~N  1                                      BHKW  1002 ␣
↪PUMP  5481331875203087055      N  5563842594211689762  5091374651838464239 ␣
↪5091374651838464239    (170.0, 45.0)
   VENT~~~4678923650983295610~QM  1                                      BHKW  1002 ␣
↪VENT  4678923650983295610     QM  5126307362398248950  5410904806390050339 ␣
↪5410904806390050339   (200.0, 110.0)
  WBLZ~~~4694700216019268978~WVB  1  Nahwärmenetz mit 1000 kW Anschlussleistu  1001 ␣
↪WBLZ  4694700216019268978    WVB  4778244458749966216  4991097791264453745 ␣
↪4991097791264453745  (354.0, -225.0)
  WBLZ~~~5262603207038486299~WES  1                                      BHKW  1002 ␣
↪WBLZ  5262603207038486299    WES  5690691957596882133  5179988968597313889 ␣
↪5179988968597313889    (90.0, 155.0)
WBLZ~~~5262603207038486299~WSPEI  1                                      BHKW  1002 ␣
↪WBLZ  5262603207038486299  WSPEI  5153847813311339683  4946584950744559030 ␣
↪4946584950744559030    (90.0, 140.0)
  WBLZ~~~5262603207038486299~WVB  1                                      BHKW  1002 ␣
↪WBLZ  5262603207038486299    WVB  5214984699859365639  5281885868749421521 ␣
↪5281885868749421521    (90.0, 150.0)
WBLZ~~~5262603207038486299~WVERL  1                                      BHKW  1002 ␣
↪WBLZ  5262603207038486299  WVERL  4722863010266870887  5476262878682325254 ␣
↪5476262878682325254    (90.0, 145.0)
>>> # ---
>>> # vKNOT
>>> # ---
>>> print(xm._getvXXXXAsOneString(vXXXX='vKNOT',end=1,dropColList=['LFKT_NAME','LF',
↪'LF_min','LF_max','PVAR_NAME','PH','PH_min','PH_max','PZON_NAME','FSTF_NAME','STOF_
↪NAME','GMIX_NAME','UTMP_NAME','2L_NAME','2L_KVR','fkHYDR','fkFQPS']))
     NAME BESCHREIBUNG IDREFERENZ                                    CONT CONT_ID␣
↪CONT_LFDNR CONT_VKNO  KTYP LFAKT QM_EIN QVAR_NAME  QM  QM_min  QM_max KVR  TE  TM  ␣
↪  XKOR     YKOR ZKOR                  pk                  tk  pXCor  pYCor  mx2Idx
0  R-K004         None         -1  Nahwärmenetz mit 1000 kW Anschlussleistu    1001  ␣
↪      -1       NaN  QKON     1      0       NaN NaN     NaN     NaN   2  10  60 ␣
↪2541539  5706361   20  4638663808856251977  4638663808856251977  799.0  152.0      ␣
↪0
>>> print(xm._getvXXXXAsOneString(vXXXX='vFWVB',dropColList=['BESCHREIBUNG',
↪'IDREFERENZ','W0','W','IRFV','LFK','TVL0','TRS0','LFKT','W_min','W_max','INDTR',
↪'TRSK','VTYP','DPHAUS','IMBG', 'pk','tk','KVR_i','TM_i', 'XKOR_i','YKOR_i','ZKOR_i',
↪'pXCor_i','pYCor_i','KVR_k','TM_k', 'XKOR_k','YKOR_k','ZKOR_k','pXCor_k', 'pYCor_k',
↪'CONT','CONT_ID', 'CONT_LFDNR']))
   W0LFK  NAME_i  NAME_k                        WBLZ  mx2Idx
0  160.0  V-K002  R-K002  [BLNZ1, BLNZ1u5, BLNZ1u5u7]       0
1  200.0  V-K004  R-K004                          []       1
2  160.0  V-K005  R-K005  [BLNZ1u5, BLNZ1u5u7, BLNZ5]       2
3  160.0  V-K007  R-K007                  [BLNZ1u5u7]       3
4  120.0  V-K003  R-K003                          []       4
```

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```
>>> # ---
>>> # vXXXX
>>> # ---
>>> xm.dataFrames['vVBEL_forTestOnly']=xm.dataFrames['vVBEL'].reset_
↪index(inplace=False) # Multiindex to Cols
>>> print(xm._getvXXXXAsOneString(vXXXX='vVBEL_forTestOnly',index=True,dropColList=[
↪'Z_i','pk_i','CONT_i','CONT_VKNO_i','Z_k','pk_k','CONT_k','CONT_VKNO_k','IDREFERENZ
↪','tk']))
   OBJTYPE                OBJID                   BESCHREIBUNG       NAME_i  NAME_k  ␣
↪          LAYR       L      D  mx2Idx
0     FWES  5638756766880678918  BHKW - Modul - 1000 kW therm.           R3     V-1  ␣
↪     [Vorlauf]       0     80       0
1     FWVB  4643800032883366034                               1       V-K002  R-K002 ␣
↪[Kundenanlagen]       0    NaN       0
2     FWVB  4704603947372595298                               3       V-K004  R-K004 ␣
↪[Kundenanlagen]       0    NaN       1
3     FWVB  5121101823283893406                               4       V-K005  R-K005 ␣
↪[Kundenanlagen]       0    NaN       2
4     FWVB  5400405917816384862                               5       V-K007  R-K007 ␣
↪[Kundenanlagen]       0    NaN       3
5     FWVB  5695730293103267172                               2       V-K003  R-K003 ␣
↪[Kundenanlagen]       0    NaN       4
6     KLAP  4801110583764519435                            None           R2      R3  ␣
↪    [Rücklauf]       0     80       0
7     PGRP  4986517622672493603                     Pumpengruppe          R-1      R3  ␣
↪            []       0    NaN       0
8     PUMP  5481331875203087055                      Umwälzpumpe          R-1      R2  ␣
↪    [Rücklauf]       0    NaN       0
9     ROHR  4613782368750024999                            None       R-K004  R-K005  ␣
↪    [Rücklauf]   88.02  107.1       0
10    ROHR  4614949065966596185                            None       V-K002  V-K003  ␣
↪     [Vorlauf]  405.96  107.1       1
11    ROHR  4637102239750163477                            None       R-K003  R-K004  ␣
↪    [Rücklauf]   83.55  107.1       2
12    ROHR  4713733238627697042                            None       V-K004  V-K005  ␣
↪     [Vorlauf]   88.02  107.1       3
13    ROHR  4769996343148550485                            None          R-L  R-K000  ␣
↪    [Rücklauf]   73.42  160.3       4
14    ROHR  4789218195240364437                            None       V-K001  V-K002  ␣
↪     [Vorlauf]  195.53  107.1       5
15    ROHR  4939422678063487923                            None          V-L  V-K000  ␣
↪     [Vorlauf]    68.6  160.3       6
16    ROHR  4945727430885351042                            None       R-K006  R-K007  ␣
↪    [Rücklauf]  109.77  107.1       7
17    ROHR  4984202422877610920                            None       V-K000  V-K001  ␣
↪     [Vorlauf]    76.4  107.1       8
18    ROHR  5037777106796980248                            None       V-K003  V-K004  ␣
↪     [Vorlauf]   83.55  107.1       9
19    ROHR  5123819811204259837                            None       V-K005  V-K006  ␣
↪     [Vorlauf]  164.91  107.1      10
20    ROHR  5266224553324203132                            None       R-K001  R-K002  ␣
↪    [Rücklauf]  195.53  107.1      11
21    ROHR  5379365049009065623                            None       R-K002  R-K003  ␣
↪    [Rücklauf]  405.96  107.1      12
22    ROHR  5611703699850694889                            None       R-K005  R-K006  ␣
↪    [Rücklauf]  164.91  107.1      13
23    ROHR  5620197984230756681                            None       V-K006  V-K007  ␣
↪     [Vorlauf]  109.77  107.1      14
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24    ROHR  5647213228462830353                          None       R-K000  R-K001  ␣
↪    [Rücklauf]    76.4  107.1      15
25    VENT  4678923650983295610                          None          V-1     V-L  ␣
↪     [Vorlauf]       0    150       0
26    VENT  4897018421024717974                          None          R-L     R-1  ␣
↪    [Rücklauf]       0    150       1
27    VENT  5525310316015533093                          None  PKON-Knoten     R-1  ␣
↪    [Rücklauf]       0     50       2
>>> # ---
>>> # vRART
>>> # ---
>>> print(xm._getvXXXXAsOneString(vXXXX='vRART',index=True,sortList=['INDSTD','NAME
↪']))
  NAME              BESCHREIBUNG                                  INDSTD_TXT  INDSTD␣
↪  DWDT WSOSTD                  pk NAME_KREF1 NAME_KREF2 NAME_SWVT
0   dp  Bezeichnung Regelungsart  Differenzdruck Druckseite, Sollwert Tabelle      55␣
↪ 1E+20      0  5552938346422332788     V-K007     R-K007      SWVT
>>> # ------
>>> # MxAdd
>>> # ------
>>> if 'vNRCV_Mx1' in xm.dataFrames:
...     del xm.dataFrames['vNRCV_Mx1'] # delete MxSync-Result to force MxSync-Call in␣
↪MxAdd
>>> oldShape=xm.dataFrames['vKNOT'].shape
>>> mx=xm.MxAdd()
>>> firstShape=xm.dataFrames['vKNOT'].shape
>>> oldShape[1]<firstShape[1]
True
>>> xm.MxAdd(mx=mx)
>>> secondShape=xm.dataFrames['vKNOT'].shape
>>> secondShape==firstShape
True
>>> xm.MxAdd(mx=mx)
>>> thirdShape=xm.dataFrames['vKNOT'].shape
>>> thirdShape==firstShape
True
>>> xm.dataFrames['vKNOT_forTestOnly']=xm.dataFrames['vKNOT'].rename(columns={'KNOT~*~
↪*~*~PH':'Druck'})
>>> if 'Druck' not in xm.dataFrames['vKNOT_forTestOnly']:
...     xm.dataFrames['vKNOT_forTestOnly'].rename(columns={'KNOT~*~*~*~H':'Druck'},
↪inplace=True)
>>> if 'Druck' not in xm.dataFrames['vKNOT_forTestOnly']:
...     xm.dataFrames['vKNOT_forTestOnly'].rename(columns={'KNOT~*~~*~PH':'Druck'},
↪inplace=True) #09
>>> f = lambda x: round(x,1) if x != None else None
>>> print(xm._getvXXXXAsOneString(vXXXX='vKNOT_forTestOnly',filterColList=['mx2Idx',
↪'KVR','NAME','Druck'],mapFunc={'Druck':f},index=True))
    mx2Idx KVR         NAME  Druck
0        0   2       R-K004    2.3
1        1   1       V-K002    4.0
2        2   1       V-K001    4.1
3        3   1       V-K000    4.1
4        4   2       R-K001    2.0
5        5   2       R-K003    2.3
6        6   2       R-K000    2.0
7        9   2       R-K005    2.3
8       11   2          R-L    2.0
```

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```
9       12   2       R-K002    2.1
10      13   1       V-K004    3.8
11      15   1       V-K005    3.8
12      16   2       R-K007    2.3
13      17   1       V-K006    3.8
14      18   2       R-K006    2.3
15      20   1       V-K003    3.8
16      21   1          V-L    4.1
17      22   1       V-K007    3.8
18       7   2           R2    4.3
19       8   1          V-1    4.1
20      10   2           R3    4.3
21      14   2  PKON-Knoten    2.0
22      19   2          R-1    2.0
>>> print(xm._getvXXXXAsOneString(vXXXX='vROHR',filterColList=['mx2Idx','L','KVR',
↪'NAME_i','NAME_k','ROHR~*~*~*~QMAV'],mapFunc={'ROHR~*~*~*~QMAV':f},sortList=['ROHR~
↪*~*~*~QMAV','NAME_i'],index=True))
    mx2Idx       L KVR  NAME_i  NAME_k  ROHR~*~*~*~QMAV
13      15    76.4   2  R-K000  R-K001            -23.0
9       11  195.53   2  R-K001  R-K002            -23.0
14       4   73.42   2     R-L  R-K000            -23.0
10      12  405.96   2  R-K002  R-K003            -19.1
2        2   83.55   2  R-K003  R-K004            -15.4
0        0   88.02   2  R-K004  R-K005             -8.5
11      13  164.91   2  R-K005  R-K006             -3.9
5        7  109.77   2  R-K006  R-K007             -3.9
8       10  164.91   1  V-K005  V-K006              3.9
12      14  109.77   1  V-K006  V-K007              3.9
3        3   88.02   1  V-K004  V-K005              8.5
7        9   83.55   1  V-K003  V-K004             15.4
1        1  405.96   1  V-K002  V-K003             19.1
6        8    76.4   1  V-K000  V-K001             23.0
4        5  195.53   1  V-K001  V-K002             23.0
15       6    68.6   1     V-L  V-K000             23.0
>>> print(xm._getvXXXXAsOneString(vXXXX='vFWVB',filterColList=['mx2Idx','NAME_i',
↪'NAME_k','FWVB~*~*~*~W'],mapFunc={'FWVB~*~*~*~W':f},sortList=['FWVB~*~*~*~W','NAME_i
↪'],index=True))
   mx2Idx  NAME_i  NAME_k  FWVB~*~*~*~W
4       4  V-K003  R-K003         120.0
0       0  V-K002  R-K002         160.0
2       2  V-K005  R-K005         160.0
3       3  V-K007  R-K007         160.0
1       1  V-K004  R-K004         200.0
>>> xm.dataFrames['vVBEL_forTestOnly2']=xm.dataFrames['vVBEL'].loc[['ROHR','FWVB'],:].
↪reset_index(inplace=False) # Multiindex to Cols
>>> xm.dataFrames['vVBEL_forTestOnly2'].rename(columns={'KNOT~*~*~*~PH_i':'Druck_i'},
↪inplace=True)
>>> if 'Druck_i' not in xm.dataFrames['vVBEL_forTestOnly2']:
...     xm.dataFrames['vVBEL_forTestOnly2'].rename(columns={'KNOT~*~*~*~H_i':'Druck_i
↪'},inplace=True)
>>> if 'Druck_i' not in xm.dataFrames['vVBEL_forTestOnly2']:
...     xm.dataFrames['vVBEL_forTestOnly2'].rename(columns={'KNOT~*~~*~PH_i':'Druck_i
↪'},inplace=True) #09
>>> f = lambda x: round(x,1) if x != None else None
>>> print(xm._getvXXXXAsOneString(vXXXX='vVBEL_forTestOnly2',filterColList=['OBJTYPE',
↪'mx2Idx','L','D','NAME_i','NAME_k','Druck_i','Q'],mapFunc={'Druck_i':f,'Q':f},
↪sortList=['OBJTYPE','NAME_i','Q'],index=True))
```

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```
   OBJTYPE  mx2Idx       L      D  NAME_i  NAME_k  Druck_i     Q
0     FWVB       0       0    NaN  V-K002  R-K002      4.0   3.9
4     FWVB       4       0    NaN  V-K003  R-K003      3.8   3.7
1     FWVB       1       0    NaN  V-K004  R-K004      3.8   6.9
2     FWVB       2       0    NaN  V-K005  R-K005      3.8   4.6
3     FWVB       3       0    NaN  V-K007  R-K007      3.8   3.9
20    ROHR      15    76.4  107.1  R-K000  R-K001      2.0 -23.0
16    ROHR      11  195.53  107.1  R-K001  R-K002      2.0 -23.0
17    ROHR      12  405.96  107.1  R-K002  R-K003      2.1 -19.1
7     ROHR       2   83.55  107.1  R-K003  R-K004      2.3 -15.4
5     ROHR       0   88.02  107.1  R-K004  R-K005      2.3  -8.5
18    ROHR      13  164.91  107.1  R-K005  R-K006      2.3  -3.9
12    ROHR       7  109.77  107.1  R-K006  R-K007      2.3  -3.9
9     ROHR       4   73.42  160.3     R-L  R-K000      2.0 -23.0
13    ROHR       8    76.4  107.1  V-K000  V-K001      4.1  23.0
10    ROHR       5  195.53  107.1  V-K001  V-K002      4.1  23.0
6     ROHR       1  405.96  107.1  V-K002  V-K003      4.0  19.1
14    ROHR       9   83.55  107.1  V-K003  V-K004      3.8  15.4
8     ROHR       3   88.02  107.1  V-K004  V-K005      3.8   8.5
15    ROHR      10  164.91  107.1  V-K005  V-K006      3.8   3.9
19    ROHR      14  109.77  107.1  V-K006  V-K007      3.8   3.9
11    ROHR       6    68.6  160.3     V-L  V-K000      4.1  23.0
>>> # ---
>>> # Clean Up LocalHeatingNetwork Xm and Mx
>>> # ---
>>> xm.delFiles()
>>> mx.delFiles()
>>> # ---
>>> # TinyWDN
>>> # ---
>>> xmlFile=os.path.join(os.path.join(path,testDir),'TinyWDN.XML')
>>> xm=Xm(xmlFile=xmlFile)
>>> # ---
>>> # GPipe
>>> # ---
>>> xmlFile=os.path.join(os.path.join(path,testDir),'GPipe.XML')
>>> xm=Xm(xmlFile=xmlFile)
```

class Xm.**Xm**(*xmlFile*, *NoH5Read=False*)

SIR 3S modelFile to pandas DataFrames.

Args:

- xmlFile (str): SIR 3S modelFile
- **NoH5Read (bool):**

  False (default):

  - An existing _and newer h5File will be read _instead of xmlFile.
  - xmlFile will _not be read (it does even not have to exist)

  True:

  - An existing h5File will be deleted.
  - xmlFile will be read.

Attributes:

- **states**
  - h5Read: True, if read from H5
- **fileNames**
  - xmlFile
  - constructed from MX during Init and Usage:
  - h5File: corresponding h5File(name) derived from xmlFile(name)
- **dataFrames**
  - dict with pandas DataFrames
  - one pandas DataFrame per SIR 3S Objecttype (i.e. KNOT, ROHR, ...)
  - keys: KNOT, ROHR, ... and vKNOT, vROHR, ...
  - **some Views as pandas DataFrames**
    * i.e. vKNOT, vROHR, ...
    * The Views are designed to deal with tedious groundwork.
    * The Views are aggregated somewhat arbitrary.
    * However: Usage of SIR 3S Modeldata is more convenient and efficient with appropriate Views.
- pXCorZero, pYCorZero

Raises: XmError

classmethod constructNewMultiindexFromCols(*df=None, mColNames=['OBJTYPE', 'pk'], mIdxNames=['OBJTYPE', 'OBJID']*)

Constructs a new Multiindex from existing cols and returns the constructed df.

Args:

- df: dataFrame without Multiindex
- mColNames: list of columns which shall be used as Multiindex; the columns must exist; the columns will be droped
- mIdxNames: list of names for the indices for the Cols above

Returns:

- df with Multiindex
- empty DataFrame is returned if an Error occurs

```
>>> d = {'OBJTYPE': ['ROHR', 'VENT'], 'pk': [123, 345], 'data': ['abc', 'def
↪']}
>>> import pandas as pd
>>> df = pd.DataFrame(data=d)
>>> from Xm import Xm
>>> df=Xm.constructNewMultiindexFromCols(df=df,mColNames=['OBJTYPE','pk'],
↪mIdxNames=['OBJTYPE','OBJID'])
>>> df['data']
OBJTYPE  OBJID
ROHR     123      abc
VENT     345      def
Name: data, dtype: object
```

classmethod constructShortestPathFromNodeList(*df=None*, *sourceCol='NAME_i'*, *targetCol='NAME_k'*, *nl=None*, *weight=None*, *query=None*, *fmask=None*)

Returns a DataFrame with Edges (one per row) implementing the shortest Path over NodeList.

Args:

- df: DataFrame with (all) Edges (one per row)
- nl: NodeList
- weight: columnName of the weight attribute

  # Der kürzeste Weg zwischen zwei Knoten in einem # zusammenhängenden Graphen ist derjenige, bei dem die # Summe der Gewichte über die durchlaufenen Kanten den # kleinstmöglichen Wert annimmt.

  # also bei konstantem Kantengewicht die kleinste Kantenanzahl # kürzeste Weglaenge: L als Gewicht # Durchflussstärkster Weg: 1 / Abs(Q) (Flüsse mit 0 oder Kanten ohne Flusswert müssen vorher eliminiert werden ... # ... birgt die Gefahr, dass es dann keinen Weg mehr gibt) # Durchflussschwächster Weg: Q

  - **examples for weight:**
    - 'L': converted to float before usage here
    - **if 'Q' is already a column in df:**
      - non Null Q-rows filtered here - then:
      - 'QAbs': constructed here
      - 'QAbsInv': constructed here
  - query: query to filter vVBEL (to filter Edges) before constructing the Graph
  - fmask: function to filter vVBEL (to filter Edges) before constructing the Graph
  - query and fmask are used both if not None

Returns:

- df: DataFrame with Edges (one per row) implementing the shortest Path over NodeList
- empty DataFrame is returned if an error occurs
- columns
  - OBJTYPE
  - OBJID
  - nextNODE
  - compNr
    - starts with 1
    - the number of the connected component
    - 1 for all edges if all nodes in NodeList are connected

```
>>> xmlFile=ms['GPipes']
>>> from Xm import Xm
>>> xm=Xm(xmlFile=xmlFile,NoH5Read=True)
>>> xm.constructShortestPathFromNodeList(df=xm.
↪getvVBELwithNodeAttributeAdded(),nl=['GL','GR'])
```

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```
  OBJTYPE                OBJID nextNODE  compNr
0    VENT  5309992331398639768       G1       1
1    ROHR  5244313507655010738      GKS       1
2    VENT  5116489323526156845      GKD       1
3    ROHR  5114681686941855110       G3       1
4    ROHR  4979507900871287244       G4       1
5    VENT  5745097345184516675       GR       1
>>> mx=xm.MxSync()
>>> xm.MxAdd(mx=mx)
>>> xm.constructShortestPathFromNodeList(df=xm.
↪getvVBELwithNodeAttributeAdded(),nl=['GL','GR'],weight='QAbsInv')   #␣
↪durchflussstaerkster Weg
  OBJTYPE                OBJID nextNODE  compNr
0    VENT  5309992331398639768       G1       1
1    ROHR  5244313507655010738      GKS       1
2    VENT  5508684139418025293      GKD       1
3    ROHR  5114681686941855110       G3       1
4    ROHR  4979507900871287244       G4       1
5    VENT  5745097345184516675       GR       1
>>> xmlFile=ms['LocalHeatingNetwork']
>>> xm=Xm(xmlFile=xmlFile,NoH5Read=True)
>>> xm.constructShortestPathFromNodeList(df=xm.
↪getvVBELwithNodeAttributeAdded(),nl=['V-L','V-K07'])
Empty DataFrame
Columns: []
Index: []
>>> xm.constructShortestPathFromNodeList(df=xm.
↪getvVBELwithNodeAttributeAdded(),nl=['V-L','V-K007'])
  OBJTYPE                OBJID nextNODE  compNr
0    ROHR  4939422678063487923   V-K000       1
1    ROHR  4984202422877610920   V-K001       1
2    ROHR  4789218195240364437   V-K002       1
3    ROHR  4614949065966596185   V-K003       1
4    ROHR  5037777106796980248   V-K004       1
5    ROHR  4713733238627697042   V-K005       1
6    ROHR  5123819811204259837   V-K006       1
7    ROHR  5620197984230756681   V-K007       1
>>> xm.constructShortestPathFromNodeList(df=xm.
↪getvVBELwithNodeAttributeAdded(),nl=['V-K007','R-K007'])
  OBJTYPE                OBJID nextNODE  compNr
0    FWVB  5400405917816384862   R-K007       1
>>> f=lambda row: True if row.KVR_i=='2' and row.KVR_k=='2' else False
>>> xm.constructShortestPathFromNodeList(df=xm.
↪getvVBELwithNodeAttributeAdded(),nl=['V-K007','R-K007'],fmask=f)
Empty DataFrame
Columns: []
Index: []
>>> xm.constructShortestPathFromNodeList(df=xm.
↪getvVBELwithNodeAttributeAdded(),nl=['R-K007','V-K007'],query="OBJTYPE not␣
↪in ['FWVB','PGRP']")
   OBJTYPE                OBJID nextNODE  compNr
0     ROHR  4945727430885351042   R-K006       1
1     ROHR  5611703699850694889   R-K005       1
2     ROHR  4613782368750024999   R-K004       1
3     ROHR  4637102239750163477   R-K003       1
4     ROHR  5379365049009065623   R-K002       1
5     ROHR  5266224553324203132   R-K001       1
```

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```
6     ROHR  5647213228462830353   R-K000       1
7     ROHR  4769996343148550485      R-L       1
8     VENT  4897018421024717974      R-1       1
9     PUMP  5481331875203087055       R2       1
10    KLAP  4801110583764519435       R3       1
11    FWES  5638756766880678918      V-1       1
12    VENT  4678923650983295610      V-L       1
13    ROHR  4939422678063487923   V-K000       1
14    ROHR  4984202422877610920   V-K001       1
15    ROHR  4789218195240364437   V-K002       1
16    ROHR  4614949065966596185   V-K003       1
17    ROHR  5037777106796980248   V-K004       1
18    ROHR  4713733238627697042   V-K005       1
19    ROHR  5123819811204259837   V-K006       1
20    ROHR  5620197984230756681   V-K007       1
```

delFiles()

Deletes Files constructed by XM during Init and Usage.

_xmlRead()

Reads the SIR 3S modelFile.

- Performs fixes and basic conversions inplace the dataFrames read from modelFile: _convertAndFix()
- Creates some Views: _vXXXX()

Raises: XmError

FromH5(*h5File=None*)

Reads all dataFrames stored in h5File into self.DataFrames.

Args:

h5File:

- (str): the h5File(name) to be read
- (None): self.h5File will be read

- Reads all keys.
- Existing keys in self.dataFrames are overwritten.

Note that after .FromH5() the content of self.dataFrames may differ from the content given by an existing self.xmlFile.

Raises: XmError

ToH5(*h5File=None*)

Stores self.dataFrames to h5File.

Args:

h5File:

- (str): the h5File(name) to be used
- (None): self.h5File will be used

- Stores all keys.

- Existing keys in h5File are overwritten.

Raises: XmError

_convertAndFix()

Performs fixes and basic conversions inplace the dataFrames read from self.xmlFile.

- Fixes and conversions here are integrity-oriented.
- Usage-oriented conversions (i.e. pd.to_numeric and base64.b64decode) are done in the ._vXXXX-methods.

Conversions:

- , > . (converted in: SWVT_ROWT, LFKT_ROWT, QVAR_ROWT, PVAR_ROWT)

Fixes:

- **No SWVT_ROWT, LFKT_ROWT, QVAR_ROWT, PVAR_ROWT?!**
  - * SWVT, LFKT, QVAR, PVAR are constructed to
- 1st Time without Value?! (fixed in: SWVT_ROWT, LFKT_ROWT, QVAR_ROWT, PVAR_ROWT)
- Template Node(s)?!
- **in new Models constructed from SIR 3S**
  - **not all Objectattributes are written?!**
    - * KMOT/TE
    - * FWVB/LFK
    - * LTGR/BESCHREIBUNG
    - * **DTRO_ROWD**
      - · AUSFALLZEIT
      - · PN
      - · REHABILITATION
      - · REPARATUR
      - · WSTEIG, WTIEFE
  - **not all Objecttypes are written?!**
    - * CONT
- Models with no PZONs . . .
- Models with no GMIXs . . .
- Models with no STOFs . . .
- empty WBLZ OBJS-BLOBs
- empty LAYR OBJS-BLOBs

Raises: XmError

_constructEmptyDf(*cols=['DummyCol1', 'DummyCol2']*)

Constructs an empty df with cols.

Args:

- cols: list of colNames

Returns: df: constructed df

Raises: XmError

getWDirModelDirModelName()

Returns (wDir,modelDir,modelName,mx1FileName).

Returns: (wDir,modelDir,modelName,mx1FileName)

wDir If wDir as given literally in .xmlFile is not a valid Dir or such a wDir relative to .xmlFile-Path exists the wDir relative is returned.

mx1FileName mx1FileName is assumed to be: .:...WD...B...V...BZ...M... .MX1 If not existing: .*.MX1 (first match is returned) If a suitable mx1File is not existing an INFO-Message is generated.

Raises: XmError

getVersion(*type='BASIS'*)

Returns VERSION-String i.e. Sir3S-90-10.

Args:

- type: BASIS or VARIANTE or BZ; the DATENEBENEn-TYPe from which the VERSION-String is requested

Returns:

- VERSION-String i.e. Sir3S-90-10; Sir3S-90-09 is returned if Attribute VERSION is not available

Raises: XmError

```
>>> xm=xms['OneLPipe']
>>> vStr=xm.getVersion()
>>> import re
>>> m=re.search('Sir(?P<Db3s>[DBdb3Ss]{2})-(?P<Major>\d+)-(?P<Minor>\d+)$',
↪vStr) # i.e. Sir3S-90-10
>>> int(m.group('Major')[0])
9
```

_getvXXXXAsOneString(*vXXXX=None, start=0, end=-1, dropColList=None, filterColList=None, mapFunc={}, sortList=None, ascending=True, fmtFunc={}, index=True, header=True*)

Returns vXXXX-Content as one String (for Doctest-Purposes).

Args:

- vXXXX: df=self.dataFrames[vXXXX]
- start
- end
- dropColList
- filterColList
- mapFunc: col:func: df[col].map(func)
- sortList

- ascending
- fmtFunc: col:func: passed to df.to_string(formatters=fmtFunc, . . .)
- index
- header

Returns:

- df.to_string(formatters=fmtFunc,index=index,header=header)

Raises: XmError

_vXXXX ( )

Creates all Views.

Views created:

- **BLOB-Data**
  - vLAYR
  - vWBLZ
  - vAGSN u. vAGSN_raw
- **Timeseries**
  - vLFKT
  - vQVAR
  - vPVAR
  - vSWVT
- **Signalmodel**
  - vRSLW
- **Miscellanea**
  - vRART
- **Hydraulicmodel**
  - **Nodes**
    - vVKNO: CONT-Nodes (also called Block-Nodes)
    - vKNOT
    - pXCorZero, pYCorZero
  - **Edges**
    - vROHR: Pipes
    - vFWVB: Housestations (district heating)
  - **all Edges (all; implemented Edges see vVBEL_edges)**
    - vVBEL
- **Annotations**
  - vNRCV
  - vGTXT

Raises: XmError

_vLAYR ()

One row per LAYR and OBJ.

Returns:

columns

LAYR (also called 'Group')

- LFDNR
- NAME

from SIR 3S OBJ BLOB collection:

- OBJTYPE: type (i.e.ROHR) of a LAYR OBJ
- OBJID: pk (or tk?!) of a LAYR OBJ

LAYR IDs

- pk, tk

ANNOTATION

- nrObjInGroup: Element Nr. in LAYR (LFDNR) - should be 1 otherwise the same OBJ occurs in the same LAYR multiple times
- nrObjtypeInGroup: Element Nr. of OBJTYPE in LAYR (LFDNR)

SORTING LFDNR,NAME,OBJTYPE,OBJID

Raises: XmError

_vWBLZ ()

One row per WBLZ and OBJ.

Returns:

columns

WBLZ

- AKTIV
- BESCHREIBUNG
- IDIM
- NAME

from SIR 3S OBJ BLOB collection:

- OBJTYPE: type (always KNOT?!)
- OBJID: pk (or tk?!)

WBLZ IDs

- pk

SORTING NAME,pk

Raises: XmError

_vAGSN ( )

One row per AGSN and OBJ.

Returns:

columns

AGSN

- LFDNR
- NAME
- AKTIV
- from SIR 3S OBJ BLOB collection:
  - OBJTYPE: type (i.e.ROHR)
  - OBJID: pk (or tk?!)

AGSN IDs

- pk, tk

Sequence

- **Model**
  - therefore nrObjIdInAgsn (see ANNOTATION below) should be the realwolrd sequence

ANNOTATION

- nrObjIdInAgsn: lfd.Nr. (in Schnittreihenfolge) Obj. (der Kante) in AGSN (AGSN is defined by LFDNR not by NAME)
- **nrObjIdTypeInAgsn: should be 1 determined by raw data**
  - nrObjIdTypeInAgsn>1 - if any - are not part of the view
  - the 1st occurance is in the view
- **Layer** 0=undef bei Netztyp 21: 1=VL, 2=RL, 0=undef wenn keine BN-Trennzeile gefunden wird, wird VL angenommen und gesetzt die BN-Trennzeile wird dem VL (1) zugerechnet
- **nextNODE: node which is connected by the edge**
  - the cut-direction is defined (per cut and comp) by edge-sequence
  - the cut node-sequence ist the (longest shortest) path between the nodes of the 1st and last edge
  - in case of 1 edge cut-direction is edge-definition and cut node-sequence is edge-definition
  - the nextNODEs are the node-sequence omitting the start-node ...
  - ... nextNODE of an edge is the node connected by this edge in cut-direction; so nextNODE might be the i-node (the source-node) of the edge
  - if edge-direction is cut-direction nextNODE is the k-node (the sink-node) of the edge
- **compNr**
  - all 1 if all edges in the cut are connected
  - otherwise the compNr (starting with 1) the edge belongs to
  - the comp-Sequence is defined by the edge-sequence

  - the nodes of the 1st and last edge in cut-definition of the comp are defining the node-Sequence of the (longest shortest) path in the comp

- **parallel Edges**
  - are omitted in the cut-Result; the 1st edge in cut-definition is in the edge
- **Abzweige**
  - are omitted in the cut-Result
  - the nodes of the 1st and last edge in cut-definition are defining the node-Sequence of the (longest shortest) path (comp-wise)
  - only edges implementing this path are in the cut-Result

Raises: XmError

```
>>> xmlFile=ms['GPipes']
>>> from Xm import Xm
>>> xm=Xm(xmlFile=xmlFile)
>>> vAGSN=xm.dataFrames['vAGSN']
>>> schnitt=vAGSN[vAGSN['NAME']=='LR']
>>> xm.dataFrames['schnitt']=schnitt.reset_index()
>>> print(xm._getvXXXXAsOneString(vXXXX='schnitt',index=True))
   index LFDNR NAME AKTIV OBJTYPE                OBJID                    pk  ␣
↪                tk  nrObjIdInAgsn  nrObjIdTypeInAgsn  Layer nextNODE compNr
0      7    14   LR   101    VENT  5309992331398639768  5625063016896368599 ␣
↪5625063016896368599              1                   1      0       G1      1
1      8    14   LR   101    ROHR  5244313507655010738  5625063016896368599 ␣
↪5625063016896368599              2                   1      0      GKS      1
2      9    14   LR   101    VENT  5508684139418025293  5625063016896368599 ␣
↪5625063016896368599              3                   1      0      GKD      1
3     10    14   LR   101    ROHR  5114681686941855110  5625063016896368599 ␣
↪5625063016896368599              4                   1      0       G3      1
4     11    14   LR   101    ROHR  4979507900871287244  5625063016896368599 ␣
↪5625063016896368599              5                   1      0       G4      1
5     12    14   LR   101    VENT  5745097345184516675  5625063016896368599 ␣
↪5625063016896368599              6                   1      0       GR      1
>>> schnitt=vAGSN[vAGSN['NAME']=='LR-Lücke']
>>> xm.dataFrames['schnitt']=schnitt.reset_index()
>>> print(xm._getvXXXXAsOneString(vXXXX='schnitt',index=True))
   index LFDNR      NAME AKTIV OBJTYPE                OBJID                  ␣
↪pk                  tk  nrObjIdInAgsn  nrObjIdTypeInAgsn  Layer nextNODE␣
↪compNr
0     13    16  LR-Lücke   101    VENT  5309992331398639768 ␣
↪5630543731618051887  5630543731618051887              1                  1 ␣
↪    0       G1      1
1     14    16  LR-Lücke   101    ROHR  5244313507655010738 ␣
↪5630543731618051887  5630543731618051887              2                  1 ␣
↪    0      GKS      1
2     15    16  LR-Lücke   101    ROHR  5114681686941855110 ␣
↪5630543731618051887  5630543731618051887              3                  1 ␣
↪    0       G3      2
3     16    16  LR-Lücke   101    ROHR  4979507900871287244 ␣
↪5630543731618051887  5630543731618051887              4                  1 ␣
↪    0       G4      2
4     17    16  LR-Lücke   101    VENT  5745097345184516675 ␣
↪5630543731618051887  5630543731618051887              5                  1 ␣
↪    0       GR      2
```

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```
>>> schnitt=vAGSN[vAGSN['NAME']=='LR-Flansch']
>>> xm.dataFrames['schnitt']=schnitt.reset_index()
>>> print(xm._getvXXXXAsOneString(vXXXX='schnitt',index=True))
   index LFDNR        NAME AKTIV OBJTYPE                OBJID
↪  pk                   tk  nrObjIdInAgsn  nrObjIdTypeInAgsn  Layer nextNODE␣
↪compNr
0     18    18  LR-Flansch   101    VENT  5309992331398639768 ␣
↪5134530907542044265  5134530907542044265              1                  1 ␣
↪    0       G1      1
1     19    18  LR-Flansch   101    ROHR  5244313507655010738 ␣
↪5134530907542044265  5134530907542044265              2                  1 ␣
↪    0      GKS      1
2     20    18  LR-Flansch   101    VENT  5508684139418025293 ␣
↪5134530907542044265  5134530907542044265              3                  1 ␣
↪    0      GKD      1
3     21    18  LR-Flansch   101    ROHR  5114681686941855110 ␣
↪5134530907542044265  5134530907542044265              4                  1 ␣
↪    0       G3      1
4     22    18  LR-Flansch   101    ROHR  4979507900871287244 ␣
↪5134530907542044265  5134530907542044265              5                  1 ␣
↪    0       G4      1
5     24    18  LR-Flansch   101    VENT  5745097345184516675 ␣
↪5134530907542044265  5134530907542044265              7                  1 ␣
↪    0       GR      1
>>> schnitt=vAGSN[vAGSN['NAME']=='LR-Parallel']
>>> xm.dataFrames['schnitt']=schnitt.reset_index()
>>> print(xm._getvXXXXAsOneString(vXXXX='schnitt',index=True))
   index LFDNR         NAME AKTIV OBJTYPE                 OBJID
↪   pk                   tk  nrObjIdInAgsn  nrObjIdTypeInAgsn  Layer␣
↪nextNODE compNr
0     25    20  LR-Parallel   101    VENT  5309992331398639768 ␣
↪4694969854935170169  4694969854935170169              1                  1 ␣
↪    0       G1      1
1     26    20  LR-Parallel   101    ROHR  5244313507655010738 ␣
↪4694969854935170169  4694969854935170169              2                  1 ␣
↪    0      GKS      1
2     27    20  LR-Parallel   101    VENT  5116489323526156845 ␣
↪4694969854935170169  4694969854935170169              3                  1 ␣
↪    0      GKD      1
3     29    20  LR-Parallel   101    ROHR  5114681686941855110 ␣
↪4694969854935170169  4694969854935170169              5                  1 ␣
↪    0       G3      1
4     30    20  LR-Parallel   101    ROHR  4979507900871287244 ␣
↪4694969854935170169  4694969854935170169              6                  1 ␣
↪    0       G4      1
5     31    20  LR-Parallel   101    VENT  5745097345184516675 ␣
↪4694969854935170169  4694969854935170169              7                  1 ␣
↪    0       GR      1
```

getvVBELwithNodeAttributeAdded(*nodeAttribute='KVR'*, *preserveMultiindex=False*)

Adds a nodeAttribute (not already in vVBEL) to vVBEL and returns the df.

Args:

- nodeAttribute (default: 'KVR'): the Node Attribute which shall be added
- preserveMultiindex (default: False): if True an existing Multiindex will be preserved; False: existing Index(Indices) will be col(s)

Returns: df (which might be incomplete, corrupt or empty if an error occurs; vVBEL is unchanged)

Raises: XmError

```
>>> xmlFile=ms['LocalHeatingNetwork']
>>> from Xm import Xm
>>> xm=Xm(xmlFile=xmlFile,NoH5Read=True)
>>> df=xm.getvVBELwithNodeAttributeAdded()
>>> xm.dataFrames['tmp']=df
>>> print(xm._getvXXXXAsOneString(vXXXX='tmp',index=True,filterColList=[
↪'OBJTYPE','OBJID','NAME_i','NAME_k','KVR_i','KVR_k']))
   OBJTYPE                OBJID       NAME_i  NAME_k KVR_i KVR_k
0     FWES  5638756766880678918           R3     V-1     2     1
1     FWVB  4643800032883366034       V-K002  R-K002     1     2
2     ROHR  5266224553324203132       R-K001  R-K002     2     2
3     ROHR  4614949065966596185       V-K002  V-K003     1     1
4     FWVB  4704603947372595298       V-K004  R-K004     1     2
5     ROHR  4637102239750163477       R-K003  R-K004     2     2
6     ROHR  4713733238627697042       V-K004  V-K005     1     1
7     FWVB  5121101823283893406       V-K005  R-K005     1     2
8     ROHR  4613782368750024999       R-K004  R-K005     2     2
9     ROHR  5123819811204259837       V-K005  V-K006     1     1
10    FWVB  5400405917816384862       V-K007  R-K007     1     2
11    ROHR  4945727430885351042       R-K006  R-K007     2     2
12    FWVB  5695730293103267172       V-K003  R-K003     1     2
13    ROHR  5379365049009065623       R-K002  R-K003     2     2
14    ROHR  5037777106796980248       V-K003  V-K004     1     1
15    KLAP  4801110583764519435           R2      R3     2     2
16    PGRP  4986517622672493603          R-1      R3     2     2
17    PUMP  5481331875203087055          R-1      R2     2     2
18    ROHR  4769996343148550485          R-L  R-K000     2     2
19    VENT  4897018421024717974          R-L     R-1     2     2
20    VENT  5525310316015533093  PKON-Knoten     R-1     2     2
21    ROHR  4789218195240364437       V-K001  V-K002     1     1
22    ROHR  4939422678063487923          V-L  V-K000     1     1
23    ROHR  4984202422877610920       V-K000  V-K001     1     1
24    ROHR  5611703699850694889       R-K005  R-K006     2     2
25    ROHR  5620197984230756681       V-K006  V-K007     1     1
26    ROHR  5647213228462830353       R-K000  R-K001     2     2
27    VENT  4678923650983295610          V-1     V-L     1     1
>>> len(df.columns.tolist())
20
>>> df.index.names
FrozenList([None])
>>> df=xm.getvVBELwithNodeAttributeAdded(preserveMultiindex=True)
>>> len(df.columns.tolist())
18
>>> df.index.names
FrozenList(['OBJTYPE', 'OBJID'])
```

vAGSN_Add(*nl=None*, *weight=None*, *Layer=0*, *AKTIV=None*, *NAME='NEU'*, *query=None*, *fmask=None*)

Adds a new User-defined Cut to the Model-defined Cuts.

Arguments:

see constructShortestPathFromNodeList:

- nl: NodeList for the Cut
- weight: columnName of the weight attribute

- query: mask to filter vVBEL (to filter Edges) before constructing the Graph
- fmask: function to filter vVBEL (to filter Edges) before constructing the Graph
- query and fmask are used both if not None

- Layer (to use in constructed cut)
- AKTIV (to use in constructed cut)
- NAME (to use in constructed cut): the cut will NOT be constructed if such a NAME already exists

Raises: XmError

```
>>> xmlFile=ms['GPipes']
>>> from Xm import Xm
>>> xm=Xm(xmlFile=xmlFile,NoH5Read=True)
>>> xm.vAGSN_Add(nl=['GL','GR'])
>>> import pandas as pd
>>> pd.set_option('display.max_columns',None)
>>> pd.set_option('display.max_rows',None)
>>> pd.set_option('display.max_colwidth',666666)
>>> pd.set_option('display.width',666666666)
>>> xm.dataFrames['vAGSN_raw']
   LFDNR         NAME AKTIV OBJTYPE                OBJID                   pk␣
↪                  tk  nrObjIdInAgsn  nrObjIdTypeInAgsn  Layer nextNODE␣
↪compNr
0      2       1 Rohr   101    ROHR  5244313507655010738  5015814781412926392␣
↪ 5015814781412926392              1                  1      0      GKS     ␣
↪1
1      6       V-Rohr   101    VENT  5309992331398639768  5396484903084432138␣
↪ 5396484903084432138              1                  1      0       G1     ␣
↪1
2      6       V-Rohr   101    ROHR  5244313507655010738  5396484903084432138␣
↪ 5396484903084432138              2                  1      0      GKS     ␣
↪1
3      8       Rohr-V   101    ROHR  4979507900871287244  4989935433418681990␣
↪ 4989935433418681990              1                  1      0       G4     ␣
↪1
4      8       Rohr-V   101    VENT  5745097345184516675  4989935433418681990␣
↪ 4989935433418681990              2                  1      0       GR     ␣
↪1
5     12      2 Rohre   101    ROHR  5114681686941855110  5748019382126004712␣
↪ 5748019382126004712              1                  1      0       G3     ␣
↪1
6     12      2 Rohre   101    ROHR  4979507900871287244  5748019382126004712␣
↪ 5748019382126004712              2                  1      0       G4     ␣
↪1
7     14           LR   101    VENT  5309992331398639768  5625063016896368599␣
↪ 5625063016896368599              1                  1      0       G1     ␣
↪1
8     14           LR   101    ROHR  5244313507655010738  5625063016896368599␣
↪ 5625063016896368599              2                  1      0      GKS     ␣
↪1
9     14           LR   101    VENT  5508684139418025293  5625063016896368599␣
↪ 5625063016896368599              3                  1      0      GKD     ␣
↪1
10    14           LR   101    ROHR  5114681686941855110  5625063016896368599␣
↪ 5625063016896368599              4                  1      0       G3     ␣
↪1
```

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```
11    14           LR   101    ROHR  4979507900871287244  5625063016896368599␣
↪ 5625063016896368599              5                  1      0       G4      ␣
↪1
12    14           LR   101    VENT  5745097345184516675  5625063016896368599␣
↪ 5625063016896368599              6                  1      0       GR      ␣
↪1
13    16     LR-Lücke   101    VENT  5309992331398639768  5630543731618051887␣
↪ 5630543731618051887              1                  1      0       G1      ␣
↪1
14    16     LR-Lücke   101    ROHR  5244313507655010738  5630543731618051887␣
↪ 5630543731618051887              2                  1      0      GKS      ␣
↪1
15    16     LR-Lücke   101    ROHR  5114681686941855110  5630543731618051887␣
↪ 5630543731618051887              3                  1      0       G3      ␣
↪2
16    16     LR-Lücke   101    ROHR  4979507900871287244  5630543731618051887␣
↪ 5630543731618051887              4                  1      0       G4      ␣
↪2
17    16     LR-Lücke   101    VENT  5745097345184516675  5630543731618051887␣
↪ 5630543731618051887              5                  1      0       GR      ␣
↪2
18    18   LR-Flansch   101    VENT  5309992331398639768  5134530907542044265␣
↪ 5134530907542044265              1                  1      0       G1      ␣
↪1
19    18   LR-Flansch   101    ROHR  5244313507655010738  5134530907542044265␣
↪ 5134530907542044265              2                  1      0      GKS      ␣
↪1
20    18   LR-Flansch   101    VENT  5508684139418025293  5134530907542044265␣
↪ 5134530907542044265              3                  1      0      GKD      ␣
↪1
21    18   LR-Flansch   101    ROHR  5114681686941855110  5134530907542044265␣
↪ 5134530907542044265              4                  1      0       G3      ␣
↪1
22    18   LR-Flansch   101    ROHR  4979507900871287244  5134530907542044265␣
↪ 5134530907542044265              5                  1      0       G4      ␣
↪1
23    18   LR-Flansch   101    VENT  5745097345184516675  5134530907542044265␣
↪ 5134530907542044265              7                  1      0       GR      ␣
↪1
24    20  LR-Parallel   101    VENT  5309992331398639768  4694969854935170169␣
↪ 4694969854935170169              1                  1      0       G1      ␣
↪1
25    20  LR-Parallel   101    ROHR  5244313507655010738  4694969854935170169␣
↪ 4694969854935170169              2                  1      0      GKS      ␣
↪1
26    20  LR-Parallel   101    VENT  5116489323526156845  4694969854935170169␣
↪ 4694969854935170169              3                  1      0      GKD      ␣
↪1
27    20  LR-Parallel   101    ROHR  5114681686941855110  4694969854935170169␣
↪ 4694969854935170169              5                  1      0       G3      ␣
↪1
28    20  LR-Parallel   101    ROHR  4979507900871287244  4694969854935170169␣
↪ 4694969854935170169              6                  1      0       G4      ␣
↪1
29    20  LR-Parallel   101    VENT  5745097345184516675  4694969854935170169␣
↪ 4694969854935170169              7                  1      0       GR      ␣
↪1
```

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```
30    21          NEU  None    VENT  5309992331398639768                 PT3S␣
↪                 PT3S              1                   1      0       G1     ␣
↪1
31    21          NEU  None    ROHR  5244313507655010738                 PT3S␣
↪                 PT3S              2                   1      0      GKS     ␣
↪1
32    21          NEU  None    VENT  5116489323526156845                 PT3S␣
↪                 PT3S              3                   1      0      GKD     ␣
↪1
33    21          NEU  None    ROHR  5114681686941855110                 PT3S␣
↪                 PT3S              4                   1      0       G3     ␣
↪1
34    21          NEU  None    ROHR  4979507900871287244                 PT3S␣
↪                 PT3S              5                   1      0       G4     ␣
↪1
35    21          NEU  None    VENT  5745097345184516675                 PT3S␣
↪                 PT3S              6                   1      0       GR     ␣
↪1
>>> # Test if the same NAME is _not constructed twice ...
>>> xm.vAGSN_Add(nl=['GL','GR'])
>>> xm.dataFrames['vAGSN_raw'].shape
(36, 12)
>>> # test if re-use works without erors ...
>>> mx=xm.MxSync()
>>> xm.MxAdd(mx=mx)
>>> # Test weight-Option ...
>>> xm.vAGSN_Add(nl=['GL','GR'],weight='QAbsInv',NAME='GL-GR w')   #␣
↪durchflussstaerksten Weg erzwingen
>>> df=xm.dataFrames['vAGSN_raw']
>>> df.query("LFDNR in [21,22] and nextNODE=='GKD'")
   LFDNR     NAME AKTIV OBJTYPE                OBJID    pk    tk ␣
↪nrObjIdInAgsn  nrObjIdTypeInAgsn  Layer nextNODE compNr
32    21      NEU  None    VENT  5116489323526156845  PT3S  PT3S             ␣
↪3                 1      0      GKD      1
38    22  GL-GR w  None    VENT  5508684139418025293  PT3S  PT3S             ␣
↪3                 1      0      GKD      1
```

_vRART ()

One row per RART.

Returns:

columns:

RART

- NAME
- BESCHREIBUNG
- INDSTD_TXT
- INDSTD (numeric)
- DWDT

RART_BZ

- WSOSTD

ID

- pk

References

- NAME_KREF1
- NAME_KREF2
- NAME_SWVT
- [NAME_RCPL] - only if RCPLs exist

sequence: Model

Raises: XmError

_OBJS(*dfName*)

Decode a column OBJS (a BLOB containing a SIR 3S OBJ collection).

Args: dfName: Name of a dataFrame with column OBJS

columns used (in self.dataFrames[dfName]):

- OBJS (BLOB): i.e.: KNOT~4668229590574507160 . . .
- pk: ID (of the row)
- None is returned if these columns are missing
- in this case no changes in column OBJS in self.dataFrames[dfName]

Returns: column OBJS in self.dataFrames[dfName] is decoded to 'XXXX~' if OBJS was None

dfOBJS: dataFrame with one row per OBJ in OBJS:

columns added (compared to self.dataFrames[dfName]):

- OBJTYPE
- OBJID

rows missing (compared to self.dataFrames[dfName]):

- rows with OBJS None

Raises: XmError

_vLFKT()

One row per Loadfactor Timeseries.

Returns:

columns

LFKT

- NAME
- BESCHREIBUNG
- INTPOL
- ZEITOPTION

SERIES

- LF: 1st Value
- LF_min
- LF_max

LFKT ID

- pk

Raises: XmError

_vNRCV ( )

One row per NRCV (NumeRiCal Value).

Returns:

columns

ANNOTATIONS

- cRefLfdNr

CONT

- CONT
- CONT_ID
- CONT_LFDNR

DP

Datapointgroup

- DPGR

Datapoint

- OBJTYPE
- fkOBJTYPE
- ATTRTYPE

Datapoint IDs

- pk_ROWS
- tk_ROWS

NRCV IDs

- pk
- tk

PLot Coordinates

- pXYLB: (X,Y): Left,Bottom

Raises: XmError

_vGTXT ( )

One row per GTXT (Graphic TeXT).

Returns:

columns

GTXT

- GRAFTEXT (the text)

CONT

- CONT

- CONT_ID
- CONT_LFDNR

GTXT IDs

- pk
- tk

PLot Coordinates

- pXYLB: (X,Y): Left,Bottom

Raises: XmError

_vSWVT()

One row per Timeseries.

Returns:

columns

SWVT

- NAME
- BESCHREIBUNG
- INTPOL
- ZEITOPTION

SERIES

- W: 1st Value
- W_min
- W_max

SWVT ID

- pk

Raises: XmError

_vRSLW(*vSWVT=None*)

One row per RSLW.

Args:

- vSWVT

Returns:

columns

RSLW

- KA
- BESCHREIBUNG
- INDWBG, INDWNO

RSLW BZ

- INDSLW, SLWKON

CONT

- CONT
- ID

SWVT

- SWVT
- BESCHREIBUNG_SWVT
- INTPOL
- ZEITOPTION

SERIES

- W: 1st Value
- W_min
- W_max

RSLW IDs

- pk
- tk

Raises: XmError

_vQVAR ( )

One row per Timeseries.

Returns:

columns

QVAR

- NAME
- BESCHREIBUNG
- INTPOL
- ZEITOPTION

SERIES

- QM: 1st Value
- QM_min
- QM_max

QVAR ID

- pk

Raises: XmError

_vPVAR ( )

One row per Timeseries.

Returns:

columns

PVAR

- NAME
- BESCHREIBUNG
- INTPOL
- ZEITOPTION

SERIES

- PH: 1st Value
- PH_min
- PH_max

QVAR ID

- pk

Raises: XmError

_vVKNO ( )

One row per Blocknode.

Returns:

columns

- NAME
- CONT (Blockname)
- fkKNOT
- fkCONT

Raises: XmError

_vKNOT (*vVKNO=None*, *vQVAR=None*, *vPVAR=None*, *vLFKT=None*)

One row per Node (KNOT).

Args:

- vVKNO
- vQVAR
- vPVAR
- vLFKT

Returns:

rows sequence: Xml

columns

KNOT

- NAME
- BESCHREIBUNG
- IDREFERENZ

CONT

- CONT
- CONT_ID

- CONT_LFDNR
- CONT_VKNO (name of the Block/Container for NAME is a Blocknode)

KNOT

- KTYP
- LFAKT (Umrechnungsfaktor)
- QM_EIN

QVAR

- QVAR_NAME
- QM, QM_min, QM_max

LFKT

- LFKT_NAME
- LF, LF_min, LF_max

PVAR

- PVAR_NAME
- PH, PH_min, PH_max

Zugehoerigkeit

- PZON_NAME
- FSTF_NAME,STOF_NAME,GMIX_NAME
- UTMP_NAME

2L

- 2L_NAME
- 2L_KVR

KNOT

- KVR
- TE
- TM

KNOT

- XKOR, YKOR, ZKOR

KNOT IDs

- pk, tk

KNOT: plot-Coordinates

- pXCor: X-pXCorZero
- pYCor: Y-pYCorZero

Raises: XmError

_vROHR(*vKNOT=None*)

One row per Pipe (ROHR).

Args: vKNOT

Returns:

rows sequence: Xml

columns

ROHR

- BESCHREIBUNG
- IDREFERENZ

ROHR

- BAUJAHR, HAL
- IPLANUNG, KENNUNG

ROHR

- L, LZU, RAU, ZAUS, ZEIN, ZUML
- JLAMBS, LAMBDA0

ROHR

- ASOLL, INDSCHALL

ROHR FW

- NAME_i_2L
- NAME_k_2L
- KVR

DTRO_ROWD

- AUSFALLZEIT, DA , DI , DN , KT , PN , REHABILITATION , REPARATUR , S , WSTEIG , WTIEFE

LTGR

- LTGR_NAME, LTGR_BESCHREIBUNG , SICHTBARKEIT , VERLEGEART

DTRO

- DTRO_NAME, DTRO_BESCHREIBUNG, E

REF

- fkSTRASSE, fkSRAT

ROHR IDs

- pk, tk

ROHR BZ

- ITRENN
- LECKSTART, LECKEND, LECKMENGE, LECKORT, LECKSTATUS

Rest

- QSVB, ZVLIMPTNZ, KANTENZV

CONT

- CONT
- CONT_ID
- CONT_LFDNR

vKNOT

KI

- NAME_i
- KVR_i, TM_i
- XKOR_i, YKOR_i, ZKOR_i

KK

- NAME_k
- KVR_k, TM_k
- XKOR_k, YKOR_k, ZKOR_k

pXCor_i, pYCor_i pXCor_k, pYCor_k

PLOT

- pXCors, pYCors
- pWAYPXCors, pWAYPYCors

Raises: XmError

_vFWVB(*vKNOT=None*, *vLFKT=None*, *vWBLZ=None*)

One row per DistrictHeatingHousestation (FWVB).

Args:

- vKNOT
- vLFKT
- wWBLZ

Returns:

columns

FWVB

- BESCHREIBUNG
- IDREFERENZ
- W0
- LFK
- W0LFK
- TVL0, TRS0

vLFKT

- LFKT
- W, W_min, W_max

FWVB contd.

- INDTR, TRSK
- VTYP
- DPHAUS, IMBG, IRFV

FWVB IDs

- pk, tk

vKNOT

Ki

- NAME_i
- KVR_i, TM_i
- XKOR_i, YKOR_i, ZKOR_i
- pXCor_i, pYCor_i

Kk

- NAME_k
- KVR_k, TM_i
- XKOR_k, YKOR_k, ZKOR_i
- pXCor_k, pYCor_i

vCONT

- CONT
- CONT_ID
- CONT_LFDNR

vWBLZ

- **['BLZ1','BLZ2',...]** list of the WBLZ-Names of the FWVB in alphabetical Order; empty list, if FWVB is not a WBLZ-Member

Raises: XmError

_vVBEL(*vKNOT=None, edges=['ROHR', 'VENT', 'FWVB', 'FWES', 'PUMP', 'KLAP', 'REGV', 'PREG', 'MREG', 'DPRG', 'PGRP'], edgesD=['', 'DN', '', 'DN', '', 'DN', 'DN', 'DN', 'DN', 'DN', ''], mColNames=['OBJTYPE', 'pk'], mIdxNames=['OBJTYPE', 'OBJID']*)

One row per Edge.

Args:

- vKNOT: df
- edges: list of strs
- edgesD: list of strs
- mColNames: list of columns which shall be used as MIndex; the columns will be droped; the columns must be delivered by _vVBEL_XXXX
- mIdxNames: list of names for the indices for the columns above

Returns: Edge-df returned Edge-df is None if an exception occurs

rows:

- sequence edges: edges

- sequence within edges: Xml

Mindices:

- OBJTYPE: str: 'ROHR','VENT',... [default a MIndex not a column]
- OBJID [default a MIndex not a column]

columns:

- LAYR
- L in m (0 if edge <> ROHR)
- D in mm (NaN if no Diameter could be determined)

columns:

- see _vVBEL_XXXX

_vVBEL_XXXX (*vKNOT=None*, *OBJTYPE=None*)

One row per Edge.

Args:

- vKNOT: df
- **OBJTYPE: str ('ROHR','VENT',...)** self.dataFrames[OBJTYPE] is used to build with vKNOT the returned Edge-df

Returns: Edge-df None is returned if an exception occurs

columns:

- OBJTYPE: str: ROHR,VENT,...
- BESCHREIBUNG
- IDREFERENZ
- pk
- tk
- NAME_i
- CONT_i
- CONT_VKNO_i
- Z_i
- pk_i
- NAME_k
- CONT_k
- CONT_VKNO_k
- Z_k
- pk_k

MxSync (*mx=None*, *ForceNoH5ReadForMx=False*)

Xm: NEW 1st Call: vNRCV_Mx1: vNRCV with MX1-Information. Some Xm-Views with MX2-Information (mx2Idx).Mx: Sir3sID Update in Mx-Object.

Args:

mx (default: None): Mx-Object

- If no Mx-Object is given the Mx-Object corresponding to the Xm-Object is constructed and returned.
- **Notes:**
  - **The Sync-Result in Xm is persisted if xm were read from H5:**
    - xm.ToH5() is called if xm.h5Read is True.
  - **The Sync-Result in Mx is persisted if mx were read from H5:**
    - mx.ToH5() is called (from __Mx1_Sir3sIDUpd) if Sir3sID-Updates occured and mx.h5Read is True.

ForceNoH5ReadForMx (deafault: False): has an Effect onlx if a new Mx-Object is constructed

- **ForceNoH5ReadForMx = True:**
  - the new Mx-Object is constructed with NoH5Read=True
- **ForceNoH5ReadForMx = False:**
  - the new Mx-Object is constructed with NoH5Read = not self.h5Read
  - if the Xm was read from H5 the Mx is constructed with NoH5Read=False
  - if the Xm was not read from H5 the Mx is constructed with NoH5Read=True

Returns: Mx-Object if no Mx-Object was given; Nothing else

Raises: XmError

```
>>> xm=xms['LocalHeatingNetwork']
>>> mx=xm.MxSync()
```

_MxSyncAddMx (*ForceNoH5ReadForMx=False*)

Mx-Object corresponding to the Xm-Object is constructed and returned.

ForceNoH5ReadForMx (deafault: False)

- **ForceNoH5ReadForMx = True:**
  - the new Mx-Object is constructed with NoH5Read=True
- **ForceNoH5ReadForMx = False:**
  - **the new Mx-Object is constructed with NoH5Read = not self.h5Read**
    - if the Xm was read from H5 the Mx is constructed with NoH5Read=False
    - if the Xm was not read from H5 the Mx is constructed with NoH5Read=True

Returns: Mx-Object

Raises: XmError

MxAdd (*mx=None*, *ForceNoH5ReadForMx=False*, *timeReq=None*, *aggReq=None*, *timeReq2nd=None*)

Add MX-Resultcolumns to some Xm-Views. NEW 1st Call: vROHRVecResults, vAGSN.

Args:

- mx, ForceNoH5ReadForMx: same Args as for MxSync; see description there
- **timeReq:**

    - TIMESTAMP
    - if None 1st TIME in Mx is used
  - **aggReq:**
    - 'TIME','TMIN','TMAX' (source: MXS) or 'MIN','MAX', . . . (source: mx.getVecAggs())
    - timeReq is considered as TIMESTAMPL
    - timeReq2nd is considered as TIMESTAMPR
  - **timeReq2nd:**
    - TIMESTAMP
    - if None last TIME in Mx is used

Views with MX2-Results added:

- in the Xm-Views below col mx2Idx must exist (i.e. MxSync mus have been called)
- mx2Idx is considered to be the last of the Model-cols
- right from mx2Idx Result-cols are added if not already existing
- already existing Result-cols are overwritten
- **mx2Idx-Views:**
  - vKNOT (KNOT. . .)
  - vROHR (ROHR. . .) - only Non-VEC-Channel-Results are added
  - vFWVB (FWVB. . .)
  - vVBEL (KNOT. . . _i and KNOT. . . _k and Q)
- NEW 1st Call:
  - **vROHRVecResults: VEC-Channel-Results for Pipe-Interior-Pts (IPts):**
    - pk
    - mx2Idx
    - IptIdx: S,0,. . . ,E - Interior Point Index; S=Start EdgeDefNode, E=End EdgeDefNode, 0=1st Ipt in EdgeDefDirection
    - one column per VEC-Channel
  - **vAGSN**
    - from vVBEL: KNOT. . . _i and KNOT. . . _k and Q
    - from vROHRVecResults: vecResults
    - **Topology:**
      - nextNODE
      - IptIdx
    - **Geometry:**
      - dx
      - x
      - xVbel

- Z (the corresponding Z_i, Z_k and ZVEC are droped)

* **Results:**
  - Q: from Q before and QMVEC for PIPEs; in Schnittrichtung; QMVEC is droped
  - **for available KNOT...#_i, KNOT...#_k and ...#VEC:**

    i.e. KNOT~*~*~*~P_i KNOT~*~*~*~P_k ROHR~*~*~*~PVEC

    P is new column

    the correspondig 3 columns are droped

Notes:

- **The Add-Result is persisted if df were read from H5:**
  - xm.ToH5() is called if xm.h5Read is True.

Returns: Mx-Object if no Mx-Object was given; Nothing else

Raises: XmError

```
>>> xm=xms['LocalHeatingNetwork']
>>> mx=xm.MxAdd()
>>> print(xm._getvXXXXAsOneString(vXXXX='vKNOT',filterColList=['BESCHREIBUNG',
↪'IDREFERENZ','NAME','KNOT~*~*~*~PH']))
                     BESCHREIBUNG IDREFERENZ         NAME  KNOT~*~*~*~PH
0                            None         -1       R-K004       2.302971
1                            None         -1       V-K002       3.985846
2                            None         -1       V-K001       4.083384
3                            None         -1       V-K000       4.121495
4                            None         -1       R-K001       2.043288
5                            None         -1       R-K003       2.283565
6                            None         -1       R-K000       2.004937
7                            None         -1       R-K005       2.309655
8                            None         -1          R-L       2.000133
9                            None         -1       R-K002       2.141440
10                           None         -1       V-K004       3.825970
11                           None         -1       V-K005       3.819467
12                           None         -1       R-K007       2.314658
13                           None         -1       V-K006       3.816599
14                           None         -1       R-K006       2.312659
15                           None         -1       V-K003       3.845104
16                           None         -1          V-L       4.125885
17                           None         -1       V-K007       3.814690
18                           None         -1           R2       4.311307
19                           None         -1          V-1       4.126019
20                           None         -1           R3       4.291591
21  Druckhaltung - 2 bar Ruhedruck         -1  PKON-Knoten       2.000000
22          Anbindung Druckhaltung         -1          R-1       2.000000
>>> print(xm._getvXXXXAsOneString(vXXXX='vROHR',filterColList=['BESCHREIBUNG',
↪'IDREFERENZ','NAME_i','NAME_k','ROHR~*~*~*~QMAV']))
   BESCHREIBUNG IDREFERENZ  NAME_i  NAME_k  ROHR~*~*~*~QMAV
0          None         -1  R-K004  R-K005        -8.509475
1          None         -1  V-K002  V-K003        19.059780
2          None         -1  R-K003  R-K004       -15.378901
3          None         -1  V-K004  V-K005         8.509476
4          None         -1  V-K001  V-K002        22.987946
5          None         -1  R-K006  R-K007        -3.928166
6          None         -1  V-K000  V-K001        22.987947
```

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```
7          None         -1  V-K003  V-K004        15.378901
8          None         -1  V-K005  V-K006         3.928167
9          None         -1  R-K001  R-K002       -22.987946
10         None         -1  R-K002  R-K003       -19.059778
11         None         -1  R-K005  R-K006        -3.928166
12         None         -1  V-K006  V-K007         3.928167
13         None         -1  R-K000  R-K001       -22.987946
14         None         -1     R-L  R-K000       -22.987946
15         None         -1     V-L  V-K000        22.987947
>>> print(xm._getvXXXXAsOneString(vXXXX='vFWVB'))
  BESCHREIBUNG IDREFERENZ   W0  LFK  W0LFK  TVL0  TRS0  LFKT      W  W_min  W_
↪max  INDTR  TRSK  VTYP  DPHAUS  IMBG  IRFV                    pk            ␣
↪      tk  NAME_i KVR_i TM_i   XKOR_i   YKOR_i ZKOR_i  pXCor_i  pYCor_i ␣
↪NAME_k KVR_k TM_k   XKOR_k   YKOR_k ZKOR_k  pXCor_k  pYCor_k              ␣
↪                      CONT CONT_ID CONT_LFDNR                        WBLZ ␣
↪mx2Idx  FWVB~*~*~*~W  FWVB~*~*~*~QM  FWVB~*~*~*~IAKTIV
0            1         -1  200  0.8  160.0    90    50  LFKT  160.0  160.0 ␣
↪160.0      1    55    14     0.7     0   0.0  4643800032883366034 ␣
↪4643800032883366034  V-K002     1   90  2541059  5706265     20    319.0   ␣
↪ 56.0  R-K002     2   60  2541059  5706265     20    319.0     56.0 ␣
↪Nahwärmenetz mit 1000 kW Anschlussleistu    1001         -1  [BLNZ1,␣
↪BLNZ1u5, BLNZ1u5u7]       0         160.0       3.928166                0.0
1            3         -1  200  1.0  200.0    90    65  LFKT  200.0  200.0 ␣
↪200.0      1    65    14     0.7     0   0.0  4704603947372595298 ␣
↪4704603947372595298  V-K004     1   90  2541539  5706361     20    799.0   ␣
↪152.0  R-K004     2   60  2541539  5706361     20    799.0    152.0 ␣
↪Nahwärmenetz mit 1000 kW Anschlussleistu    1001         -1               ␣
↪          []       1         200.0       6.869426                0.0
2            4         -1  200  0.8  160.0    90    60  LFKT  160.0  160.0 ␣
↪160.0      1    60    14     0.7     0   0.0  5121101823283893406 ␣
↪5121101823283893406  V-K005     1   90  2541627  5706363     20    887.0   ␣
↪154.0  R-K005     2   60  2541627  5706363     20    887.0    154.0 ␣
↪Nahwärmenetz mit 1000 kW Anschlussleistu    1001         -1  [BLNZ1u5,␣
↪BLNZ1u5u7, BLNZ5]       2         160.0       4.581308                0.0
3            5         -1  200  0.8  160.0    90    55  LFKT  160.0  160.0 ␣
↪160.0      1    55    14     0.7     0   0.0  5400405917816384862 ␣
↪5400405917816384862  V-K007     1   90  2541899  5706325     20   1159.0   ␣
↪116.0  R-K007     2   60  2541899  5706325     20   1159.0    116.0 ␣
↪Nahwärmenetz mit 1000 kW Anschlussleistu    1001         -1               ␣
↪ [BLNZ1u5u7]       3         160.0       3.928166                0.0
4            2         -1  200  0.6  120.0    90    60  LFKT  120.0  120.0 ␣
↪120.0      1    62    14     0.7     0   0.0  5695730293103267172 ␣
↪5695730293103267172  V-K003     1   90  2541457  5706345     20    717.0   ␣
↪136.0  R-K003     2   60  2541457  5706345     20    717.0    136.0 ␣
↪Nahwärmenetz mit 1000 kW Anschlussleistu    1001         -1               ␣
↪          []       4         120.0       3.680879                0.0
>>> print(xm._getvXXXXAsOneString(vXXXX='vVBEL',filterColList=['BESCHREIBUNG',
↪'IDREFERENZ','NAME_i','NAME_k','Q']))
                                                BESCHREIBUNG IDREFERENZ      ␣
↪NAME_i  NAME_k            Q
OBJTYPE OBJID                                                                ␣
↪
FWES    5638756766880678918  BHKW - Modul - 1000 kW therm.         -1        ␣
↪  R3     V-1      22.9879
FWVB    4643800032883366034                                 1         -1       V-
↪K002  R-K002      3.92817
        4704603947372595298                                 3         -1       V-
↪K004  R-K004      6.86943
```

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```
        5121101823283893406                                4          -1        V-
↪K005  R-K005      4.58131
        5400405917816384862                                5          -1        V-
↪K007  R-K007      3.92817
        5695730293103267172                                2          -1        V-
↪K003  R-K003      3.68088
KLAP    4801110583764519435                             None          -1         ␣
↪  R2      R3      22.9879
PGRP    4986517622672493603                      Pumpengruppe         -1         ␣
↪ R-1      R3         None
PUMP    5481331875203087055                       Umwälzpumpe         -1         ␣
↪ R-1      R2      22.9879
ROHR    4613782368750024999                             None          -1        R-
↪K004  R-K005     -8.50947
        4614949065966596185                             None          -1        V-
↪K002  V-K003      19.0598
        4637102239750163477                             None          -1        R-
↪K003  R-K004     -15.3789
        4713733238627697042                             None          -1        V-
↪K004  V-K005      8.50948
        4769996343148550485                             None          -1         ␣
↪ R-L  R-K000     -22.9879
        4789218195240364437                             None          -1        V-
↪K001  V-K002      22.9879
        4939422678063487923                             None          -1         ␣
↪ V-L  V-K000      22.9879
        4945727430885351042                             None          -1        R-
↪K006  R-K007     -3.92817
        4984202422877610920                             None          -1        V-
↪K000  V-K001      22.9879
        5037777106796980248                             None          -1        V-
↪K003  V-K004      15.3789
        5123819811204259837                             None          -1        V-
↪K005  V-K006      3.92817
        5266224553324203132                             None          -1        R-
↪K001  R-K002     -22.9879
        5379365049009065623                             None          -1        R-
↪K002  R-K003     -19.0598
        5611703699850694889                             None          -1        R-
↪K005  R-K006     -3.92817
        5620197984230756681                             None          -1        V-
↪K006  V-K007      3.92817
        5647213228462830353                             None          -1        R-
↪K000  R-K001     -22.9879
VENT    4678923650983295610                             None          -1         ␣
↪ V-1     V-L      22.9879
        4897018421024717974                             None          -1         ␣
↪ R-L     R-1      22.9879
        5525310316015533093                             None          -1  PKON-
↪Knoten     R-1  2.19997e-06
>>> print(xm._getvXXXXAsOneString(vXXXX='vROHRVecResults'))
                   pk  mx2Idx IptIdx  ROHR~*~*~*~SVEC  ROHR~*~*~*~PVEC ␣
↪ROHR~*~*~*~RHOVEC  ROHR~*~*~*~TVEC  ROHR~*~*~*~MVEC  ROHR~*~*~*~ZVEC
0   4613782368750024999       0      S         0.000000         3.302971      ␣
↪   983.700012             60.0        -2.363743             20.0
1   4613782368750024999       0      E        88.019997         3.309655      ␣
↪   983.700012             60.0        -2.363743             20.0
```

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```
2   4614949065966596185      1      S         0.000000         4.985845 ␣
↪   965.700012             90.0         5.294384             20.0
3   4614949065966596185      1      E       405.959991         4.845104 ␣
↪   965.700012             90.0         5.294384             20.0
4   4637102239750163477      2      S         0.000000         3.283566 ␣
↪   983.700012             60.0        -4.271917             20.0
5   4637102239750163477      2      E        83.550003         3.302971 ␣
↪   983.700012             60.0        -4.271917             20.0
6   4713733238627697042      3      S         0.000000         4.825970 ␣
↪   965.700012             90.0         2.363743             20.0
7   4713733238627697042      3      E        88.019997         4.819468 ␣
↪   965.700012             90.0         2.363743             20.0
8   4789218195240364437      5      S         0.000000         5.083383 ␣
↪   965.700012             90.0         6.385540             20.0
9   4789218195240364437      5      E       195.529999         4.985845 ␣
↪   965.700012             90.0         6.385540             20.0
10  4945727430885351042      7      S         0.000000         3.312659 ␣
↪   983.700012             60.0        -1.091157             20.0
11  4945727430885351042      7      E       109.769997         3.314658 ␣
↪   983.700012             60.0        -1.091157             20.0
12  4984202422877610920      8      S         0.000000         5.121495 ␣
↪   965.700012             90.0         6.385540             20.0
13  4984202422877610920      8      E        76.400002         5.083383 ␣
↪   965.700012             90.0         6.385540             20.0
14  5037777106796980248      9      S         0.000000         4.845104 ␣
↪   965.700012             90.0         4.271917             20.0
15  5037777106796980248      9      E        83.550003         4.825970 ␣
↪   965.700012             90.0         4.271917             20.0
16  5123819811204259837     10      S         0.000000         4.819468 ␣
↪   965.700012             90.0         1.091157             20.0
17  5123819811204259837     10      E       164.910004         4.816599 ␣
↪   965.700012             90.0         1.091157             20.0
18  5266224553324203132     11      S         0.000000         3.043288 ␣
↪   983.700012             60.0        -6.385540             20.0
19  5266224553324203132     11      E       195.529999         3.141441 ␣
↪   983.700012             60.0        -6.385540             20.0
20  5379365049009065623     12      S         0.000000         3.141441 ␣
↪   983.700012             60.0        -5.294383             20.0
21  5379365049009065623     12      E       405.959991         3.283566 ␣
↪   983.700012             60.0        -5.294383             20.0
22  5611703699850694889     13      S         0.000000         3.309655 ␣
↪   983.700012             60.0        -1.091157             20.0
23  5611703699850694889     13      E       164.910004         3.312659 ␣
↪   983.700012             60.0        -1.091157             20.0
24  5620197984230756681     14      S         0.000000         4.816599 ␣
↪   965.700012             90.0         1.091157             20.0
25  5620197984230756681     14      E       109.769997         4.814690 ␣
↪   965.700012             90.0         1.091157             20.0
26  5647213228462830353     15      S         0.000000         3.004937 ␣
↪   983.700012             60.0        -6.385540             20.0
27  5647213228462830353     15      E        76.400002         3.043288 ␣
↪   983.700012             60.0        -6.385540             20.0
28  4769996343148550485      4      S         0.000000         3.000133 ␣
↪   983.700012             60.0        -6.385540             20.0
29  4769996343148550485      4      E        73.419998         3.004937 ␣
↪   983.700012             60.0        -6.385540             20.0
30  4939422678063487923      6      S         0.000000         5.125884 ␣
↪   965.700012             90.0         6.385541             20.0
```

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```
31  4939422678063487923        6       E         68.599998          5.121495
↪   965.700012              90.0          6.385541              20.0
>>> xm.MxAdd(mx=mx,aggReq='TMAX')
>>> dfTMax=xm.dataFrames['vROHRVecResults'].copy()
>>> xm.MxAdd(mx=mx,aggReq='TMIN')
>>> dfTMin=xm.dataFrames['vROHRVecResults'].copy()
>>> xm.MxAdd(mx=mx,aggReq='MAX')
>>> dfMax=xm.dataFrames['vROHRVecResults'].copy()
>>> xm.MxAdd(mx=mx,aggReq='MIN')
>>> dfMin=xm.dataFrames['vROHRVecResults'].copy()
>>> import pandas as pd
>>> decimals=pd.Series([6],index=['ROHR~*~*~*~PVEC'])
>>> dfTMax.round(decimals=decimals).equals(dfMax.round(decimals=decimals))
True
>>> dfTMin.round(decimals=decimals).equals(dfMin.round(decimals=decimals))
True
```

_MxAddvVBEL (*dfSource*)

(Re-)constructing vVBEL with MX2-Results.

Arguments: dfSource

Result:

View with MX2-Results added:

- in the Xm-View below col mx2Idx must exist
- mx2Idx is considered to be the last of the Model-cols
- right from mx2Idx Result-cols are added if not already existing
- already existing Result-cols are overwritten
- **mx2Idx-View:**
  - ...
  - vVBEL (KNOT..._i and KNOT..._k and Q)

Raises: XmError

```
>>> xm=xms['GPipes']
>>> mx=xm.MxSync()
>>> xm.MxAdd(mx=mx)
>>> print(xm._getvXXXXAsOneString(vXXXX='vVBEL',filterColList=['BESCHREIBUNG
↪','IDREFERENZ','tk NAME_i','CONT_i',' CONT_VKNO_i','Z_i','pk_i']))
                              BESCHREIBUNG             IDREFERENZ   CONT_i   Z_
↪i                 pk_i
OBJTYPE OBJID                                                                 ␣
↪
ROHR    4979507900871287244  _Split_Split  3S4979507900871287244  M-1-0-1   ␣
↪0  4731210032713520411
        5114681686941855110        _Split  3S5114681686941855110  M-1-0-1   ␣
↪0  5709889458254995435
        5244313507655010738           NaN  3S5244313507655010738  M-1-0-1   ␣
↪0  5256558483525770176
        5694016449043789006           NaN  3S5694016449043789006  M-1-0-1   ␣
↪0  4731210032713520411
VENT    5116489323526156845           NaN  3S5508684139418025293  M-1-0-1 ␣
↪100  4683988347517083361
```

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```
        5309992331398639768           NaN  3S5309992331398639768  M-1-0-1
↪0  5046108271210239718
        5508684139418025293           NaN  3S5508684139418025293  M-1-0-1
↪100  4683988347517083361
        5745097345184516675           NaN  3S5745097345184516675  M-1-0-1
↪0  5308591811899364960
>>> print(xm._getvXXXXAsOneString(vXXXX='vVBEL',filterColList=['BESCHREIBUNG
↪','IDREFERENZ','tk NAME_k','CONT_k',' CONT_VKNO_k','Z_k','pk_k']))
                              BESCHREIBUNG             IDREFERENZ   CONT_k  Z_
↪k                 pk_k
OBJTYPE OBJID                                                                
↪
ROHR    4979507900871287244  _Split_Split  3S4979507900871287244  M-1-0-1
↪0  5441322867018839631
        5114681686941855110        _Split  3S5114681686941855110  M-1-0-1
↪0  5441322867018839631
        5244313507655010738           NaN  3S5244313507655010738  M-1-0-1
↪100  4683988347517083361
        5694016449043789006           NaN  3S5694016449043789006  M-1-0-1
↪0  5061043246189134395
VENT    5116489323526156845           NaN  3S5508684139418025293  M-1-0-1
↪0  5709889458254995435
        5309992331398639768           NaN  3S5309992331398639768  M-1-0-1
↪0  5256558483525770176
        5508684139418025293           NaN  3S5508684139418025293  M-1-0-1
↪0  5709889458254995435
        5745097345184516675           NaN  3S5745097345184516675  M-1-0-1
↪0  4731210032713520411
>>> print(xm._getvXXXXAsOneString(vXXXX='vVBEL',filterColList=['LAYR','L','D
↪','mx2Idx']))
                            LAYR           L    D  mx2Idx
OBJTYPE OBJID
ROHR    4979507900871287244   []       10000  500       2
        5114681686941855110   []       10000  500       1
        5244313507655010738   []      160000  500       0
        5694016449043789006   []  100.498688  450       3
VENT    5116489323526156845   []           0  666       3
        5309992331398639768   []           0  800       0
        5508684139418025293   []           0  800       2
        5745097345184516675   []           0  800       1
```

```
>>> print(xm._getvXXXXAsOneString(vXXXX='vVBEL',filterColList=['KNOT~*~*~*~
↪RHON_i','KNOT~*~*~*~H_i','KNOT~*~*~*~LFAKTAKT_i','KNOT~*~*~*~P_i','KNOT~*~
↪*~*~PH_i','KNOT~*~*~*~PH_EIN_i','KNOT~*~*~*~QM_i','KNOT~*~*~*~RHO_i','KNOT~
↪*~*~*~T_i','KNOT~*~*~*~EH_i']))
                             KNOT~*~*~*~RHON_i  KNOT~*~*~*~H_i  KNOT~*~*~*~
↪LFAKTAKT_i  KNOT~*~*~*~P_i  KNOT~*~*~*~PH_i  KNOT~*~*~*~PH_EIN_i  KNOT~*~*~
↪*~QM_i  KNOT~*~*~*~RHO_i  KNOT~*~*~*~T_i  KNOT~*~*~*~EH_i
OBJTYPE OBJID                                                                
↪                                                                            
↪
ROHR    4979507900871287244                0.83    10106.165039              
↪     1.0       11.106165        10.106165            10.106165          0.
↪00000          7.888546       49.020111     13046.647461
        5114681686941855110                0.83    16375.773438              
↪     1.0       17.375772        16.375772            16.375772          0.
↪00000         12.826390       40.874298     13008.437500
```

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```
        5244313507655010738                0.83    39973.992188                ␣
↪     1.0       40.973991        39.973991            39.973991          0.
↪00000         31.891241       39.985443     12777.368164
        5694016449043789006                0.83    10106.165039                ␣
↪     1.0       11.106165        10.106165            10.106165          0.
↪00000          7.888546       49.020111     13046.647461
VENT    5116489323526156845                0.83    16440.441406                ␣
↪     1.0       17.440441        16.440441            16.440441          0.
↪00000         12.874356       40.910797     13111.193359
        5309992331398639768                0.83    40000.000000                ␣
↪     1.0       41.000000        40.000000            40.000000     118257.
↪53125         31.911810       40.000000     12777.444336
        5508684139418025293                0.83    16440.441406                ␣
↪     1.0       17.440441        16.440441            16.440441          0.
↪00000         12.874356       40.910797     13111.193359
        5745097345184516675                0.83    10000.000000                ␣
↪     1.0       11.000000        10.000000            10.000000    -118257.
↪53125          7.812713       48.967743     13034.717773
>>> print(xm._getvXXXXAsOneString(vXXXX='vVBEL',filterColList=['KNOT~*~*~*~
↪HMAX_INST_i','KNOT~*~*~*~HMIN_INST_i','KNOT~*~*~*~PMAX_INST_i','KNOT~*~*~*~
↪PMIN_INST_i','KNOT~*~*~*~IAKTIV_i','KNOT~*~*~*~PDAMPF_i']))
                           KNOT~*~*~*~HMAX_INST_i  KNOT~*~*~*~HMIN_INST_i ␣
↪KNOT~*~*~*~PMAX_INST_i  KNOT~*~*~*~PMIN_INST_i  KNOT~*~*~*~IAKTIV_i  KNOT~
↪*~*~*~PDAMPF_i
OBJTYPE OBJID                                                              ␣
↪                                                                          ␣
↪
ROHR    4979507900871287244            10106.165039            10106.165039 ␣
↪             11.106165               11.106165                   0.0     ␣
↪          0.0
        5114681686941855110            16375.773438            16375.773438 ␣
↪             17.375772               17.375772                   0.0     ␣
↪          0.0
        5244313507655010738            39973.992188            39973.992188 ␣
↪             40.973991               40.973991                   0.0     ␣
↪          0.0
        5694016449043789006            10106.165039            10106.165039 ␣
↪             11.106165               11.106165                   0.0     ␣
↪          0.0
VENT    5116489323526156845            16440.441406            16440.441406 ␣
↪             17.440441               17.440441                   0.0     ␣
↪          0.0
        5309992331398639768            40000.000000            40000.000000 ␣
↪             41.000000               41.000000                   0.0     ␣
↪          0.0
        5508684139418025293            16440.441406            16440.441406 ␣
↪             17.440441               17.440441                   0.0     ␣
↪          0.0
        5745097345184516675            10000.000000            10000.000000 ␣
↪             11.000000               11.000000                   0.0     ␣
↪          0.0
```

_MxAddvROHRVecResults (*dfSource*)

(Re-)constructing vROHRVecResults.

Arguments:

- dfSource

Result:

- **vROHRVecResults: VEC-Channel-Results for Pipe-Interior-Pts (IPts):**
  - pk
  - mx2Idx
  - IptIdx: S,0,...,E - Interior Point Index; S=Start EdgeDefNode, E=End EdgeDefNode, 0=1st Ipt in EdgeDefDirection
  - one column per VEC-Channel

Raises: XmError

```
>>> xm=xms['GPipes']
>>> mx=xm.MxSync()
>>> mxVecsFileDataLst=mx.getMxsVecsFileData()
>>> mxVecsFileData=mxVecsFileDataLst[0]
>>> # construct MultiIndex Start ... ---
>>> colsToBeUnpacked=mxVecsFileData.columns.tolist() # all columns
>>> arrays=[[mxVecsFileData.index[0]]*len(colsToBeUnpacked),colsToBeUnpacked]
>>> tuples = list(zip(*arrays))
>>> import pandas as pd
>>> mIndex = pd.MultiIndex.from_tuples(tuples, names=['Timestamp', 'Sir3sID
↪'])
>>> # construct MultiIndex End ... ---
>>> dfUnpacked=mx.unPackMxsVecsFileDataDf(mxVecsFileData,mIndex,
↪returnMultiIndex=False)
>>> xm._MxAddvROHRVecResults(dfSource=dfUnpacked)
>>> print(xm._getvXXXXAsOneString(vXXXX='vROHRVecResults',sortList=['ROHR~*~
↪*~*~PHVEC','ROHR~*~*~*~SVEC'],ascending=False))
                    pk  mx2Idx IptIdx  ROHR~*~*~*~SVEC  ROHR~*~*~*~TVEC ␣
↪ROHR~*~*~*~ZVEC  ROHR~*~*~*~PVEC  ROHR~*~*~*~MVEC  ROHR~*~*~*~RHOVEC  ROHR~
↪*~*~*~PHVEC  ROHR~*~*~*~QMVEC
8   5244313507655010738       0      S         0.000000        39.985443    ␣
↪       0.000        40.973988        27.264917          31.891241        ␣
↪39.973988       3077.763672
9   5244313507655010738       0      0      5000.000000        40.407806    ␣
↪       3.125        40.453373        27.264917          31.443642        ␣
↪39.453373       3121.575439
10  5244313507655010738       0      1     10000.000000        40.397430    ␣
↪       6.250        39.925545        27.264917          30.999212        ␣
↪38.925545       3166.328857
11  5244313507655010738       0      2     15000.000000        40.387299    ␣
↪       9.375        39.390327        27.264917          30.549555        ␣
↪38.390327       3212.934082
12  5244313507655010738       0      3     20000.000000        40.377533    ␣
↪      12.500        38.847401        27.264917          30.094410        ␣
↪37.847401       3261.526123
13  5244313507655010738       0      4     25000.000000        40.368195    ␣
↪      15.625        38.296429        27.264917          29.633537        ␣
↪37.296429       3312.250732
14  5244313507655010738       0      5     30000.000000        40.359253    ␣
↪      18.750        37.737045        27.264917          29.166674        ␣
↪36.737045       3365.269043
15  5244313507655010738       0      6     35000.000000        40.350830    ␣
↪      21.875        37.168861        27.264917          28.693537        ␣
↪36.168861       3420.760010
16  5244313507655010738       0      7     40000.000000        40.342926    ␣
↪      25.000        36.591457        27.264917          28.213825        ␣
↪35.591457       3478.922119
```

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```
17  5244313507655010738       0      8     45000.000000        40.335602
↪       28.125        36.004372        27.264917          27.727211
↪35.004372       3539.977539
18  5244313507655010738       0      9     50000.000000        40.328918
↪       31.250        35.407112        27.264917          27.233337
↪34.407112       3604.174561
19  5244313507655010738       0     10     55000.000000        40.322968
↪       34.375        34.799137        27.264917          26.731815
↪33.799137       3671.793457
20  5244313507655010738       0     11     60000.000000        40.317780
↪       37.500        34.179863        27.264917          26.222223
↪33.179863       3743.149414
21  5244313507655010738       0     12     65000.000000        40.313477
↪       40.625        33.548645        27.264917          25.704098
↪32.548645       3818.601318
22  5244313507655010738       0     13     70000.000000        40.310181
↪       43.750        32.904778        27.264917          25.176931
↪31.904778       3898.557129
23  5244313507655010738       0     14     75000.000000        40.308014
↪       46.875        32.247482        27.264917          24.640163
↪31.247482       3983.484131
24  5244313507655010738       0     15     80000.000000        40.307098
↪       50.000        31.575897        27.264917          24.093176
↪30.575897       4073.921387
25  5244313507655010738       0     16     85000.000000        40.307587
↪       53.125        30.889069        27.264917          23.535278
↪29.889069       4170.492676
26  5244313507655010738       0     17     90000.000000        40.309753
↪       56.250        30.185926        27.264917          22.965700
↪29.185926       4273.925781
27  5244313507655010738       0     18     95000.000000        40.313782
↪       59.375        29.465275        27.264917          22.383583
↪28.465275       4385.075684
28  5244313507655010738       0     19    100000.000000        40.320038
↪       62.500        28.725761        27.264917          21.787949
↪27.725761       4504.953613
29  5244313507655010738       0     20    105000.000000        40.328827
↪       65.625        27.965855        27.264917          21.177694
↪26.965855       4634.768066
30  5244313507655010738       0     21    110000.000000        40.340607
↪       68.750        27.183802        27.264917          20.551556
↪26.183802       4775.974121
31  5244313507655010738       0     22    115000.000000        40.355988
↪       71.875        26.377584        27.264917          19.908077
↪25.377584       4930.345703
32  5244313507655010738       0     23    120000.000000        40.375671
↪       75.000        25.544865        27.264917          19.245565
↪24.544865       5100.068359
33  5244313507655010738       0     24    125000.000000        40.400543
↪       78.125        24.682890        27.264917          18.562040
↪23.682890       5287.872559
34  5244313507655010738       0     25    130000.000000        40.431854
↪       81.250        23.788412        27.264917          17.855148
↪22.788412       5497.221191
35  5244313507655010738       0     26    135000.000000        40.471252
↪       84.375        22.857525        27.264917          17.122063
↪21.857525       5732.586426
```

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```
36  5244313507655010738       0     27    140000.000000        40.520874
↪       87.500        21.885471        27.264917          16.359346
↪20.885471       5999.854980
37  5244313507655010738       0     28    145000.000000        40.583862
↪       90.625        20.866375        27.264917          15.562735
↪19.866375       6306.970215
38  5244313507655010738       0     29    150000.000000        40.664673
↪       93.750        19.792791        27.264917          14.726833
↪18.792791       6664.956543
39  5244313507655010738       0     30    155000.000000        40.770111
↪       96.875        18.655102        27.264917          13.844651
↪17.655102       7089.647949
40  5244313507655010738       0      E    160000.000000        40.910797
↪      100.000        17.440462        27.264917          12.906846
↪16.440462       7604.778320
5   5114681686941855110       1      S         0.000000        40.874298
↪        0.000        17.375792        27.264917          12.826405
↪16.375792       7652.472168
6   5114681686941855110       1      0      5000.000000        43.653412
↪        0.000        16.058306        27.264917          11.750417
↪15.058306       8353.209961
7   5114681686941855110       1      E     10000.000000        43.972473
↪        0.000        14.612197        27.264917          10.651937
↪13.612197       9214.634766
4   4979507900871287244       2      E     10000.000000        43.972504
↪        0.000        14.612185       -27.264917          10.620180
↪13.612185      -9242.188477
3   4979507900871287244       2      0      5000.000000        48.149719
↪        0.000        12.987305       -27.264917           9.317566
↪11.987305     -10534.264648
2   4979507900871287244       2      S         0.000000        49.020142
↪        0.000        11.106176       -27.264917           7.919193
↪10.106176     -12394.407227
0   5694016449043789006       3      S         0.000000        10.000000
↪        0.000        11.106160         0.000000           6.933107
↪10.106160          0.000000
1   5694016449043789006       3      E       100.498688        10.000000
↪        0.000         1.000000         0.000000           0.791800
↪0.000000          0.000000
>>> print(xm._getvXXXXAsOneString(vXXXX='vROHRVecResults'))
                    pk  mx2Idx IptIdx  ROHR~*~*~*~SVEC  ROHR~*~*~*~TVEC
↪ROHR~*~*~*~ZVEC  ROHR~*~*~*~PVEC  ROHR~*~*~*~MVEC  ROHR~*~*~*~RHOVEC  ROHR~
↪*~*~*~PHVEC  ROHR~*~*~*~QMVEC
0   5694016449043789006       3      S         0.000000        10.000000
↪        0.000        11.106160         0.000000           6.933107
↪10.106160          0.000000
1   5694016449043789006       3      E       100.498688        10.000000
↪        0.000         1.000000         0.000000           0.791800
↪0.000000          0.000000
2   4979507900871287244       2      S         0.000000        49.020142
↪        0.000        11.106176       -27.264917           7.919193
↪10.106176     -12394.407227
3   4979507900871287244       2      0      5000.000000        48.149719
↪        0.000        12.987305       -27.264917           9.317566
↪11.987305     -10534.264648
4   4979507900871287244       2      E     10000.000000        43.972504
↪        0.000        14.612185       -27.264917          10.620180
↪13.612185      -9242.188477
```

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```
5   5114681686941855110       1      S         0.000000        40.874298
↪       0.000        17.375792        27.264917          12.826405
↪16.375792       7652.472168
6   5114681686941855110       1      0      5000.000000        43.653412
↪       0.000        16.058306        27.264917          11.750417
↪15.058306       8353.209961
7   5114681686941855110       1      E     10000.000000        43.972473
↪       0.000        14.612197        27.264917          10.651937
↪13.612197       9214.634766
8   5244313507655010738       0      S         0.000000        39.985443
↪       0.000        40.973988        27.264917          31.891241
↪39.973988       3077.763672
9   5244313507655010738       0      0      5000.000000        40.407806
↪       3.125        40.453373        27.264917          31.443642
↪39.453373       3121.575439
10  5244313507655010738       0      1     10000.000000        40.397430
↪       6.250        39.925545        27.264917          30.999212
↪38.925545       3166.328857
11  5244313507655010738       0      2     15000.000000        40.387299
↪       9.375        39.390327        27.264917          30.549555
↪38.390327       3212.934082
12  5244313507655010738       0      3     20000.000000        40.377533
↪      12.500        38.847401        27.264917          30.094410
↪37.847401       3261.526123
13  5244313507655010738       0      4     25000.000000        40.368195
↪      15.625        38.296429        27.264917          29.633537
↪37.296429       3312.250732
14  5244313507655010738       0      5     30000.000000        40.359253
↪      18.750        37.737045        27.264917          29.166674
↪36.737045       3365.269043
15  5244313507655010738       0      6     35000.000000        40.350830
↪      21.875        37.168861        27.264917          28.693537
↪36.168861       3420.760010
16  5244313507655010738       0      7     40000.000000        40.342926
↪      25.000        36.591457        27.264917          28.213825
↪35.591457       3478.922119
17  5244313507655010738       0      8     45000.000000        40.335602
↪      28.125        36.004372        27.264917          27.727211
↪35.004372       3539.977539
18  5244313507655010738       0      9     50000.000000        40.328918
↪      31.250        35.407112        27.264917          27.233337
↪34.407112       3604.174561
19  5244313507655010738       0     10     55000.000000        40.322968
↪      34.375        34.799137        27.264917          26.731815
↪33.799137       3671.793457
20  5244313507655010738       0     11     60000.000000        40.317780
↪      37.500        34.179863        27.264917          26.222223
↪33.179863       3743.149414
21  5244313507655010738       0     12     65000.000000        40.313477
↪      40.625        33.548645        27.264917          25.704098
↪32.548645       3818.601318
22  5244313507655010738       0     13     70000.000000        40.310181
↪      43.750        32.904778        27.264917          25.176931
↪31.904778       3898.557129
23  5244313507655010738       0     14     75000.000000        40.308014
↪      46.875        32.247482        27.264917          24.640163
↪31.247482       3983.484131
```

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```
24  5244313507655010738       0     15     80000.000000        40.307098
↪      50.000        31.575897        27.264917          24.093176
↪30.575897       4073.921387
25  5244313507655010738       0     16     85000.000000        40.307587
↪      53.125        30.889069        27.264917          23.535278
↪29.889069       4170.492676
26  5244313507655010738       0     17     90000.000000        40.309753
↪      56.250        30.185926        27.264917          22.965700
↪29.185926       4273.925781
27  5244313507655010738       0     18     95000.000000        40.313782
↪      59.375        29.465275        27.264917          22.383583
↪28.465275       4385.075684
28  5244313507655010738       0     19    100000.000000        40.320038
↪      62.500        28.725761        27.264917          21.787949
↪27.725761       4504.953613
29  5244313507655010738       0     20    105000.000000        40.328827
↪      65.625        27.965855        27.264917          21.177694
↪26.965855       4634.768066
30  5244313507655010738       0     21    110000.000000        40.340607
↪      68.750        27.183802        27.264917          20.551556
↪26.183802       4775.974121
31  5244313507655010738       0     22    115000.000000        40.355988
↪      71.875        26.377584        27.264917          19.908077
↪25.377584       4930.345703
32  5244313507655010738       0     23    120000.000000        40.375671
↪      75.000        25.544865        27.264917          19.245565
↪24.544865       5100.068359
33  5244313507655010738       0     24    125000.000000        40.400543
↪      78.125        24.682890        27.264917          18.562040
↪23.682890       5287.872559
34  5244313507655010738       0     25    130000.000000        40.431854
↪      81.250        23.788412        27.264917          17.855148
↪22.788412       5497.221191
35  5244313507655010738       0     26    135000.000000        40.471252
↪      84.375        22.857525        27.264917          17.122063
↪21.857525       5732.586426
36  5244313507655010738       0     27    140000.000000        40.520874
↪      87.500        21.885471        27.264917          16.359346
↪20.885471       5999.854980
37  5244313507655010738       0     28    145000.000000        40.583862
↪      90.625        20.866375        27.264917          15.562735
↪19.866375       6306.970215
38  5244313507655010738       0     29    150000.000000        40.664673
↪      93.750        19.792791        27.264917          14.726833
↪18.792791       6664.956543
39  5244313507655010738       0     30    155000.000000        40.770111
↪      96.875        18.655102        27.264917          13.844651
↪17.655102       7089.647949
40  5244313507655010738       0      E    160000.000000        40.910797
↪     100.000        17.440462        27.264917          12.906846
↪16.440462       7604.778320
>>> xm=xms['LocalHeatingNetwork']
>>> mx=xm.MxSync()
>>> xm.MxAdd(mx=mx)
>>> print(xm._getvXXXXAsOneString(vXXXX='vROHRVecResults',filterColList=[
↪'mx2Idx','IptIdx','ROHR~*~*~*~SVEC'],index=True))
    mx2Idx IptIdx  ROHR~*~*~*~SVEC
```

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```
0        0      S         0.000000
1        0      E        88.019997
2        1      S         0.000000
3        1      E       405.959991
4        2      S         0.000000
5        2      E        83.550003
6        3      S         0.000000
7        3      E        88.019997
8        5      S         0.000000
9        5      E       195.529999
10       7      S         0.000000
11       7      E       109.769997
12       8      S         0.000000
13       8      E        76.400002
14       9      S         0.000000
15       9      E        83.550003
16      10      S         0.000000
17      10      E       164.910004
18      11      S         0.000000
19      11      E       195.529999
20      12      S         0.000000
21      12      E       405.959991
22      13      S         0.000000
23      13      E       164.910004
24      14      S         0.000000
25      14      E       109.769997
26      15      S         0.000000
27      15      E        76.400002
28       4      S         0.000000
29       4      E        73.419998
30       6      S         0.000000
31       6      E        68.599998
```

_MxAddvAGSN ( )

(Re-)constructing vAGSN.

Result:

- **vAGSN**
  - from vVBEL: KNOT. . . _i and KNOT. . . _k and Q
  - from vROHRVecResults: vecResults
  - **Topology:**
    - nextNODE
    - IptIdx
  - **Geometry:**
    - dx
    - x
    - xVbel
    - Z (the corresponding Z_i, Z_k and ZVEC are droped)
  - **Results:**

* Q: from Q before and QMVEC for PIPEs; in Schnittrichtung; QMVEC is droped
* **for available KNOT…#_i, KNOT…#_k and …#VEC:**
  - i.e. KNOT~*~*~*~P_i KNOT~*~*~*~P_k ROHR~*~*~*~PVEC
  - P is new column
  - the correspondig 3 source-columns are droped

Raises: XmError

```
>>> xm=xms['GPipes']
>>> mx=xm.MxSync()
>>> xm.MxAdd(mx=mx)
>>> vAGSN=xm.dataFrames['vAGSN']
>>> schnitt=vAGSN[vAGSN['NAME']=='LR']
>>> xm.dataFrames['schnitt']=schnitt
>>> print(xm._getvXXXXAsOneString(vXXXX='schnitt',filterColList=['OBJTYPE',
↪'NAME_i','NAME_k','IptIdx','nextNODE','x','PH','Q'],index=True))
    OBJTYPE NAME_i NAME_k IptIdx nextNODE         x        PH        Q
79     VENT     GL     G1      S       G1       0.0        40   118258
80     VENT     GL     G1      E       G1       0.0    39.974   118258
81     ROHR     G1    GKS      S      GKS       0.0    39.974  3077.76
82     ROHR     G1    GKS      0      GKS    5000.0   39.4534  3121.58
83     ROHR     G1    GKS      1      GKS   10000.0   38.9255  3166.33
84     ROHR     G1    GKS      2      GKS   15000.0   38.3903  3212.93
85     ROHR     G1    GKS      3      GKS   20000.0   37.8474  3261.53
86     ROHR     G1    GKS      4      GKS   25000.0   37.2964  3312.25
87     ROHR     G1    GKS      5      GKS   30000.0    36.737  3365.27
88     ROHR     G1    GKS      6      GKS   35000.0   36.1689  3420.76
89     ROHR     G1    GKS      7      GKS   40000.0   35.5915  3478.92
90     ROHR     G1    GKS      8      GKS   45000.0   35.0044  3539.98
91     ROHR     G1    GKS      9      GKS   50000.0   34.4071  3604.17
92     ROHR     G1    GKS     10      GKS   55000.0   33.7991  3671.79
93     ROHR     G1    GKS     11      GKS   60000.0   33.1799  3743.15
94     ROHR     G1    GKS     12      GKS   65000.0   32.5486   3818.6
95     ROHR     G1    GKS     13      GKS   70000.0   31.9048  3898.56
96     ROHR     G1    GKS     14      GKS   75000.0   31.2475  3983.48
97     ROHR     G1    GKS     15      GKS   80000.0   30.5759  4073.92
98     ROHR     G1    GKS     16      GKS   85000.0   29.8891  4170.49
99     ROHR     G1    GKS     17      GKS   90000.0   29.1859  4273.93
100    ROHR     G1    GKS     18      GKS   95000.0   28.4653  4385.08
101    ROHR     G1    GKS     19      GKS  100000.0   27.7258  4504.95
102    ROHR     G1    GKS     20      GKS  105000.0   26.9659  4634.77
103    ROHR     G1    GKS     21      GKS  110000.0   26.1838  4775.97
104    ROHR     G1    GKS     22      GKS  115000.0   25.3776  4930.35
105    ROHR     G1    GKS     23      GKS  120000.0   24.5449  5100.07
106    ROHR     G1    GKS     24      GKS  125000.0   23.6829  5287.87
107    ROHR     G1    GKS     25      GKS  130000.0   22.7884  5497.22
108    ROHR     G1    GKS     26      GKS  135000.0   21.8575  5732.59
109    ROHR     G1    GKS     27      GKS  140000.0   20.8855  5999.85
110    ROHR     G1    GKS     28      GKS  145000.0   19.8664  6306.97
111    ROHR     G1    GKS     29      GKS  150000.0   18.7928  6664.96
112    ROHR     G1    GKS     30      GKS  155000.0   17.6551  7089.65
113    ROHR     G1    GKS      E      GKS  160000.0   16.4405  7604.78
114    VENT    GKS    GKD      S      GKD  160000.0   16.4404   118258
115    VENT    GKS    GKD      E      GKD  160000.0   16.3758   118258
116    ROHR    GKD     G3      S       G3  160000.0   16.3758  7652.47
```

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```
117    ROHR    GKD     G3      0       G3  165000.0  15.0583  8353.21
118    ROHR    GKD     G3      E       G3  170000.0  13.6122  9214.63
119    ROHR     G4     G3      S       G4  170000.0  13.6122  9242.19
120    ROHR     G4     G3      0       G4  175000.0  11.9873  10534.3
121    ROHR     G4     G3      E       G4  180000.0  10.1062  12394.4
122    VENT     GR     G4      S       GR  180000.0  10.1062   118258
123    VENT     GR     G4      E       GR  180000.0       10   118258
```

_Xm__Mx1_Sir3sIDUpd(*mx*, *checkAllChannels=True*)

Update NAME1,2 and Sir3sID in mx.mx1Df and mx.df.

Args: mx: Mx-Object checkAllChannels: if False only Channels with empty NAME1 are updated; default: True: all Channels are checked and updated if necessary

Notes:

The following OBJTYPEs are covered:

- KNOT
- WBLZ
- RXXXX
- all Channels in vVBEL

mx.ToH5() is called if Sir3sID-Updates occured and mx.h5Read is True.

Raises: XmError

_Xm__Mx1_Sir3sIDUpd_ObjTypeNode(*mx=None*, *dfUpd=None*, *dfNAME1=None*, *NAME1Col='NAME'*)

Update Sir3sID and NAME1 in mx.mx1Df and mx.df for Channels in dfUpd.

Args: mx: Mx-Object dfUpd: df with OBJTYPE,NAME1,NAME2,OBJTYPE_PK,ATTRTYPE,Sir3sID to be updated dfNAME1: df with NAME1-Information NAME1Col: col in dfNAME1 with NAME1-Information

Returns: nOfSir3sIDsUpdated

Note: only wrong Channels are updated

Raises: XmError

_Xm__Mx1_vNRCV(*mx*)

vNRCV_Mx1 (vNRCV with Mx1-Information).

Args: mx: Mx-Object

self.dataFrames['vNRCV_Mx1']

index

- reindex

FILTERed

- existing MX-Channels only
- cRefLfdNr: 1st references only

SORTed

- Sir3sID

columns NEW

- Sir3sID

Raises: XmError

_Xm__Mx2_vFWVB(*mx*)

Mx2-Information into vFWVB.

Args: mx: Mx-Object

self.dataFrames['vFWVB']

columns NEW

- mx2Idx

Raises: XmError

_Xm__Mx2_vKNOT(*mx*)

Mx2-Information into vKNOT.

Args: mx: Mx-Object

self.dataFrames['vKNOT']

columns NEW

- mx2Idx

Raises: XmError

_Xm__Mx2_vROHR(*mx*)

Mx2-Information into vROHR.

Args: mx: Mx-Object

self.dataFrames['vROHR']

columns NEW

- mx2NoPts
- mx2Idx

Raises: XmError

_Xm__Mx2_vVBEL(*mx, edges=['ROHR', 'VENT', 'FWVB', 'FWES', 'PUMP', 'KLAP', 'REGV', 'PREG', 'MREG', 'DPRG', 'PGRP']*)

Mx2-Information into vVBEL.

Args: mx: Mx-Object

self.dataFrames['vVBEL']:

columns NEW

- mx2Idx
- **Notes:**
  - for all edges mx2Idx is taken from mx.mx2Df

Raises: XmError

_Xm__MxAddForOneDf(*dfTarget=None, dfSource=None, multiIndexKey=None, testStr='testStr'*)

Add MX2-Resultdata from dfSource as cols to returned dfTarget.

Args: dfTarget: df with col mx2Idx dfSource: df with mx2Idx-corresponding index and cols (containing MX2-Resultdata) to be added multiIndexKey: value for 1st Index if dfTarget is Multiindexed - i.e. 'XXXX'

Notes:

- all cols from dfSource are added at the end of dfTarget in dfSource-sequence
- the cols can already exist in dfTarget
- if so, _all cols must already exist . . .
- . . . the dfTarget-sequence should but must be not necessary the dfSource-sequence

Returns: dfTarget

Raises: XmError

CHAPTER

FOUR

RM

```
>>> # ---
>>> # SETUP
>>> # ---
>>> import os
>>> import logging
>>> logger = logging.getLogger('PT3S.Rm')
>>> # ---
>>> # path
>>> # ---
>>> if __name__ == "__main__":
...   try:
...      dummy=__file__
...      logger.debug("{0:s}{1:s}{2:s}".format('DOCTEST: __main__ Context: ','path =␣
↪os.path.dirname(__file__)'," ."))
...      path = os.path.dirname(__file__)
...   except NameError:
...      logger.debug("{0:s}{1:s}{2:s}".format('DOCTEST: __main__ Context: ","path =
↪'.' because __file__ not defined and: "," from Rm import Rm"))
...      path = '.'
...      from Rm import Rm
... else:
...    path = '.'
...    logger.debug("{0:s}{1:s}".format('Not __main__ Context: ","path = '.' ."))
>>> try:
...    from PT3S import Mx
... except ImportError:
...    logger.debug("{0:s}{1:s}".format("DOCTEST: from PT3S import Mx: ImportError: ",
↪"trying import Mx maybe pip install -e . is active ..."))
...    import Mx
>>> try:
...    from PT3S import Xm
... except ImportError:
...    logger.debug("{0:s}{1:s}".format("DOCTEST: from PT3S import Xm: ImportError: ",
↪"trying import Xm maybe pip install -e . is active ..."))
...    import Xm
>>> # ---
>>> # testDir
>>> # ---
>>> # globs={'testDir':'testdata'}
>>> try:
...    dummy= testDir
... except NameError:
...    testDir='testdata'
>>> # ---
```

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```
>>> # dotResolution
>>> # ---
>>> # globs={'dotResolution':''}
>>> try:
...     dummy= dotResolution
... except NameError:
...     dotResolution=''
>>> import pandas as pd
>>> import matplotlib.pyplot as plt
>>> pd.set_option('display.max_columns',None)
>>> pd.set_option('display.width',666666666)
>>> # ---
>>> # LocalHeatingNetwork SETUP
>>> # ---
>>> xmlFile=os.path.join(os.path.join(path,testDir),'LocalHeatingNetwork.XML')
>>> xm=Xm.Xm(xmlFile=xmlFile)
>>> mx1File=os.path.join(path,os.path.join(testDir,'WDLocalHeatingNetwork\B1\V0\BZ1\M-
↪1-0-1'+dotResolution+'.MX1'))
>>> mx=Mx.Mx(mx1File=mx1File,NoH5Read=True,NoMxsRead=True)
>>> mx.setResultsToMxsFile(NewH5Vec=True)
5
>>> xm.MxSync(mx=mx)
>>> rm=Rm(xm=xm,mx=mx)
>>> # ---
>>> # Plot 3Classes False
>>> # ---
>>> plt.close('all')
>>> ppi=72 # matplotlib default
>>> dpi_screen=2*ppi
>>> fig=plt.figure(dpi=dpi_screen,linewidth=1.)
>>> timeDeltaToT=mx.df.index[2]-mx.df.index[0]
>>> # 3Classes und FixedLimits sind standardmaessig Falsch; RefPerc ist␣
↪standardmaessig Wahr
>>> # die Belegung von MCategory gemaess FixedLimitsHigh/Low erfolgt immer ...
>>> pFWVB=rm.pltNetDHUS(timeDeltaToT=timeDeltaToT,pFWVBMeasureCBFixedLimitHigh=0.80,
↪pFWVBMeasureCBFixedLimitLow=0.66,pFWVBGCategory=['BLNZ1u5u7'],pVICsDf=pd.DataFrame({
↪'Kundenname': ['VIC1'],'Knotenname': ['V-K007']}))
>>> # ---
>>> # Check pFWVB Return
>>> # ---
>>> f=lambda x: "{0:8.5f}".format(x)
>>> print(pFWVB[['Measure','MCategory','GCategory','VIC']].to_string(formatters={
↪'Measure':f}))
   Measure MCategory  GCategory   VIC
0  0.80971       Top  BLNZ1u5u7   NaN
1  0.66692    Middle              NaN
2  0.66043    Middle  BLNZ1u5u7   NaN
3  0.65551    Bottom  BLNZ1u5u7  VIC1
4  0.68548    Middle              NaN
>>> # ---
>>> # Print
>>> # ---
>>> (wD,fileName)=os.path.split(xm.xmlFile)
>>> (base,ext)=os.path.splitext(fileName)
>>> plotFileName=wD+os.path.sep+base+'.'+'pdf'
>>> if os.path.exists(plotFileName):
...    os.remove(plotFileName)
```

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```
>>> plt.savefig(plotFileName,dpi=2*dpi_screen)
>>> os.path.exists(plotFileName)
True
>>> # ---
>>> # Plot 3Classes True
>>> # ---
>>> plt.close('all')
>>> # FixedLimits wird automatisch auf Wahr gesetzt wenn 3Classes Wahr ...
>>> pFWVB=rm.pltNetDHUS(timeDeltaToT=timeDeltaToT,pFWVBMeasure3Classes=True,
↪pFWVBMeasureCBFixedLimitHigh=0.80,pFWVBMeasureCBFixedLimitLow=0.66)
>>> # ---
>>> # LocalHeatingNetwork Clean Up
>>> # ---
>>> if os.path.exists(mx.h5File):
...     os.remove(mx.h5File)
>>> if os.path.exists(mx.mxsZipFile):
...     os.remove(mx.mxsZipFile)
>>> if os.path.exists(mx.h5FileVecs):
...     os.remove(mx.h5FileVecs)
>>> if os.path.exists(plotFileName):
...     os.remove(plotFileName)
```

`Rm.pltNetFigAx`(*pDf*, ***kwds*)

Erzeugt eine für die Netzdarstellung verzerrungsfreie Axes-Instanz.

- verwendet gcf() (will return an existing figure if one is open, or it will make a new one if there is no active figure)
- an already existing figure might be created this way: fig=plt.figure(dpi=2*72,linewidth=1.)
- errechnet die verzerrungsfreie Darstellung unter Berücksichtigung einer zukünftigen horizontalen Farblegende
- erzeugt eine Axes-Instanz
- setzt Attribute der Axes-Instanz
- setzt Attribute der Figure-Instanz

Args: pDf: dataFrame

Coordinates:

- pXCor_i: colName in pDf (default: 'pXCor_i'): x-Start Coordinate of all Edges to be plotted
- pYCor_i: colName in pDf (default: 'pYCor_i'): y-Start Coordinate of all Edges to be plotted
- pXCor_k: colName in pDf (default: 'pXCor_k'): x-End Coordinate of all Edges to be plotted
- pYCor_k: colName in pDf (default: 'pYCor_k'): y-End Coordinate of all Edges to be plotted

Colorlegend:

- CBFraction: fraction of original axes to use for colorbar (default: 0.05)
- CBHpad: fraction of original axes between colorbar and new image axes (default: 0.0275)

Figure:

- pltTitle: title [not suptitle] (default: 'pltNetFigAx')
- figFrameon: figure frame (background): displayed or invisible (default: True)

- figEdgecolor: edge color of the Figure rectangle (default: 'black')
- figFacecolor: face color of the Figure rectangle (default: 'white')

`Rm.pltNetNodes`(*pDf*, ***kwds*)

Scatters NODEs on gca().

Args: pDf: dataFrame

NODE: Size (Attribute)

- pAttribute: colName (default: 'Attribute') in pDf
- pSizeFactor: (deafault: 1.)
- scatter Sy-Area in pts^2 = pSizeFactor * Attribute

NODE: Color (Measure)

- pMeasure: colName (default: 'Measure') in pDf
- pMeasureColorMap (default: plt.cm.autumn)
- pMeasureAlpha (default: 0.9)
- pMeasureClip (default: False)
- CBFixedLimits (default: True)
- CBFixedLimitLow (default: 0.)
- CBFixedLimitHigh (default: 1.)

NODE: 3Classes

- pMeasure3Classes (default: True)
- pMCategory: colName (default: 'MCategory') in pDf
- pMCatTopTxt (default: 'Top')
- pMCatMidTxt (default: 'Middle')
- pMCatBotTxt (default: 'Bottom')
- pMCatTopColor (default: 'palegreen')
- pMCatTopAlpha (default: 0.9)
- pMCatTopClip (default: False)
- pMCatMidColorMap (default: plt.cm.autumn)
- pMCatMidAlpha (default: 0.9)
- pMCatMidClip (default: False)
- pMCatBotColor (default: 'violet')
- pMCatBotAlpha (default: 0.9)
- pMCatBotClip (default: False)

NODE:

- pXCor: colName (default: 'pXCor_i') in pDf
- pYCor: colName (default: 'pYCor_i') in pDf

Returns: (pcN, vmin, vmax)

- pcN: die mit Farbskala gezeichneten Symbole
- vmin/vmax: die für die Farbskala verwendeten Extremalwerte

`Rm.`**`pltNetPipes`**(*pDf*, ***kwds*)

Plots Lines with Marker on gca().

Args: pDf: dataFrame

PIPE-Line:

- pAttribute: column in pDf (default: 'Attribute')
- pAttributeLs (default: '-')
- pAttributeSizeFactor: plot linewidth in pts = pAttributeSizeFactor (default: 1.0) * Attribute
- pAttributeColorMap (default: plt.cm.binary)
- pAttributeColorMapUsageStart (default: 1./3; Wertebereich: [0,1])
  - Farbskala nach vorh. min./max. Wert
  - die Farbskala wird nur ab UsageStart genutzt
  - d.h. Werte die eine "kleinere" Farbe hätten, bekommen die Farbe von UsageStart

PIPE-Marker:

- pMeasure: column in pDf (default: 'Measure')
- pMeasureMarker (default: '.')
- pMeasureSizeFactor: plot markersize in pts = pMeasureSizeFactor (default: 1.0) * Measure
- pMeasureColorMap (default: plt.cm.cool)
- pMeasureColorMapUsageStart (default: 0.; Wertebereich: [0,1])
  - Farbskala nach vorh. min./max. Wert
  - die Farbskala wird nur ab UsageStart genutzt
  - d.h. Werte die eine "kleinere" Farbe hätten, bekommen die Farbe von UsageStart

PIPE:

- pWAYPXCors: column in pDf (default: 'pWAYPXCors')
- pWAYPYCors: column in pDf (default: 'pWAYPYCors')
- pClip (default: False)

`Rm.`**`pltNetLegendColorbar`**(*pc*, *pDf*, ***kwds*)

Erzeugt eine Axes cax für den Legendenbereich aus ax (=gca()) und zeichnet auf cax die Farblegende (die Farbskala mit allen Eigenschaften).

Args: pc: (eingefaerbte) PathCollection (aus pltNetNodes); wird für die Erzeugung der Farbskala zwingend benoetigt pDf: dataFrame (default: None)

Measure:

- pMeasure: colName in pDf (default: 'Measure')
- pMeasureInPerc: Measure wird interpretiert in Prozent [0-1] (default: True)
- pMeasure3Classes (default: False d.h. Measure wird nicht in 3 Klassen dargestellt)

CBFixedLimits (Ticks):

- CBFixedLimits (default: False d.h. Farbskala nach vorh. min./max. Wert)
- CBFixedLimitLow (default: .10)
- CBFixedLimitHigh (default: .95)

Label:

- pMeasureUNIT (default: '[]')
- pMeasureTYPE (default: '')

CB

- CBFraction: fraction of original axes to use for colorbar (default: 0.05)
- CBHpad: fraction of original axes between colorbar and new image axes (default: 0.0275)
- CBLabelPad (default: -50)
- CBTicklabelsHPad (default: 0.)
- CBAspect: ratio of long to short dimension (default: 10.)
- CBShrink: fraction by which to shrink the colorbar (default: 0.3)
- CBAnchorHorizontal: horizontaler Fußpunkt der colorbar in Plot-% (default: 0.)
- CBAnchorVertical: vertikaler Fußpunkt der colorbar in Plot-% (default: 0.2)

Return: cax

`Rm.`**`pltNetLegendColorbar3Classes`**(*pDf*, ***kwds*)

Zeichnet auf gca() die ergaenzenden Legendeninformationen bei 3 Klassen.

- scatters the Top-Symbol
- scatters the Bot-Symbol
- the "Mid-Symbol" is the (already existing) colorbar with (already existing) ticks and ticklabels

Args:

pDf: dataFrame

Category:

- pMCategory: colName in pDf (default: 'MCategory')
- pMCatTopText
- pMCatMidText
- pMCatBotText

CBLegend (3Classes) - Parameterization of the representative Symbols

- CBLe3cTopVPad (default: 1+1*1/4)
- CBLe3cMidVPad (default: .5)
- CBLe3cBotVPad (default: 0-1*1/4)
  - 1 is the height of the Colorbar
  - the VPads (the vertical Sy-Positions) are defined in cax.transAxes Coordinates
  - cax is the Colorbar Axes
- CBLe3cSySize=10**2 (Sy-Area in pts^2)

- CBLe3cSyType='o'

Color:

- pMCatBotColor='violet'
- pMCatTopColor='palegreen'

Returns: (bbTop, bbMid, bbBot): the boundingBoxes of the 3Classes-Symbols

class Rm.**Rm**(*xm=None*, *mx=None*)

classmethod pltNetPipes(*pDf*, ***kwds*)

Plots colored PIPES.

Args:

DATA:

pDf: dataFrame

- query: query to filter pDf; default: None; Exp.: ='CONT_ID == '1001'"
- fmask: function to filter pDf; default: None; Exp.: =lambda row: True if row.KVR_i=='2' and row.KVR_k=='2' else False
- query and fmask are used both (query 1st) if not None
- sort_values_by: list of colNames defining the plot order; default: None
- sort_values_ascending; default: False (d.h. kleine zuletzt und damit i.d.R. auch dünne über dicke)

AXES: pAx: Axes to be plotted on; if not specified: gca() is used

Colorlegend:

- CBFraction in % (default: 5)
- CBHpad (default: 0.05)
- CBLabel (default: pAttribute/pAttributeFunc)

PIPE-Attribute:

- pAttribute: column in pDf (default: 'Attribute')
- **pAttributeFunc:**
  - function to be used to construct a new col to be plotted
  - if pAttributeFunc is not None pAttribute is not used: pAttribute is set to 'pAttributeFunc'
  - the new constructed col is named 'pAttributeFunc'; this name can be used in sort_values_by

PIPE-Color:

- pAttributeColorMap (default: plt.cm.cool)
- Farbskalamapping:
-
- pAttributeColorMapMin (default: pAttribute.min()); ordnet der kleinsten Farbe einen Wert zu

- pAttributeColorMapMax (default: pAttribute.max()); ordnet der größten Farbe einen Wert zu
- Standard: Farbskala wird (wenn die Werte die vorstehend parametrierten Min/Max erreichen) voll ausgenutzt
- **pAttributeColorMapUsageStart (default: 0.; Wertebereich: [0,1[)**
  - hier: die Farbskala wird unten nur ab UsageStart genutzt …
  - … d.h. Werte die eine "kleinere" Farbe hätten, bekommen die Farbe von UsageStart
- **pAttributeColorMapUsageEnd (default: 1.; Wertebereich:]0,1])**
  - hier: die Farbskala wird oben nur bis UsageEnd genutzt …
  - … d.h. Werte die eine "größere" Farbe hätten, bekommen die Farbe von UsageEnd

PIPE-Color 2nd:

- um "unwichtige" Bereiche zu "dimmen":
- Beispiele:
- räumlich: nicht-Schnitt Bereiche; Bestand (2nd) vs. Ausbau; Zonen unwichtig (2nd) vs. Zonen wichtig
- "Ampel": "uninteressante" Wertebereiche
- es wird dieselbe Spalte pAttribute/pAttributeFunc für die 2. Farbskala verwendet
- es wird derselbe Linienstil (pAttributeLs) für die 2. Farbskala verwendet
- es wird dieselbe Dicke pAttrLineSize (pAttribute/pAttributeFunc) für die 2. Farbskala verwendet
- nur die Farbskala ist anders sowie ggf. das Farbskalamapping
- pAttributeColorMapFmask: function to filter pDf to decide to plot with colorMap; default: =lambda row: True
- pAttributeColorMap2ndFmask: function to filter pDf to decide to plot with colorMap2nd; default: =lambda row: False
- mit den beiden Funktionsmasken kann eine Filterung zusätzlich zu query und fmask realisiert werden
- die Funktionsmasken sollten schnittmengenfrei sein - wenn nicht: 2nd überschreibt
- pAttributeColorMap2nd (default: plt.cm.binary)
- Farbskalamapping:
-
- pAttributeColorMap2ndMin (default: pAttributeColorMapMin)
- pAttributeColorMap2ndMax (default: pAttributeColorMapMax)
- pAttributeColorMap2ndUsageStart (default: 0.; Wertebereich: [0,1[)
- pAttributeColorMap2ndUsageEnd (default: 1.; Wertebereich:]0,1])

PIPE-Linestyle:

- pAttributeLs (default: '-')
- same for all colors if mutliple colors are specified

PIPE-Linesize:

- pAttrLineSize: column in pDf; if not specified: pAttribute/pAttributeFunc
- pAttrLineSizeFactor (>0): plot linewidth in pts = pAttrLineSizeFactor (default: ...) * fabs(pAttrLineSize)
- ...: 1./(pDf[pAttrLineSize].std()*2.)
- same for all colors if mutliple colors are specified

PIPE-Geometry:

- pWAYPXCors: column in pDf (default: 'pWAYPXCors')
- pWAYPYCors: column in pDf (default: 'pWAYPYCors')
- pClip (default: True)

```
>>> import pandas as pd
>>> import matplotlib.pyplot as plt
>>> import matplotlib.gridspec as gridspec
>>> import math
>>> # ---
>>> xm=xms['DHNetwork']
>>> mx=mxs['DHNetwork']
>>> # ---
>>> plt.close()
>>> size_DINA3quer=(16.5, 11.7)
>>> plt.rc('figure',figsize=size_DINA3quer)
>>> plt.rc('figure',dpi=72)
>>> plt.rc('savefig',dpi=72*2)
>>> fig=plt.figure()
>>> gs = gridspec.GridSpec(4, 2)
>>> # ---
>>> vROHR=xm.dataFrames['vROHR']
>>> # ---
>>> import Rm
>>> # ---
>>> # Attribute with neg. Values
>>> # --------------------------
>>> axNfd = fig.add_subplot(gs[0])
>>> Rm.Rm.pltNetPipes(vROHR
...     ,query="CONT_ID == '1001'"
...     ,fmask=lambda row: True if row.KVR_i=='2' and row.KVR_k=='2'␣
↪else False
...     ,pAx=axNfd
...     ,pAttribute='ROHR~*~*~*~QMAV'
...     )
>>> # ---
>>> # Function as Attribute
>>> # --------------------------
>>> axNfd = fig.add_subplot(gs[1])
>>> Rm.Rm.pltNetPipes(vROHR
...     ,query="CONT_ID == '1001'"
...     ,fmask=lambda row: True if row.KVR_i=='2' and row.KVR_k=='2'␣
↪else False
...     ,pAx=axNfd
...     ,pAttributeFunc=lambda row: math.fabs(row['ROHR~*~*~*~QMAV'])
...     )
>>> # --------------------------
```

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```
>>> # ---
>>> # Farbskalamapping u. -eigenschaften u. Plotorder groß auf klein
>>> # --------------------------
>>> axNfd = fig.add_subplot(gs[2])
>>> Rm.Rm.pltNetPipes(vROHR
...     ,query="CONT_ID == '1001'"
...     ,fmask=lambda row: True if row.KVR_i=='2' and row.KVR_k=='2'␣
↪else False
...     ,pAx=axNfd
...     ,pAttributeFunc=lambda row: math.fabs(row['ROHR~*~*~*~QMAV'])
...     ,pAttributeColorMapMin=0.
...     ,pAttributeColorMapMax=1600.
...     ,CBLabel='Q [t/h]'
...     ,sort_values_by=['pAttributeFunc']
...     ,sort_values_ascending=True
...     )
>>> # --------------------------
>>> # ---
>>> # Farbskalamapping u. -eigenschaften u. Plotorder klein auf groß
>>> # --------------------------
>>> axNfd = fig.add_subplot(gs[3])
>>> Rm.Rm.pltNetPipes(vROHR
...     ,query="CONT_ID == '1001'"
...     ,fmask=lambda row: True if row.KVR_i=='2' and row.KVR_k=='2'␣
↪else False
...     ,pAx=axNfd
...     ,pAttributeFunc=lambda row: math.fabs(row['ROHR~*~*~*~QMAV'])
...     ,pAttributeColorMapMin=0.
...     ,pAttributeColorMapMax=1600.
...     ,CBLabel='Q [t/h]'
...     ,sort_values_by=['pAttributeFunc']
...     )
>>> # --------------------------
>>> # ---
>>> # Farbskalamapping u. -eigenschaften u. Plotorder  groß auf klein /␣
↪2nd Color
>>> # --------------------------
>>> vAGSN=xm.dataFrames['vAGSN']
>>> hpRL=vAGSN[(vAGSN['LFDNR']=='1') & (vAGSN['Layer']==2)]
>>> pDf=pd.merge(vROHR
...     ,hpRL[hpRL.IptIdx=='S']
...     ,how='left'
...     ,left_on='pk'
...     ,right_on='OBJID'
...     ,suffixes=('','_AGSN')).filter(items=vROHR.columns.tolist()+[
↪'OBJID'])
>>> axNfd = fig.add_subplot(gs[4])
>>> Rm.Rm.pltNetPipes(pDf
...     ,query="CONT_ID == '1001'"
...     ,fmask=lambda row: True if row.KVR_i=='2' and row.KVR_k=='2'␣
↪else False
...     ,pAx=axNfd
...     ,pAttributeFunc=lambda row: math.fabs(row['ROHR~*~*~*~QMAV'])
...     ,pAttributeColorMapMin=0.
...     ,pAttributeColorMapMax=1600.
...     ,CBLabel='Q [t/h]'
...     ,sort_values_by=['pAttributeFunc']
```

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```
...     ,sort_values_ascending=True
...     ,pAttributeColorMapFmask=lambda row: True if not pd.isnull(row.
↪OBJID) else False
...     ,pAttributeColorMap2ndFmask=lambda row: True if pd.isnull(row.
↪OBJID) else False
...     )
>>> # --------------------------
>>> # ---
>>> # Farbskalamapping u. -eigenschaften u. Plotorder  groß auf klein /␣
↪2nd Color / UsageLimits
>>> # --------------------------
>>> axNfd = fig.add_subplot(gs[5])
>>> Rm.Rm.pltNetPipes(pDf
...     ,query="CONT_ID == '1001'"
...     ,fmask=lambda row: True if row.KVR_i=='2' and row.KVR_k=='2'␣
↪else False
...     ,pAx=axNfd
...     ,pAttributeFunc=lambda row: math.fabs(row['ROHR~*~*~*~QMAV'])
...     ,pAttributeColorMapMin=0.
...     ,pAttributeColorMapMax=1600.
...     ,CBLabel='Q [t/h]'
...     ,sort_values_by=['pAttributeFunc']
...     ,sort_values_ascending=True
...     ,pAttributeColorMapFmask=lambda row: True if not pd.isnull(row.
↪OBJID) else False
...     ,pAttributeColorMap2ndFmask=lambda row: True if pd.isnull(row.
↪OBJID) else False
...     ,pAttributeColorMap2ndUsageStart=.5/5. # nicht zu weiß
...     ,pAttributeColorMap2ndUsageEnd=2.5/5. # nicht zu schwarz
...     ,pAttributeColorMapUsageStart=5/16.
...     ,pAttributeColorMapUsageEnd=11/16.
...     )
>>> # --------------------------
>>> gs.tight_layout(fig)
>>> plt.savefig('pltNetPipes.pdf',format='pdf')
```

pltNetDHUS(***kwds*)

Plot: Net: DistrictHeatingUnderSupply.

Args (optional):

TIMEs (als TIMEDELTA zu Szenariumbeginn):

- timeDeltaToRef: Reference Scenariotime (for MeasureInRefPerc-Calculations) (default: pd.to_timedelta('0 seconds'))
- timeDeltaToT: Scenariotime (default: pd.to_timedelta('0 seconds'))

FWVB

- **pFWVBFilterFunction: Filterfunction to be applied to FWVB to determine the FWVB to be plotted**
  - **default: lambda df: (df.CONT_ID.astype(int).isin([1001])) & (df.W0LFK>0)**
    - **CONT_IDisIn: [1001]**
      - um zu vermeiden, dass FWVB aus Bloecken gezeichnet werden (unwahrscheinlich, dass es solche gibt)

        * **W0LFK>0:**
            · um zu vermeiden, dass versucht wird, FWVB mit der Soll-Leistung 0 zu zeichnen (pFWVBAttribute default is 'W0LFK')

FWVB Attribute (Size, z-Order)

- pFWVBAttribute: columnName (default: 'W0LFK')
  - the column must be able to be converted to a float
  - the conversion is done before FilterFunction
  - **see ApplyFunction and NaNValue for conversion details:**
    * **pFWVBAttributeApplyFunction: Function to be applied to column pFWVBAttribute**
      · default: lambda x: pd.to_numeric(x,errors='coerce')
    * **pFWVBAttributeApplyFunctionNaNValue: Value for NaN-Values produced by pFWVBAttribu**
      · default: 0
      · .fillna(pFWVBAttributeApplyFunktionNaNValue).astype(float) is called after ApplyFunction
- pFWVBAttributeAsc: z-Order (default: False d.h. "kleine auf große")
- pFWVBAttributeRefSize: scatter Sy-Area in pts^2 of for RefSizeValue (default: 10**2)
  - corresponding RefSizeValue is Attribute.std() or Attribute.mean() if Attribute.std() is < 1

FWVB (plot only large (small, medium) FWVB . . .)

- quantil_pFWVBAttributeHigh <= (default: 1.)
- quantil_pFWVBAttributeLow >= (default: .0)
- default: all FWVB are plotted
- note that Attribute >0 is a precondition

FWVB Measure (Color)

- pFWVBMeasure (default: 'FWVB~*~*~*~W')
  - float() must be possible
- pFWVBMeasureInRefPerc (default: True d.h. Measure wird verarbeitet in Prozent T zu Ref)
  - 0-1
  - if refValue is 0 than refPerc-Result is set to 1
- pFWVBMeasureAlpha/Colormap/Clip
- 3Classes
  - **pFWVBMeasure3Classes (default: False)**
    * **False:**
      · Measure wird nicht in 3 Klassen dargestellt

        - die Belegung von MCategory gemaess FixedLimitsHigh/Low erfolgt dennoch
    - CatTexts (werden verwendet wenn 3Classes Wahr gesetzt ist)
        - für CBLegend (3Classes) als _zusätzliche Beschriftung rechts
        - als Texte für die Spalte MCategory in return pFWVB
        - pMCatTopText
        - pMCatMidText
        - pMCatBotText
    - CatAttribs (werden verwendet wenn 3Classes Wahr gesetzt ist)
        - für die Knotendarstellung
        - pMCatTopAlpha/Color/Clip
        - pMCatMidAlpha/Colormap/Clip
        - pMCatBotAlpha/Color/Clip
- CBFixedLimits
    - pFWVBMeasureCBFixedLimits (default: False d.h. Farbskala nach vorh. min./max. Wert)
        - wird Wahr gesetzt sein, wenn 3Classes Wahr gesetzt ist
        - damit die mittlere Farbskala den Klassengrenzen "gehorcht"
    - pFWVBMeasureCBFixedLimitLow (default: .10)
    - pFWVBMeasureCBFixedLimitHigh (default: .95)

CB

- CBFraction: fraction of original axes to use for colorbar (default: 0.05)
- CBHpad: fraction of original axes between colorbar and new image axes (default: 0.0275)
- CBLabelPad (default: -50)
- CBTicklabelsHPad (default: 0.)
- CBAspect: ratio of long to short dimension (default: 10.)
- CBShrink: fraction by which to shrink the colorbar (default: .3)
- CBAnchorHorizontal: horizontaler Fußpunkt der colorbar in Plot-% (default: 0.)
- CBAnchorVertical: vertikaler Fußpunkt der colorbar in Plot-% (default: 0.2)

CBLegend (3Classes) - Parameterization of the representative Symbols

- CBLe3cTopVPad (default: 1+1*1/4)
- CBLe3cMidVPad (default: .5)
- CBLe3cBotVPad (default: 0-1*1/4)
    - 1 is the height of the Colorbar
    - the VPads (the vertical Sy-Positions) are defined in cax.transAxes Coordinates
    - cax is the Colorbar Axes

- CBLe3cSySize=10**2 (Sy-Area in pts^2)
- CBLe3cSyType='o'

ROHR

- **pROHRFilterFunction: Filterfunction to be applied to PIPEs to determine the PIPEs to be plotted**
  - **default: lambda df: (df.KVR.astype(int).isin([2])) & (df.CONT_ID.astype(int).isin([1001])) & (df.**
    - **KVRisIn: [2]**
      - 1: supply-line
      - 2: return-line
    - **CONT_IDisIn: [1001]**
      - um zu vermeiden, dass Rohre aus Bloecken gezeichnet werden (deren Koordinaten nicht zu den Koordinaten von Rohren aus dem Ansichtsblock passen)
    - **DI>0:**
      - um zu vermeiden, dass versucht wird, Rohre mit dem Innendurchmesser 0 zu zeichnen (pROHRAttribute default is 'DI')

ROHR (PIPE-Line: Size and Color, z-Order)

- **pROHRAttribute: columnName (default: 'DI')**
  - the column must be able to be converted to a float
  - the conversion is done before FilterFunction
  - **see ApplyFunction and NaNValue for conversion details:**
    - **pROHRAttributeApplyFunction: Function to be applied to column pROHRAttribute**
      - default: lambda x: pd.to_numeric(x,errors='coerce')
    - **pROHRAttributeApplyFunctionNaNValue: Value for NaN-Values produced by pROHRAttri**
      - default: 0
      - .fillna(pROHRAttributeApplyFunktionNaNValue).astype(float) is called after ApplyFunction
- pROHRAttributeAsc: z-Order (default: False d.h. "kleine auf grosse")
- pROHRAttributeLs (default: '-')
- pROHRAttributeRefSize: plot linewidth in pts for RefSizeValue (default: 1.0)
  - corresponding RefSizeValue is Attribute.std() or Attribute.mean() if Attribute.std() is < 1
- pROHRAttributeColorMap (default: plt.cm.binary)
- pROHRAttributeColorMapUsageStart (default: 1./3; Wertebereich: [0,1])
  - Farbskala nach vorh. min./max. Wert
  - die Farbskala wird nur ab UsageStart genutzt

  - d.h. Werte die eine "kleinere" Farbe haetten, bekommen die Farbe von UsageStart

ROHR (plot only large (small, medium) pipes . . .)

- quantil_pROHRAttributeHigh <= (default: 1.)
- quantil_pROHRAttributeLow >= (default: .75)
- default: only the largest 25% are plotted
- note that Attribute >0 is a precondition

ROHR (PIPE-Marker: Size and Color)

- pROHRMeasure columnName (default: 'ROHR~*~*~*~QMAV')
- pROHRMeasureApplyFunction: Function to be applied to column pROHRMeasure (default: lambda x: math.fabs(x))
- pROHRMeasureMarker (default: '.')
- pROHRMeasureRefSize: plot markersize for RefSizeValue in pts (default: 1.0)
  - corresponding RefSizeValue is Measure.std() or Measure.mean() if Measure.std() is < 1
  - if pROHRMeasureRefSize is None: plot markersize will be plot linewidth
- pROHRMeasureColorMap (default: plt.cm.cool)
- pROHRMeasureColorMapUsageStart (default: 0.; Wertebereich: [0,1])
  - Farbskala nach vorh. min./max. Wert
  - die Farbskala wird nur ab UsageStart genutzt
  - d.h. Werte die eine "kleinere" Farbe hätten, bekommen die Farbe von UsageStart

NRCVs - NumeRiCal Values to be displayed

- **pFIGNrcv: List of Sir3sID RegExps to be displayed (i.e. ['KNOT~PKON-Knoten~S*~S+~QM']) defau**
  the 1st Match is used if a RegExp matches more than 1 Channel

  further Examples for RegExps (and corresponding Texts):

  - WBLZ~WärmeblnzGes~S*~S+~WES (Generation)
  - WBLZ~WärmeblnzGes~S*~S+~WVB (Load)
  - WBLZ~WärmeblnzGes~S*~S+~WVERL (Loss)

  WBLZ~[S]+~S*~S+~S+: Example for a RegExp matching all Channels with OBJTYPE WBLZ
- pFIGNrcvTxt: corresponding (same length required!) List of Texts (i.e. ['Kontrolle DH']) default: None
- **pFIGNrcvFmt (i.e. '{:12s}: {:8.2f} {:6s}')**
  - Text (from pFIGNrcvTxt)
  - Value
  - UNIT (determined from Channel-Data)
- **pFIGNrcvPercFmt (i.e. ' {:6.1f}%')**
  - ValueInRefPercent
  - if refValue==0: 100%

- pFIGNrcvXStart (.5 default)
- pFIGNrcvYStart (.5 default)

Category - User Heat Balances to be displayed

- pFWVBGCategory: List of Heat Balances to be displayed (i.e. ['BLNZ1u5u7']) default: None
- pFWVBGCategoryUnit: Unit of all these Balances (default: '[kW]'])
- pFWVBGCategoryXStart (.1 default)
- pFWVBGCategoryYStart (.9 default)
- **pFWVBGCategoryCatFmt (i.e. '{:12s}: {:6.1f} {:4s}')**
  - Category NAME
  - Category Load
  - pFWVBGCategoryUnit
- **pFWVBGCategoryPercFmt (i.e. ' {:6.1f}%')**
  - Last Ist/Soll
- **pFWVBGCategory3cFmt (i.e. ' {:5d}/{:5d}/{:5d}')**
  - NOfTops
  - NOfMids
  - NOfBots

VICs - VeryImportantCustomers whose Values to be displayed

- **pVICsDf: DataFrame with VeryImportantCustomers (Text & Specification)**

  columns expected:

  - Kundenname (i.e. 'VIC1') - Text
  - Knotenname (i.e. 'V-K007') - Specification by Supply-Node

  i.e.: pd.DataFrame({'Kundenname': ['VIC1'],'Knotenname': ['V-K007']})

- **pVICsPercFmt (i.e. '{:12s}: {:6.1f}%')**
  - Kundenname
  - Load in Percent to Reference
- **pVICsFmt (i.e. '{:12s}: {:6.1f} {:6s}')**
  - Kundenname
  - Load
  - pFWVBGCategoryUnit
- pVICsXStart (.5 default)
- pVICsYStart (.1 default)

Figure:

- pltTitle: title [not suptitle] (default: 'pltNetFigAx')

- figFrameon: figure frame (background): displayed or invisible (default: True)
- figEdgecolor: edge color of the Figure rectangle (default: 'black')
- figFacecolor: face color of the Figure rectangle (default: 'white')

Returns:

pFWVB

- **columns changed (compared to vFWVB):**
  - pFWVBAttribute (wg. z.B. pFWVBAttributeApplyFunction und .astype(float))
- **columns added (compared to vFWVB):**
  - Measure (in % zu Ref wenn pFWVBMeasureInRefPer=True)
  - MeasureRef (Wert von Measure im Referenzzustand)
  - MeasureOrig (Wert von Measure)
  - **MCategory: str (Kategorisierung von Measure mit FixedLimitHigh/Low-Werten):**
    - TopText or
    - MidText or
    - BotText
  - GCategory: list (non-empty only if req. GCategories are a subset of the available Categories and object belongs to a req. Category)
  - VIC (filled with Kundenname from pVICsDf)
- **rows (compared to vFWVB):**
  - pFWVB enthaelt dieselben Objekte wie vFWVB
  - aber: die geplotteten Objekte sind ggf. nur eine Teilmenge (wg. z.B. pFWVB-FilterFunction)

exception Rm.**RmError**(*value*)

CHAPTER

FIVE

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